

SMA 2102: CALCULUS II

©Francis O. Ochieng

francokech@gmail.com

YouTube Channel: Prof. Oketch Maths Lab*

(<https://bit.ly/3S0ua7t>)

Department of Pure and Applied Mathematics
Jomo Kenyatta University of Agriculture and Technology

Course Description

- Parametric and implicit differentiation involving higher order derivatives. Applications to equations of tangent line and normal line.
- Curve sketching and asymptotes.
- Hyperbolic functions: their definitions, their differentiation and integration.
- Techniques of integration: powers of trigonometric functions, standard substitutions including trigonometric and hyperbolic functions, integration by parts, integration via partial fraction, integration via t -substitution.
- Solutions of first order ordinary differential equations by separation of variables method.
- Applications of integration to: kinematics including simple harmonic motion, economics, arc length, plane and surface area, and volume in Cartesian coordinates.
- Numerical integration: mid-ordinate rule, trapezoidal rule and Simpson's rule.
- Complex numbers: Argand diagrams, arithmetic operators and geometric representation. De Moivre's theorem and application to trigonometric identities, roots of complex numbers.

References

- [1] Calculus: Early Transcendentals (8th Edition) by James Stewart
- [2] Calculus with Analytic Geometry by Roland E. Larson, Robert P. Hostetler and Bruce H. Edwards; 5th edition
- [3] Calculus and Analytical Geometry (9th edition) by George B. Thomas and Ross L. Finney
- [4] Advanced Engineering Mathematics (10th ed.) by Erwin Kreyszig
- [5] Calculus by Larson Hostellem

LECTURE 1

*<https://www.youtube.com/channel/UC3AD2-agUgS0SXtkyXFle9A>

1 Parametric and Implicit Differentiation

1.1 Parametric differentiation

Consider the curve defined by the functions

$$x = x(t), \quad y = y(t) \quad (1)$$

If the functions in equation (1) have the same domain, then the set of points (x, y) for t in this domain is called the parametric representation of the curve given by equation (1). The first derivative, $\frac{dy}{dx}$ (or y') is given by

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

Since $\frac{dy}{dx}$ is found in terms of the parameter t and $\frac{d^2y}{dx^2}$ (or y'') requires differentiation with respect to x , we use the chain rule of differentiation. Thus,

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dt} \left(\frac{dy}{dx} \right) \times \frac{dt}{dx} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{dx/dt}$$

Similarly, $\frac{d^3y}{dx^3}$ (or y''') is given by

$$\frac{d^3y}{dx^3} = \frac{d}{dx} \left(\frac{d^2y}{dx^2} \right) = \frac{d}{dt} \left(\frac{d^2y}{dx^2} \right) \times \frac{dt}{dx} = \frac{\frac{d}{dt} \left(\frac{d^2y}{dx^2} \right)}{dx/dt},$$

etc.

Example(s):

1. If $x = t^3 + 1$, $y = t^2 - t + 4$, find $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$ and $\frac{d^3y}{dx^3}$ in terms of t .

Solution

$\frac{dx}{dt} = 3t^2$, $\frac{dy}{dt} = 2t - 1$. Therefore, chain rule yields

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2t - 1}{3t^2}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \cdot \frac{dt}{dx} = \frac{d}{dt} \left(\frac{2t - 1}{3t^2} \right) \cdot \left(\frac{1}{3t^2} \right) = \frac{3t^2(2) - (2t - 1)(6t)}{9t^4} \cdot \left(\frac{1}{3t^2} \right) = \frac{2(1 - t)}{9t^5}$$

$$\begin{aligned} \frac{d^3y}{dx^3} &= \frac{d}{dt} \left(\frac{d^2y}{dx^2} \right) \cdot \frac{dt}{dx} = \frac{d}{dt} \left(\frac{2 - 2t}{9t^5} \right) \cdot \left(\frac{1}{3t^2} \right) = \left(\frac{9t^5(-2) - (2 - 2t)(45t^4)}{81t^{10}} \right) \cdot \left(\frac{1}{3t^2} \right) \\ &= \frac{8t - 10}{27t^8} \end{aligned}$$

2. The parametric equations of a curve are $x = e^t$, $y = \sin t$. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ as functions of t .

Hence, show that $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = 0$.

Solution

$\frac{dx}{dt} = e^t$, $\frac{dy}{dt} = \cos t$. Therefore, chain rule yields

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\cos t}{e^t} = e^{-t} \cos t$$

$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \div \frac{dx}{dt} = \frac{d}{dt} (e^{-t} \cos t) \div e^t = \frac{-e^{-t} \sin t - e^{-t} \cos t}{e^t} = e^{-2t} (-\sin t - \cos t)$$

Now,

$$\begin{aligned} x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y &= e^{2t} [e^{-2t} (-\sin t - \cos t)] + e^t [e^{-t} \cos t] + \sin t = -\sin t - \cos t + \cos t + \sin t \\ &= 0 \end{aligned}$$

3. Given the curve $x = \sin 2\theta$, $y = \cos 2\theta$. Find $\frac{dy}{dx}$, $\frac{d^2 y}{dx^2}$ and $\frac{d^3 y}{dx^3}$ as functions of θ .

Solution

$\frac{dx}{d\theta} = 2 \cos 2\theta$, $\frac{dy}{d\theta} = -2 \sin 2\theta$. Therefore, chain rule yields

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy/d\theta}{dx/d\theta} = \frac{-2 \sin 2\theta}{2 \cos 2\theta} = -\tan 2\theta \\ \frac{d^2 y}{dx^2} &= \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{d\theta} (-\tan 2\theta) \cdot \frac{d\theta}{dx} = (-2 \sec^2 2\theta) \cdot \left(\frac{1}{2 \cos 2\theta} \right) = -\sec^3 2\theta \\ \frac{d^3 y}{dx^3} &= \frac{d}{dx} \left(\frac{d^2 y}{dx^2} \right) = \frac{d}{d\theta} (-\sec^3 2\theta) \cdot \frac{d\theta}{dx} = (-6 \sec^3 2\theta \tan 2\theta) \cdot \left(\frac{1}{2 \cos 2\theta} \right) = -3 \sec^4 2\theta \tan 2\theta \end{aligned}$$

4. If $x = A(t + \sin^{-1} t)$, $y = A(1 - \sqrt{1 - t^2})$ for some constant A , show that $\frac{dy}{dx} = \frac{t}{1 + \sqrt{1 - t^2}}$.
Hence, find the value of $\frac{d^2 y}{dx^2}$ at $t = \frac{1}{\sqrt{2}}$.

Solution

Recall that if $p = \sin^{-1} t$, then $\sin p = t$. Differentiating with respect to t on both sides yields

$$\cos p \frac{dp}{dt} = 1 \quad \Rightarrow \quad \frac{dp}{dt} = \frac{1}{\cos p} = \frac{1}{\sqrt{1 - \sin^2 p}} = \frac{1}{\sqrt{1 - t^2}}.$$

Now,

$$\begin{aligned} \frac{dx}{dt} &= A \left(1 + \frac{1}{\sqrt{1 - t^2}} \right) = A \left(\frac{1 + \sqrt{1 - t^2}}{\sqrt{1 - t^2}} \right) \\ \frac{dy}{dt} &= A \left(0 - \frac{1}{2} (1 - t^2)^{-1/2} (-2t) \right) = \frac{At}{\sqrt{1 - t^2}} \\ \therefore \frac{dy}{dx} &= \frac{dy/dt}{dx/dt} = \frac{At}{\sqrt{1 - t^2}} \cdot \frac{\sqrt{1 - t^2}}{A(1 + \sqrt{1 - t^2})} = \frac{t}{1 + \sqrt{1 - t^2}} \end{aligned}$$

Then,

$$\begin{aligned} \frac{d^2 y}{dx^2} &= \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{t}{1 + \sqrt{1 - t^2}} \right) = \frac{d}{dt} \left(\frac{t}{1 + \sqrt{1 - t^2}} \right) \frac{dt}{dx} \\ &= \frac{(1 + \sqrt{1 - t^2})(1) - \frac{t}{2} (1 - t^2)^{-1/2} (-2t)}{(1 + \sqrt{1 - t^2})^2} \cdot \frac{\sqrt{1 - t^2}}{A(1 + \sqrt{1 - t^2})} \\ &= \frac{\sqrt{1 - t^2} \left[(1 + \sqrt{1 - t^2}) + \frac{t^2}{\sqrt{1 - t^2}} \right]}{A(1 + \sqrt{1 - t^2})^3} = \frac{\sqrt{1 - t^2} + 1 - t^2 + t^2}{A(1 + \sqrt{1 - t^2})^3} = \frac{1}{A(1 + \sqrt{1 - t^2})^2} \end{aligned}$$

Therefore, at $t = \frac{1}{\sqrt{2}}$, we have

$$\begin{aligned} \frac{d^2y}{dx^2} \Big|_{t=\frac{1}{\sqrt{2}}} &= \frac{1}{A \left(1 + \sqrt{1 - \left(\frac{1}{\sqrt{2}}\right)^2}\right)^2} = \frac{1}{A \left(1 + \sqrt{1 - \frac{1}{2}}\right)^2} = \frac{1}{A \left(1 + \sqrt{\frac{1}{2}}\right)^2} \\ &= \frac{1}{A \left(1 + \frac{1}{\sqrt{2}}\right)^2} = \frac{1}{A \left(\frac{\sqrt{2} + 1}{\sqrt{2}}\right)^2} = \frac{2}{A (\sqrt{2} + 1)^2} = \frac{2(\sqrt{2} - 1)^2}{A (\sqrt{2} + 1)^2 (\sqrt{2} - 1)^2} \\ &= \frac{2(\sqrt{2} - 1)^2}{A(2 - 1)^2} = \frac{2}{A}(3 - 2\sqrt{2}) \end{aligned}$$

5. If $x = A[\cos t + \ln(\tan(t/2))]$ and $y = A \sin t$ for some constant A , find $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$.

Solution

$$\begin{aligned} \frac{dx}{dt} &= A \left[-\sin t + \frac{1}{\tan(t/2)} \cdot \sec^2(t/2) \cdot \frac{1}{2} \right] = A \left[-\sin t + \frac{\cos(t/2)}{2 \sin(t/2) \cos^2(t/2)} \right] \\ &= A \left[-\sin t + \frac{1}{\sin t} \right] = A \left[\frac{1 - \sin^2 t}{\sin t} \right] = \frac{A \cos^2 t}{\sin t} \\ \frac{dy}{dt} &= A \cos t \\ \therefore \frac{dy}{dx} &= \frac{dy/dt}{dx/dt} = \frac{A \cos t}{\frac{A \cos^2 t}{\sin t}} = A \cos t \cdot \frac{\sin t}{A \cos^2 t} = \tan t \end{aligned}$$

Then,

$$\begin{aligned} \frac{d^2y}{dx^2} &= \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} (\tan t) = \frac{d}{dt} (\tan t) \frac{dt}{dx} \\ &= \sec^2 t \cdot \frac{\sin t}{A \cos^2 t} = \frac{1}{\cos^2 t} \cdot \frac{\sin t}{A \cos^2 t} = \frac{\sin t}{A \cos^4 t} \end{aligned}$$

Therefore, at $t = \frac{\pi}{4}$, we have

$$\frac{d^2y}{dx^2} \Big|_{t=\frac{\pi}{4}} = \frac{\sin(\pi/4)}{A \cos^4(\pi/4)} = \frac{1/\sqrt{2}}{A(1/\sqrt{2})^4} = \frac{1/\sqrt{2}}{A/4} = \frac{4}{A\sqrt{2}} = \frac{4\sqrt{2}}{2A} = \frac{2\sqrt{2}}{A}$$

6. If $x = \tan^{-1}(\frac{2t}{1-t^2})$ and $y = \sin^{-1}(\frac{2t}{1+t^2})$, show that $\frac{dy}{dx} = 1$.

Solution

Given, $x = \tan^{-1}(\frac{2t}{1-t^2})$. Taking tangent on both sides yields $\tan x = \frac{2t}{1-t^2}$. Differentiating with respect to t on both sides yields $\sec^2 x \frac{dx}{dt} = \frac{(1-t^2)(2) - (2t)(-2t)}{(1-t^2)^2}$. Making $\frac{dx}{dt}$ the subject yields

$$\begin{aligned} \frac{dx}{dt} &= \frac{1}{\sec^2 x} \cdot \frac{(1-t^2)(2) - (2t)(-2t)}{(1-t^2)^2} = \frac{1}{1 + \tan^2 x} \cdot \frac{2 - 2t^2 + 4t^2}{(1-t^2)^2} = \frac{1}{1 + \frac{4t^2}{(1-t^2)^2}} \cdot \frac{2 - 2t^2 + 4t^2}{(1-t^2)^2} \\ &= \frac{(1-t^2)^2}{(1-t^2)^2 + 4t^2} \cdot \frac{2(1+t^2)}{(1-t^2)^2} = \frac{2(1+t^2)}{(1-t^2)^2 + 4t^2} = \frac{2(1+t^2)}{(1+t^2)^2} \\ &= \frac{2}{1+t^2} \end{aligned}$$

Given $y = \sin^{-1}\left(\frac{2t}{1+t^2}\right)$. Taking sine on both sides yields $\sin y = \frac{2t}{1+t^2}$. Differentiating with respect to t on both sides yields $\cos y \frac{dy}{dt} = \frac{(1+t^2)(2) - (2t)(2t)}{(1+t^2)^2}$. Making $\frac{dy}{dt}$ the subject yields

$$\begin{aligned} \frac{dy}{dt} &= \frac{1}{\cos y} \cdot \frac{(1+t^2)(2) - (2t)(2t)}{(1+t^2)^2} = \frac{1}{\sqrt{1-\sin^2 y}} \cdot \frac{2+2t^2-4t^2}{(1+t^2)^2} = \frac{1}{\sqrt{1-\frac{4t^2}{(1+t^2)^2}}} \cdot \frac{2+2t^2-4t^2}{(1+t^2)^2} \\ &= \frac{(1+t^2)}{\sqrt{(1+t^2)^2-4t^2}} \cdot \frac{2(1-t^2)}{(1+t^2)\sqrt{(1-t^2)^2}} = \frac{2(1-t^2)}{(1+t^2)(1-t^2)} \\ &= \frac{2}{1+t^2} \end{aligned}$$

Therefore, chain rule yields

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2/(1+t^2)}{2/(1+t^2)} = 1.$$

7. Differentiate $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ with respect to $\tan^{-1} x$.

Solution

Let $u = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ and $v = \tan^{-1} x$. We need to find $\frac{du}{dv}$. Now,

Given $u = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right) \Rightarrow \tan u = \frac{\sqrt{1+x^2}-1}{x}$. Differentiating with respect to x on both sides yields

$$\sec^2 u \frac{du}{dx} = \frac{x\left(\frac{x}{\sqrt{1+x^2}}\right) - (\sqrt{1+x^2}-1)(1)}{x^2}.$$

Making $\frac{du}{dx}$ the subject yields

$$\begin{aligned} \frac{du}{dx} &= \frac{1}{\sec^2 u} \cdot \frac{x\left(\frac{x}{\sqrt{1+x^2}}\right) - (\sqrt{1+x^2}-1)(1)}{x^2} = \frac{1}{1+\tan^2 u} \cdot \frac{\frac{x^2}{\sqrt{1+x^2}} - (\sqrt{1+x^2}-1)}{x^2} \\ &= \frac{1}{1+\frac{(\sqrt{1+x^2}-1)^2}{x^2}} \cdot \frac{\frac{x^2}{\sqrt{1+x^2}} - (\sqrt{1+x^2}-1)}{x^2} = \frac{x^2}{x^2 + (\sqrt{1+x^2}-1)^2} \cdot \frac{\sqrt{1+x^2}-1}{x^2\sqrt{1+x^2}} \\ &= \frac{\sqrt{1+x^2}-1}{[x^2 + (\sqrt{1+x^2}-1)^2]\sqrt{1+x^2}} = \frac{\sqrt{1+x^2}-1}{[2+2x^2-2\sqrt{1+x^2}]\sqrt{1+x^2}} \\ &= \frac{\sqrt{1+x^2}-1}{2[(1+x^2)\sqrt{1+x^2} - (1+x^2)]} = \frac{\sqrt{1+x^2}-1}{2(1+x^2)(\sqrt{1+x^2}-1)} \\ &= \frac{1}{2(1+x^2)} \end{aligned}$$

Also, given $v = \tan^{-1} x \Rightarrow \tan v = x$. Differentiating with respect to x on both sides yields $\sec^2 v \frac{dv}{dx} = 1$. Making $\frac{dv}{dx}$ the subject yields

$$\frac{dv}{dx} = \frac{1}{\sec^2 v} = \frac{1}{1+\tan^2 v} = \frac{1}{1+x^2}$$

Therefore, chain rule yields

$$\frac{du}{dv} = \frac{du/dx}{dv/dx} = \frac{\frac{1}{2(1+x^2)}}{\frac{1}{1+x^2}} = \frac{1}{2(1+x^2)} \cdot \frac{(1+x^2)}{1} = \frac{1}{2}$$

Exercise:

1. Given that $x = \sec \theta + \tan \theta$ and $y = \operatorname{cosec} \theta + \cot \theta$, show that $x + \frac{1}{x} = 2 \sec \theta$ and $y + \frac{1}{y} = 2 \operatorname{cosec} \theta$.

Find $\frac{dx}{d\theta}$ and $\frac{dy}{d\theta}$ in terms of θ , and hence show that $\frac{dy}{dx} = -\left(\frac{1+y^2}{1+x^2}\right)$.

2. (a) If $x = \frac{1+t}{1-2t}$ and $y = \frac{1+2t}{1-t}$, find the value of $\frac{dy}{dx}$ at $t = 0$.

(b) If $x = \sin^2 t$ and $y = \ln(\cos t)$, find the value of $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$. [ans: $= -2$]

3. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ given

(a) $x = t - \sin t$ and $y = 1 - \sin t$

(b) $y = \sin^{-1} 2t$ and $x = \cos^{-1} 2t$

LECTURE 2

1.2 Implicit differentiation

An equation $f(x, y) = c$, on certain restricted ranges of the variables x and y , is said to be defined implicitly for some constant c . For instance, the equation $xy + x - 2y = 1$ is implicit since y is not the subject. To find $\frac{dy}{dx}$ (or y'), follow these steps:

- i) Differentiate x normally
- ii) Apply direct chain rule in differentiating y
- iii) Collect like terms and make $\frac{dy}{dx}$ the subject

Example(s):

1. Find y' given that $xy + x - 2y = 1$.

Solution

Differentiating the given equation with respect to x , we get

$$xy' + y + 1 - 2y' = 0 \quad \Rightarrow \quad y' = \frac{y+1}{2-x}$$

2. Find y' when $x = \sqrt{5}$ given that $4x^2 + 9y^2 = 36$.

Solution

Differentiating the given equation with respect to x , we get

$$8x + 18yy' = 0 \quad \Rightarrow \quad y' = \frac{-8x}{18y} = \frac{-4x}{9y}.$$

When $x = \sqrt{5}$ we have $4(\sqrt{5})^2 + 9y^2 - 36 = 0 \quad \Rightarrow \quad 20 + 9y^2 - 36 = 0 \quad \Rightarrow \quad y^2 = \frac{16}{9}.$

Therefore, $y = \pm \frac{4}{3}$.

□ At the point $(\sqrt{5}, 4/3)$, $y' = \frac{-4\sqrt{5}}{9(4/3)} = \frac{-\sqrt{5}}{3}.$

$$\square \text{ At the point } (\sqrt{5}, -4/3), y' = \frac{-4\sqrt{5}}{9(-4/3)} = \frac{\sqrt{5}}{3}.$$

Exercise:

1. Find $\frac{dy}{dx}$ for the following implicit functions

(a) $x^2y - xy^2 + x^2 + y^2 = 0.$

[ans: $y' = \frac{y^2 - 2xy - 2x}{x^2 - 2xy + 2y}$]

(b) $x^5 + y^5 = 5x^2y^2.$

[ans:]

(c) $\cos(x + y) = \sin x \sin y.$

[ans:]

1.2.1 Higher order derivatives of implicit functions

Higher order derivatives of implicit functions may be obtained in either of the two ways.

- $\square$ **Method I:** differentiate implicitly the derivatives of one lower order and replace y' by the relation previously found.
- $\square$ **Method II:** differentiate implicitly both sides of the given equation as many times as is necessary to produce the required derivative and eliminate all derivatives of lower order. This procedure is recommended only when a derivative of higher order at a given point is required.

Example(s):

1. Find y' and y'' given that $x^2 - xy + y^2 = 3.$

Solution

Differentiating the given equation with respect to x , we get

$$2x - xy' - y + 2yy' = 0 \Rightarrow y' = \frac{2x - y}{x - 2y}$$

Differentiating further with respect to x yields

$$\begin{aligned} y'' &= \frac{d}{dx} \left(\frac{2x - y}{x - 2y} \right) = \frac{(x - 2y)(2 - y') - (2x - y)(1 - 2y')}{(x - 2y)^2} \\ &= \frac{2x - xy' - 4y + 2yy' - 2x + 4xy' + y - 2yy'}{(x - 2y)^2} = \frac{3xy' - 3y}{(x - 2y)^2} = \frac{3x \left[\frac{2x - y}{x - 2y} \right] - 3y}{(x - 2y)^2} \\ &= \frac{3x [2x - y] - 3y[x - 2y]}{(x - 2y)^3} = \frac{6x^2 - 3xy - 3xy + 6y^2}{(x - 2y)^3} = \frac{6x^2 - 6xy + 6y^2}{(x - 2y)^3} \\ &= \frac{6(x^2 - xy + y^2)}{(x - 2y)^3} = \frac{6(3)}{(x - 2y)^3} = \frac{18}{(x - 2y)^3} \\ &\therefore y'' = \frac{18}{(x - 2y)^3} \end{aligned}$$

2. Find the value of y'' at the point $(-1, 1)$ on the curve $x^2y + 3y - 4 = 0.$

Solution

Differentiating the given equation with respect to x , we get

$$2xy + x^2y' + 3y' = 0. \text{ Putting } x = -1, y = 1 \text{ yields } y' = \frac{1}{2}$$

Differentiating further with respect to x yields

$$2xy' + 2y + x^2y'' + 2xy' + 3y'' = 0 \Rightarrow 4xy' + 2y + x^2y'' + 3y'' = 0$$

Putting $x = -1, y = 1$ and $y' = \frac{1}{2}$ yields $y'' = 0.$

3. Find y' and y'' at $x = 1$ given $x^3y + xy^3 = 2$. [ans: $y' = -1, y'' = 0$]

Solution

Putting $x = 1$ in the given equation we get

$$y + y^3 = 2 \Rightarrow y^3 + y - 2 = 0 \Rightarrow (y - 1)(y^2 + y + 2) = 0 \Rightarrow y = 1$$

So, we need to solve for y' and y'' at the point $(1, 1)$. Differentiating the given equation with respect to x , we get

$$x^3y' + 3x^2y + y^3 + 3xy^2y' = 0. \text{ Putting } x = 1, y = 1 \text{ yields } y' = -1$$

Differentiating further with respect to x yields

$$x^3y'' + 3x^2y' + 3x^2y' + 6xy + 3y^2y' + 3xy^2y'' + 3y^2y' + 6xy(y')^2 = 0$$

Putting $x = 1, y = 1$ and $y' = -1$ yields $y'' = 0$.

Exercise:

1. Find y' and y'' given

(a) $x + xy + y = 2$.

(b) $x^3 - 3xy + y^3 = 1$

(c) $2e^{-x} + e^y = 3e^{x-y}$

(d) $x^{2/3} + y^{2/3} = 5$

(e) $xy^2 + x^2y = 2$

(f) $\sqrt{xy} = x - 2y$

(g) $y^2 = \frac{x^2 - 9}{x^2 + 9}$

(h) $x^2 + 2xy + 3y^2 = 1$

[ans: $y' = -\frac{x+y}{x+3y}$ and $y'' = -\frac{2}{(x+3y)^3}$]

(i) $\cos^2 y + \sin^2 x = 1$

2. Find y', y'' and y''' at

(a) the point $(2, 1)$ on $x^2 - y^2 - x = 1$.

(b) the point $(1, 1)$ on $x^3 + 3x^2y - 6xy^2 + 2y^3 = 0$.

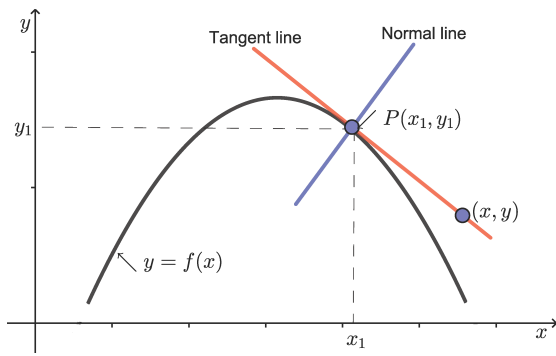
1.3 Applications of differentiation

1.3.1 Equations of tangent and normal lines

Let $P(x_1, y_1)$ be a point on the curve $y = f(x)$, as shown below. The equation of a *tangent line* to the curve at point P is given by

$$\frac{y - y_1}{x - x_1} = m \Rightarrow y = m(x - x_1) + y_1,$$

where $m = \left. \frac{dy}{dx} \right|_{(x_1, y_1)}$ is the gradient of the tangent line.



A *normal line* to the curve $y = f(x)$ is a line through P which is perpendicular to the tangent line at point P. If the gradient of the tangent line at P is m , then the gradient of a normal to a curve at P is $-\frac{1}{m}$ since the product of their gradients is -1 . Thus, the equation of the normal line at P is given by

$$y = -\frac{1}{m}(x - x_1) + y_1$$

Example(s):

- Find the equation of the tangent line and normal line to the curve $y^2 = 4x$ at point $(4, 4)$.

Solution

Given the curve $y^2 = 4x$. Clearly, the point $(4, 4)$ lies on the curve. Differentiating the given curve with respect to x yields $2ydy = 4dx \Rightarrow \frac{dy}{dx} = \frac{4}{2y} = \frac{2}{y}$. The gradient of the tangent line to the curve at point $(4, 4)$ is given by $m = \frac{dy}{dx}\bigg|_{(4,4)} = \frac{2}{4} = \frac{1}{2}$. Hence, the equation of the tangent line at point $(4, 4)$ is

$$y = \frac{1}{2}(x - 4) + 4 \Rightarrow y = \frac{1}{2}x + 2$$

The gradient of the normal line to the curve at point $(4, 4)$ is given by $-\frac{1}{m} = -2$. Hence, the equation of the normal line at point $(4, 4)$ is

$$y = -2(x - 4) + 4 \Rightarrow y = -2x + 12$$

- Find the equation of the normal line to the curve $xy = 4$ at $x = 3$.

Solution

Substituting $x = 3$ into the given curve $xy = 4$ yields $y = \frac{4}{3}$. Thus, the point of contact is $(3, 4/3)$.

Differentiating the given curve with respect to x yields $xdy + ydx = 0 \Rightarrow \frac{dy}{dx} = -\frac{y}{x}$. The gradient of the tangent line to the curve at point $(3, 4/3)$ is given by $m = \frac{dy}{dx}\bigg|_{(3, 4/3)} = -\frac{4/3}{3} = -\frac{4}{9}$.

The gradient of the normal line at point $(3, 4/3)$ is $\frac{9}{4}$. Hence, the equation of the normal line at point $(3, 4/3)$ is

$$y = \frac{9}{4}(x - 3) + \frac{4}{3} \Rightarrow 12y - 27x + 65 = 0$$

Exercise:

- Find the equation of the tangent line to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point (x_1, y_1) .

Solution

Given $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Differentiating yields $\frac{2xdx}{a^2} + \frac{2ydy}{b^2} = 0 \Rightarrow \frac{dy}{dx} = -\frac{b^2x}{a^2y}$. The gradient of the tangent line to the curve at point (x_1, y_1) is given by $m = \frac{dy}{dx}\bigg|_{(x_1, y_1)} = -\frac{b^2x_1}{a^2y_1}$. Hence, the

equation of the tangent line at point (x_1, y_1) is

$$y = -\frac{b^2 x_1}{a^2 y_1}(x - x_1) + y_1 \Rightarrow a^2 y_1(y - y_1) + b^2 x_1(x - x_1) = 0 \Rightarrow a^2 y y_1 + b^2 x x_1 = b^2 x_1^2 + a^2 y_1^2$$

Dividing through by $a^2 b^2$ yields

$$\frac{x x_1}{a^2} + \frac{y y_1}{b^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1$$

Therefore, $\frac{x x_1}{a^2} + \frac{y y_1}{b^2} = 1$.

2. Use implicit differentiation to find an equation of the line tangent to the curves

(a) $xy = 6e^{2x-3y}$ at point $(3, 2)$

(b) $\frac{1}{x+1} + \frac{1}{y+1} = 1$ at point $(1, 1)$

(c) $x^2 + (y - x)^3 = 0$ at $x = 1$.

[ans: $6y - 5x = 13$]

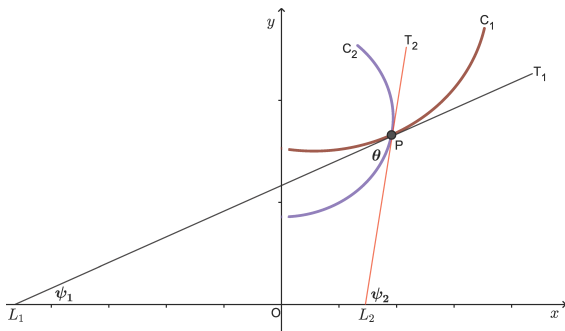
3. Find the equation of the tangent line and the normal line at the point for which $t = 2$, given

that the parametric equations of a curve are $x = \frac{3t}{1+t}, y = \frac{t^2}{1+t}$ [ans: tangent: $y = \frac{8}{3}x - 4$, normal: $y = -\frac{3}{8}x + \frac{25}{12}$]

4. Find the equations of the tangent lines to the curve $9x^2 + 16y^2 = 52$ that are parallel to the line $9x - 8y = 1$. [ans: $y = \frac{9}{8}x + \frac{13}{4}k$ where k is a parameter]

1.3.2 The intersection of two curves

The angle between two curves is the angle between the tangents to the two curves at the point of intersection. Let C_1 and C_2 be two plane curves intersecting each other at P and let PT_1 and PT_2 be tangents to C_1 and C_2 respectively at point P .



Angle T_1PT_2 is the angle of intersection of the curves C_1 and C_2 . Suppose PT_1 and PT_2 intersect the x -axis (OX) at L_1 and L_2 respectively. If angle $T_1L_1X = \psi_1$ and angle $T_2L_2X = \psi_2$. Then, $\theta = \psi_2 - \psi_1$. Now, the slope of the tangent to C_1 at P is $m_1 = \tan \psi_1$ and the slope of the tangent to C_2 at P is $m_2 = \tan \psi_2$. Thus,

$$\tan \theta = \tan(\psi_2 - \psi_1) = \frac{\tan \psi_2 - \tan \psi_1}{1 + \tan \psi_1 \tan \psi_2} = \frac{m_2 - m_1}{1 + m_1 m_2} \Rightarrow \theta = \tan^{-1} \left(\frac{m_2 - m_1}{1 + m_1 m_2} \right)$$

If $\theta = \pm \frac{\pi}{2}$, then C_1 and C_2 intersect orthogonally. Thus C_1 and C_2 are orthogonal if $m_1 m_2 = -1$.

Example(s):

1. Find the angle of intersection of the following curves.

(a) $x^2 + y^2 = 8$ and $x^2 = 2y$ at $(2, 2)$.

Solution

From $x^2 + y^2 = 8$. Differentiating yields $2x dx + 2y dy = 0 \Rightarrow \frac{dy}{dx} = -\frac{x}{y}$. So,

$$m_1 = \left. \frac{dy}{dx} \right|_{(2,2)} = -\frac{2}{2} = -1$$

From $x^2 = 2y$. Differentiating yields $2x dx = 2dy \Rightarrow \frac{dy}{dx} = x$. So,

$$m_2 = \left. \frac{dy}{dx} \right|_{(2,2)} = 2$$

Hence, if θ is the angle of intersection of the two curves then

$$\tan \theta = \frac{m_2 - m_1}{1 + m_1 m_2} = \frac{2 + 1}{1 + (-1)(2)} = -3 \Rightarrow \theta = \tan^{-1}(-3) = 108.4^\circ$$

(b) $y^2 = 4ax$ and $x^2 = 4by$ at a point different from the origin (0,0), where $a \neq b$.

Solution

At the point of intersection, the two curves are equal. Solving the two curves simultaneously: from $y^2 = 4ax$ we have $x = \frac{y^2}{4a}$. Substituting into $x^2 = 4by$ yields $\frac{y^4}{16a^2} = 4by \Rightarrow \frac{y^4}{16a^2} - 4by = 0 \Rightarrow y \left(\frac{y^3}{16a^2} - 4b \right) = 0$. Hence, $y = 0$ or $\left(\frac{y^3}{16a^2} - 4b \right) = 0 \Rightarrow y = 4a^{2/3}b^{1/3}$. When $y = 0, x = 0$ and when $y = 4a^{2/3}b^{1/3}, x = 4a^{1/3}b^{2/3}$. So, the points of intersection of the two curves are (0,0) and $(4a^{1/3}b^{2/3}, 4a^{2/3}b^{1/3})$. But (0,0) is unwanted. Therefore, the curves intersect at $P(4a^{1/3}b^{2/3}, 4a^{2/3}b^{1/3})$.

From $y^2 = 4ax$. Differentiating yields $2y dy = 4a dx \Rightarrow \frac{dy}{dx} = \frac{2a}{y}$. So,

$$m_1 = \left. \frac{dy}{dx} \right|_P = \frac{2a}{4a^{2/3}b^{1/3}} = \frac{a^{1/3}}{2b^{1/3}}$$

From $x^2 = 4by$. Differentiating yields $2x dx = 4b dy \Rightarrow \frac{dy}{dx} = \frac{x}{2b}$. So,

$$m_2 = \left. \frac{dy}{dx} \right|_P = \frac{4a^{1/3}b^{2/3}}{2b} = \frac{2a^{1/3}}{b^{1/3}}$$

Hence, if θ is the angle of intersection of the two curves then

$$\tan \theta = \frac{m_2 - m_1}{1 + m_1 m_2} = \frac{\frac{2a^{1/3}}{b^{1/3}} - \frac{a^{1/3}}{2b^{1/3}}}{1 + \frac{a^{1/3}}{2b^{1/3}} \cdot \frac{2a^{1/3}}{b^{1/3}}} = \frac{3}{2} \left[\frac{a^{1/3}b^{1/3}}{a^{2/3} + b^{2/3}} \right] \Rightarrow \theta = \tan^{-1} \left(\frac{3}{2} \left[\frac{a^{1/3}b^{1/3}}{a^{2/3} + b^{2/3}} \right] \right)$$

2. Show that the curves $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $\frac{x^2}{a_1^2} + \frac{y^2}{b_1^2} = 1$ will intersect orthogonally if $a^2 - b^2 = a_1^2 - b_1^2$.

Solution

At the point of intersection, the two curves are equal. Solving the two curves simultaneously: from $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ we have

$$x^2 = a^2 \left(1 - \frac{y^2}{b^2} \right) \Rightarrow x = a \sqrt{1 - \frac{y^2}{b^2}}.$$

Substituting in $\frac{x^2}{a_1^2} + \frac{y^2}{b_1^2} = 1$ yields $\frac{a^2}{a_1^2} \left(1 - \frac{y^2}{b^2} \right) + \frac{y^2}{b_1^2} = 1 \Rightarrow \left(\frac{1}{b_1^2} - \frac{a^2}{a_1^2 b^2} \right) y^2 = 1 - \frac{a^2}{a_1^2} \Rightarrow$

$y^2 = \frac{a_1^2 - a^2}{a_1^2} \cdot \frac{a_1^2 b_1^2 b^2}{a_1^2 b^2 - a^2 b_1^2}$. Hence,

$$y^2 = \frac{b^2 b_1^2 (a_1^2 - a^2)}{a_1^2 b^2 - a^2 b_1^2} \Rightarrow y = \frac{bb_1 \sqrt{a_1^2 - a^2}}{\sqrt{a_1^2 b^2 - a^2 b_1^2}}$$

Back substitution yields

$$x = a\sqrt{1 - \frac{y^2}{b^2}} = a\sqrt{1 - \frac{b_1^2(a_1^2 - a^2)}{a_1^2b^2 - a^2b_1^2}} = a\sqrt{\frac{a_1^2b^2 - a^2b_1^2 - a_1^2b_1^2 + a^2b_1^2}{a_1^2b^2 - a^2b_1^2}} = \frac{aa_1\sqrt{b^2 - b_1^2}}{\sqrt{a_1^2b^2 - a^2b_1^2}}$$

Therefore, the two curves intersect at point $P(x, y)$ where

$$x = \frac{aa_1\sqrt{b^2 - b_1^2}}{\sqrt{a_1^2b^2 - a^2b_1^2}} \text{ and } y = \frac{bb_1\sqrt{a_1^2 - a^2}}{\sqrt{a_1^2b^2 - a^2b_1^2}}$$

From $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Differentiating yields $\frac{2xdx}{a^2} + \frac{2ydy}{b^2} = 0 \Rightarrow \frac{dy}{dx} = \frac{-b^2x}{a^2y}$. So,

$$m_1 = \left. \frac{dy}{dx} \right|_P = \frac{-aa_1b^2\sqrt{b^2 - b_1^2}}{a^2bb_1\sqrt{a_1^2 - a^2}} = \frac{-a_1b\sqrt{b^2 - b_1^2}}{ab_1\sqrt{a_1^2 - a^2}}$$

From $\frac{x^2}{a_1^2} + \frac{y^2}{b_1^2} = 1$. Differentiating yields $\frac{2xdx}{a_1^2} + \frac{2ydy}{b_1^2} = 0 \Rightarrow \frac{dy}{dx} = \frac{-b_1^2x}{a_1^2y}$. So,

$$m_2 = \left. \frac{dy}{dx} \right|_P = \frac{-aa_1b_1^2\sqrt{b^2 - b_1^2}}{a_1^2b_1b\sqrt{a_1^2 - a^2}} = \frac{-ab_1\sqrt{b^2 - b_1^2}}{a_1b\sqrt{a_1^2 - a^2}}$$

Hence, the two curves will intersect orthogonally if $m_1m_2 = -1$.

$$\Rightarrow \frac{-a_1b\sqrt{b^2 - b_1^2}}{ab_1\sqrt{a_1^2 - a^2}} \cdot \frac{-ab_1\sqrt{b^2 - b_1^2}}{a_1b\sqrt{a_1^2 - a^2}} = -1 \Rightarrow \frac{b^2 - b_1^2}{a_1^2 - a^2} = -1 \Rightarrow a^2 - b^2 = a_1^2 - b_1^2$$

LECTURE 3

2 Curve Sketching

To sketch the graph of a function $y = f(x)$, follow these steps.

- i) Identify the turning points and classify the points as maximum, minimum or points of inflection by either the sign test (or first derivative test) or by the second derivative test.
- ii) Identify the x and y -intercepts.
- iii) Find the asymptotes: vertical, horizontal, and slant asymptotes.
- iv) Mark the above points on an unscaled Cartesian plane, along with the asymptotes, and use free-hand to join the points.

2.1 Intercepts

- i) To determine the x -intercept, put $y = 0$ in the given equation and solve for x .
- ii) To determine the y -intercept, put $x = 0$ in the given equation and solve for y .

2.2 Turning points

At every turning point, the slope is zero, i.e., $y'(x) = 0$. Let $(a, y(a))$ be the turning/critical point of the function $y = f(x)$. To classify the critical point, we compute the second derivative, $y''(x)$, and use the second derivative test:

- i) If $y''(a) > 0$, then point $(a, y(a))$ is a minimum point (concave up).
- ii) If $y''(a) < 0$, then $(a, y(a))$ is a maximum point (concave down).
- iii) If $y''(a) = 0$, then $(a, y(a))$ is a point of inflection.

2.3 Asymptotes

An asymptote to a curve is a straight line to which the curve approaches as the distance x from the origin increases. It can also be thought of as a tangent to the curve at infinity, i.e., the curve touches the asymptote at two coincident points at infinity. The asymptotes to be determined are: vertical asymptotes, horizontal asymptotes, and slant (or oblique) asymptotes.

□ **Vertical asymptotes:** the line $x = a$ is a vertical asymptote of the graph of the function $y = f(x)$ if either

$$\lim_{x \rightarrow a^-} f(x) = +\infty \quad (\text{or } -\infty) \quad \text{or} \quad \lim_{x \rightarrow a^+} f(x) = +\infty \quad (\text{or } -\infty)$$

→ Note: for rational functions such as $f(x) = \frac{p(x)}{q(x)}$, the vertical asymptotes are given by the values of x for which $q(x) = 0$ (i.e., the denominator is zero). For example, given the rational function $f(x) = \frac{x^2}{4 - x^2}$, the vertical asymptotes are the values of x for which $4 - x^2 = 0$, i.e., $x = 2$ and $x = -2$.

→ Alternatively: to find *vertical asymptotes* of the function $y = f(x)$, follow these steps.

- i) If $f(x)$ is a rational function, cross multiply the given function through by the denominator to get rid of the fractions.
- ii) Equate to zero the coefficient of the highest power of y and solve for x .

□ **Horizontal asymptotes:** the line $y = b$ is a horizontal asymptote of the graph of the function $y = f(x)$ if

$$\lim_{x \rightarrow \infty} f(x) = b$$

For example, given $f(x) = \frac{x^2}{4 - x^2}$, then the horizontal asymptote is given by

$$y = \lim_{x \rightarrow \infty} \left(\frac{x^2}{4 - x^2} \right) = -1,$$

i.e., the line $y = -1$.

→ Alternatively: to find *horizontal asymptotes* of the function $y = f(x)$, follow these steps.

- i) If $f(x)$ is a rational function, cross multiply the given function through by the denominator to get rid of the fractions.
- ii) Equate to zero the coefficient of the highest power of x and solve for y .

□ **Slant (or oblique) asymptotes:** if the function $f(x) = \frac{p(x)}{q(x)}$ can be rewritten to take the form $f(x) = (mx + c) + g(x)$ via long division in case the degree of x in $p(x)$ is greater than that in $q(x)$, then the part that represents the straight line equation ($y = mx + c$) is called the slant asymptote of the graph of the function $y = f(x)$. For example, given $f(x) = \frac{x^2 + x - 1}{x - 1}$. By long division, we have

$$\begin{array}{r} x + 2 \\ x - 1 \overline{) x^2 + x - 1} \\ \underline{-x^2 + x} \\ 2x - 1 \\ \underline{-2x + 2} \\ 1 \end{array}$$

Therefore, the given function can be written as $f(x) = (x + 2) + \frac{1}{x - 1}$. Hence, the line $y = x + 2$ is a slant asymptote for the graph of $y = f(x)$.

→ Alternatively: to find *slant asymptotes* of the function $y = f(x)$, follow these steps.

- i) If $f(x)$ is a rational function, cross multiply the given function through by the denominator to get rid of the fractions.
- ii) Substitute $y = mx + c$ into the given function and simplify by grouping the like terms together.
- iii) Equate to zero the coefficients of the two highest powers of x and solve the resulting equations simultaneously to determine the values of m and c .

→ Note:

- i) The graph of $y = f(x)$ will not intersect any of the above asymptotes since the asymptotes are the break points for the function $y = f(x)$.
- ii) The asymptotes are not part of the graph of the function $y = f(x)$ but they are tools necessary for sketching the graph.

Example(s):

1. Sketch the graph of the function $y = \frac{x^2}{4 - x^2}$.

Solution

i) Turning Points:

At a critical point, $y'(x) = 0$. Now, given $y = \frac{x^2}{4 - x^2}$. Differentiating with respect to x , we get $y'(x) = \frac{2x(4 - x^2) - x^2(-2x)}{(4 - x^2)^2} = \frac{8x}{(4 - x^2)^2}$. Therefore, the critical points are given by

$$\frac{8x}{(4 - x^2)^2} = 0 \quad \Rightarrow \quad 8x = 0 \quad \Rightarrow \quad x = 0$$

When $x = 0$, we have $y = 0$. Hence, the critical/turning point is (0,0). To classify the critical point, we first compute y'' and test as follows.

$$y''(x) = \frac{8(4 - x^2)^2 + 32x(4 - x^2)}{(4 - x^2)^4}$$

Now, $y''(0) = \frac{8(4-0^2)^2 + 32(0)(4-x^2)}{(4-0^2)^4} = 0.5 > 0 \Rightarrow (0,0)$ is a minimum point (concave up).

ii) **Intercepts:**

(a) **x -intercepts:** put $y = 0 \Rightarrow 0 = \frac{x^2}{4-x^2} \Rightarrow x^2 = 0 \Rightarrow x = 0$. So, x -intercept is the point $(0,0)$.

(b) **y -intercepts:** put $x = 0 \Rightarrow y = \frac{0^2}{4-0^2} = 0$. So, y -intercept is the point $(0,0)$.

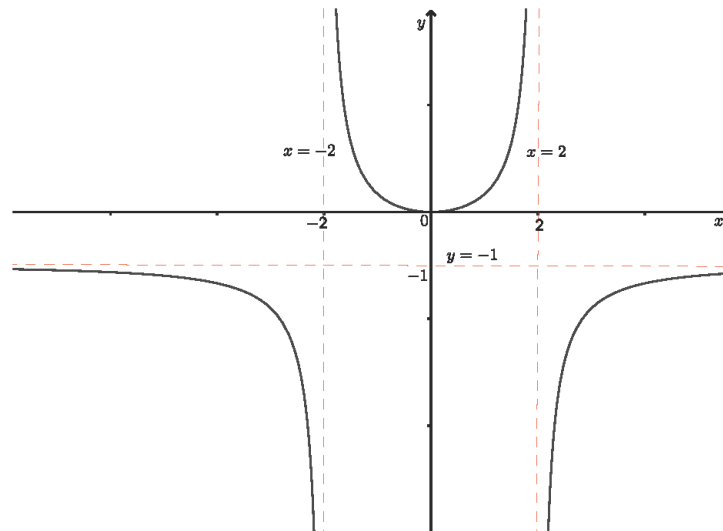
iii) **Asymptotes:** given $f(x) = \frac{x^2}{4-x^2}$.

(a) **vertical asymptotes:** equate to zero the denominator, i.e., $4-x^2 = 0 \Rightarrow x = 2$ and $x = -2$.

(b) **horizontal asymptotes:** $y = \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{x^2}{4-x^2} = -1$. Therefore, $y = -1$ is the horizontal asymptote.

(c) **slant asymptote:** No slant asymptote since the degree of the polynomial in the numerator and denominator are the same.

iv) **Sketch:**



2. Sketch the graph of the function $y = \frac{x^2 + 1}{x}$.

Solution

i) **Turning Points:**

At a critical point, $y'(x) = 0$. Now, given $y = \frac{x^2 + 1}{x}$. Differentiating with respect to x , we get $y'(x) = \frac{x^2 - 1}{x^2}$. Therefore, the critical points are given by

$$\frac{x^2 - 1}{x^2} = 0 \Rightarrow x^2 - 1 = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$$

When $x = -1, y = -2$, and when $x = 1, y = 2$. Hence, the critical/turning points are: $(-1, -2)$ and $(1, 2)$. To classify the critical points, we first compute $y''(x)$ and test as follows.

$$y''(x) = \frac{2}{x^3}$$

Now, $y''(-1) = \frac{2}{(-1)^3} = -2 < 0 \Rightarrow (-1, -2)$ is a maximum point (concave down), and
 $y''(1) = \frac{2}{(1)^3} = 2 > 0 \Rightarrow (1, 2)$ is a minimum point (concave up).

ii) **Intercepts:**

(a) **x -intercepts:** put $y = 0 \Rightarrow 0 = \frac{x^2 + 1}{x} \Rightarrow x^2 + 1 = 0$. No real roots. So, no x -intercept.

(b) **y -intercepts:** put $x = 0 \Rightarrow y = \frac{0^2 + 1}{0} = \infty$ (undefined). So, no y -intercept.

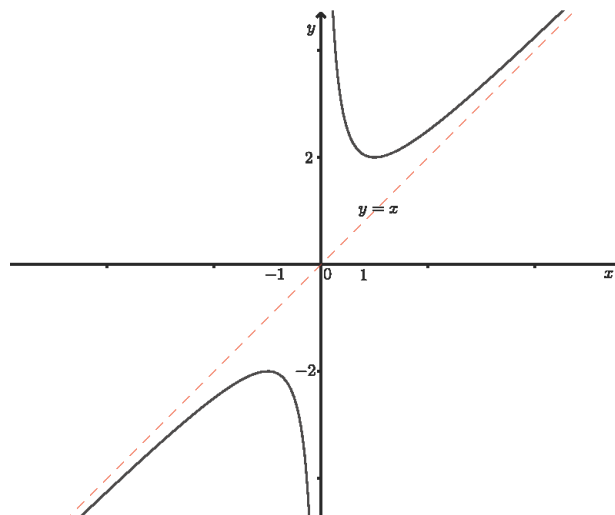
iii) **Asymptotes:** given $f(x) = \frac{x^2 + 1}{x}$.

(a) **vertical asymptotes:** equate to zero the denominator, i.e., $x = 0$ is a vertical asymptote.

(b) **horizontal asymptotes:** $y = \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{x^2 + 1}{x} = \infty$. Therefore, no horizontal asymptote.

(c) **slant asymptote:** By long division, the given function can be written as $f(x) = x + \frac{1}{x}$. Therefore, the line $y = x$ is a slant asymptote.

iv) **Sketch:**



3. Sketch the graph of $x^2y - 5y - x^3 = 0$.

Solution

i) **Turning Points:**

At a critical point, $y'(x) = 0$. Now, given $x^2y - 5y - x^3 = 0 \Rightarrow y = \frac{x^3}{x^2 - 5}$.

Differentiating with respect to x we get $y'(x) = \frac{(x^2 - 5)(3x^2) - x^3(2x)}{(x^2 - 5)^2} = \frac{x^4 - 15x^2}{(x^2 - 5)^2}$.

Therefore, the critical points are given by

$$\frac{x^4 - 15x^2}{(x^2 - 5)^2} = 0 \Rightarrow x^4 - 15x^2 = 0 \Rightarrow x^2(x^2 - 15) = 0 \Rightarrow x = 0, x = \pm\sqrt{15} = \pm 3.9$$

When $x = 0, y = 0$, when $x = -3.9, y = -5.8$, and when $x = 3.9, y = 5.8$. Hence, the critical/turning points are: $(0, 0)$, $(-3.9, -5.8)$ and $(3.9, 5.8)$. To classify the critical points, we first compute $y''(x)$ and test as follows.

$$y''(x) = \frac{(x^2 - 5)^2(4x^3 - 30x) - (x^4 - 15x^2)(2)(x^2 - 5)(2x)}{(x^2 - 5)^4}$$

Now, $y''(0) = 0 \Rightarrow (0,0)$ is an inflection point, $y''(-\sqrt{15}) = 0.3\sqrt{15} > 0 \Rightarrow (-3.9, -5.8)$ is a minimum point (concave up), and $y''(\sqrt{15}) = -0.3\sqrt{15} < 0 \Rightarrow (3.9, 5.8)$ is a maximum point (concave down).

ii) **Intercepts:**

(a) **x -intercepts:** put $y = 0 \Rightarrow 0 = \frac{x^3}{x^2 - 5} \Rightarrow x = 0$. Point is $(0,0)$.

(b) **y -intercepts:** put $x = 0 \Rightarrow y = \frac{0^3}{0^2 - 5} \Rightarrow y = 0$. Point is $(0,0)$.

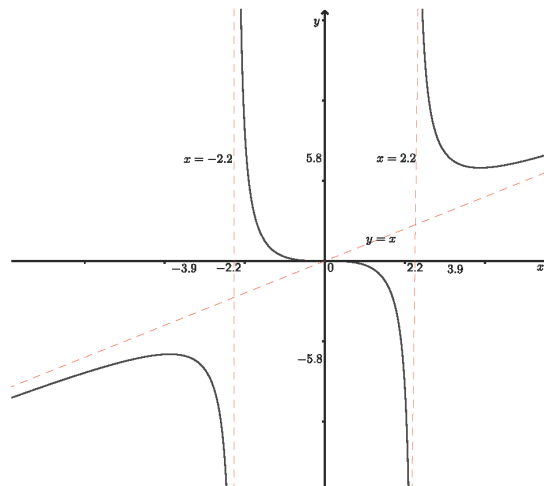
iii) **Asymptotes:** Making y the subject, we get $y = \frac{x^3}{x^2 - 5}$. Therefore, $f(x) = \frac{x^3}{x^2 - 5}$.

(a) **vertical asymptotes:** equate to zero the denominator, i.e., $x^2 - 5 = 0 \Rightarrow x = \pm\sqrt{5} = \pm 2.2$. Thus, $x = -2.2$ and $x = 2.2$ are the vertical asymptotes.

(b) **horizontal asymptotes:** $y = \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{x^3}{x^2 - 5} = \infty$. Therefore, no horizontal asymptote.

(c) **slant asymptote:** By long division, the given function can be written as $f(x) = x + \frac{5x}{x^2 - 5}$. Therefore, the line $y = x$ is a slant asymptote.

iv) **Sketch:**



4. Sketch the graph of $y^2 = \frac{x^4}{x^2 - 4}$.

Solution

i) **Turning Points:**

At a critical point, $y' = 0$. Now, given $y^2 = \frac{x^4}{x^2 - 4} \Rightarrow y^2(x^2 - 4) = x^4$. Differentiating with respect to x we get $2yy'(x^2 - 4) + 2xy^2 - 4x^3 = 0 \Rightarrow y' = \frac{4x^3 - 2xy^2}{2y(x^2 - 4)}$. Therefore, the critical points are given by

$$\frac{4x^3 - 2xy^2}{2y(x^2 - 4)} = 0 \Rightarrow 4x^3 - 2x \left(\frac{x^4}{x^2 - 4} \right) = 0 \Rightarrow x = 0, x = \pm\sqrt{8} = \pm 2.81$$

When $x = 0, y = 0$, when $x = -2.81, y = \pm 4$, and when $x = 2.81, y = \pm 4$. Hence, the critical/turning points are: $(0,0), (-2.81, -4), (-2.81, 4), (2.81, -4)$ and $(2.81, 4)$. To classify the critical points, we first compute $y''(x)$ and test as follows.

$$2(y')^2(x^2 - 4) + 2yy''(x^2 - 4) + 8xyy' + 2y^2 - 12x^2 = 0$$

$$y'' = \frac{12x^2 - 2y^2 - 8xyy' - 2(y')^2(x^2 - 4)}{2y(x^2 - 4)}$$

Now, y'' is not defined at point (0,0), hence (0,0) is not a turning point. Clearly, the points (-2.81,-4) and (2.81,-4) are the maximum points (concave down) while the points (-2.81,4) and (2.81,4) are the minimum points (concave up).

ii) **Intercepts:**

(a) **x -intercepts:** put $y = 0 \Rightarrow 0 - x^4 = 0 \Rightarrow x = 0$. Point is (0,0).

(b) **y -intercepts:** put $x = 0 \Rightarrow y^2(-4) = 0 \Rightarrow y = 0$. Point is (0,0).

iii) **Asymptotes:** Making y the subject, we get $y = \frac{x^2}{\sqrt{x^2 - 4}}$. Therefore, $f(x) = \frac{x^2}{\sqrt{x^2 - 4}}$.

(a) **vertical asymptotes:** equate to zero the denominator, i.e., $\sqrt{x^2 - 4} = 0$. Solving yields $x = \pm 2$. Thus, the lines $x = -2$ and $x = 2$ are the vertical asymptotes.

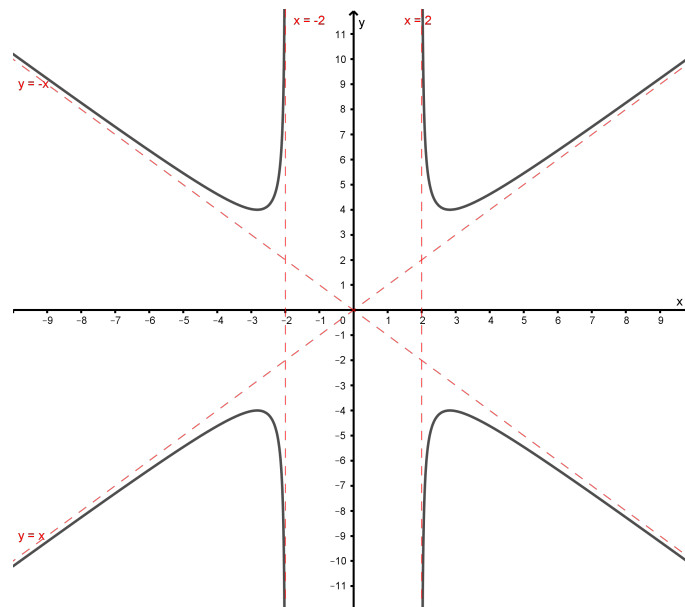
(b) **horizontal asymptotes:** $y = \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{x^2}{\sqrt{x^2 - 4}} = \infty$. Therefore, no horizontal asymptote.

(c) **slant asymptote:** The given function can be written as $y^2(x^2 - 4) - x^4 = 0$. Substituting $y = mx + c$ and simplifying yields

$$(m^2 - 1)x^4 + 2mcx^3 + (c^2 - 4m^2)x^2 - 8mcx - 4c^2 = 0$$

Equating to zero the coefficients of x^4 and x^3 yields $m^2 - 1 = 0$ and $2mc = 0$. Solving yields $m = \pm 1$ and $c = 0$. Therefore, the lines $y = -x$ and $y = x$ are the slant asymptotes.

iv) **Sketch:**



Exercise:

1. Sketch the graph of the given functions, stating the x - and y -intercepts, horizontal, vertical and slant asymptotes.

(a) $f(x) = \frac{2x^2}{x^2 - 1}$

(b) $f(x) = \frac{x^2 - 16}{x^2 - 9}$

(c) $f(x) = x^3 - 6x^2 + 9x$

$$\begin{aligned}
\text{(d)} \quad f(x) &= \frac{x^2 + x - 2}{x - 1} \\
\text{(e)} \quad f(x) &= \frac{2x^2 - 7x + 9}{x - 2} \\
\text{(f)} \quad f(x) &= \frac{x^2 + 4}{x^2 - 4} \\
\text{(g)} \quad f(x) &= x^3 - 3x^2 \\
\text{(h)} \quad f(x) &= \frac{(x + 2)(x + 3)}{x + 1}
\end{aligned}$$

2. Discuss and sketch the graph of the following functions

$$\begin{aligned}
\text{(a)} \quad y &= \frac{1}{x} + \frac{x}{4}. \\
\text{(b)} \quad y &= \frac{12}{x} + 3x + 1. \\
\text{(c)} \quad y &= \frac{(x + 2)(x - 3)}{x + 1}. \\
\text{(d)} \quad y &= \frac{4}{3}x^3 - 2x^2. \\
\text{(e)} \quad x^2(x^2 + 2) &= y^3(x + 5). \quad [\text{hint: asymptotes are } x = -5, y = x - \frac{5}{3}] \\
\text{(f)} \quad y &= \frac{x^3}{6} - x^2 + \frac{3x}{2} + 1.
\end{aligned}$$

3. A manufacturer of motorcycles estimates that if x thousands Kenya shillings are spent on advertising, then for $x > 0$

$$M(x) = 2300 + \frac{25}{x} - \frac{500}{x^2}$$

motorcycles will be sold. Discuss and sketch the curve of $M(x)$.

LECTURE 4

3 Hyperbolic Functions

Hyperbolic sine and hyperbolic cosine of a real number x are denoted by $\sinh(x)$ and $\cosh(x)$, respectively, and are defined in terms of the exponential function as

$$\sinh x = \frac{e^x - e^{-x}}{2} \quad \text{and} \quad \cosh x = \frac{e^x + e^{-x}}{2}$$

Thus, the hyperbolic tangent of x is obtained from $\sinh x$ and $\cosh x$ as:

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Similarly, the reciprocal hyperbolic functions (i.e., hyperbolic secant, hyperbolic cosecant, and hyperbolic cotangent) of x are given as

$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}, \quad \operatorname{cosech} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}, \quad \coth x = \frac{1}{\tanh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

3.1 Hyperbolic identities

Definition 3.1 (Osborne's rule). *The usual trigonometric identities will hold for hyperbolic functions so long as you change the sign whenever there is a product of sines.*

For instance,

$$\square \text{ From } \cos^2 x + \sin^2 x = 1, \text{ we have } \boxed{\cosh^2 x - \sinh^2 x = 1}.$$

$$\square \text{ From } \cos 2x = \cos^2 x - \sin^2 x, \text{ we have } \boxed{\cosh 2x = \cosh^2 x + \sinh^2 x}.$$

$$\square \text{ From } 1 + \tan^2 x = \sec^2 x, \text{ we have } \boxed{1 - \tanh^2 x = \operatorname{sech}^2 x}.$$

$$\square \text{ From } 1 + \cot^2 x = \operatorname{cosec}^2 x, \text{ we have } 1 - \coth^2 x = -\operatorname{cosech}^2 x \Rightarrow \boxed{\coth^2 x - 1 = \operatorname{cosech}^2 x}.$$

In general, the following addition formulae hold for hyperbolic functions

$$\sinh(x \pm y) = \sinh(x) \cosh(y) \pm \cosh(x) \sinh(y)$$

$$\cosh(x \pm y) = \cosh(x) \cosh(y) \pm \sinh(x) \sinh(y)$$

For instance, if $x = y$ then $\sinh(2x) = 2 \sinh(x) \cosh(x)$ and $\cosh(2x) = \cosh^2(x) + \sinh^2(x)$, etc.

Example(s):

1. Prove that $\cosh^2 x - \sinh^2 x = 1$.

Solution

$$\begin{aligned} \cosh^2 x - \sinh^2 x &= \left(\frac{e^x + e^{-x}}{2} \right)^2 - \left(\frac{e^x - e^{-x}}{2} \right)^2 \\ &= \frac{1}{4} (e^{2x} + 2 + e^{-2x}) - \frac{1}{4} (e^{2x} - 2 + e^{-2x}) \\ &= \frac{1}{4} (4) = 1 \end{aligned}$$

3.2 Derivatives of hyperbolic functions

$$\begin{aligned} \frac{d}{dx}(\sinh x) &= \cosh x, & \frac{d}{dx}(\cosh x) &= \sinh x, & \frac{d}{dx}(\tanh x) &= \operatorname{sech}^2 x, \\ \frac{d}{dx}(\operatorname{cosech} x) &= -\operatorname{cosech} x \coth x, & \frac{d}{dx}(\operatorname{sech} x) &= -\operatorname{sech} x \tanh x, & \frac{d}{dx}(\coth x) &= -\operatorname{cosech}^2 x, \end{aligned}$$

Example(s):

1. Show that $\frac{d}{dx}(\sinh x) = \cosh x$.

Solution

$$\frac{d}{dx}(\sinh x) = \frac{d}{dx} \left(\frac{e^x - e^{-x}}{2} \right) = \frac{1}{2} (e^x + e^{-x}) = \cosh x$$

2. Find $\frac{dy}{dx}$ given the following functions.

(a) $y = \sinh 3x$.

Solution

Let $u = 3x$ so that $y = \sinh u$. Therefore, $\frac{du}{dx} = 3$ and $\frac{dy}{du} = \cosh u$. By back substitution, we have $\frac{dy}{dx} = \cosh(3x)$. Thus, the chain rule of differentiation yields

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = 3 \cosh 3x$$

(b) $y = \cosh(x/2)$.

Solution

$$\frac{dy}{dx} = \sinh(x/2) \cdot \frac{d}{dx}(x/2) = \frac{1}{2} \sinh(x/2)$$

(c) $y = \tanh(1 + x^2)$.

Solution

$$\frac{dy}{dx} = \text{sech}^2(1 + x^2) \cdot \frac{d}{dx}(1 + x^2) = 2x \text{sech}^2(1 + x^2)$$

(d) $y = \coth(1/x)$.

Solution

$$\frac{dy}{dx} = -\text{cosech}^2(1/x) \cdot \frac{d}{dx}(1/x) = \frac{1}{x^2} \text{cosech}^2(1/x)$$

(e) $y = x \text{sech}(x^2)$.

Solution

$$\begin{aligned} \frac{dy}{dx} &= x \frac{d}{dx}[\text{sech}(x^2)] + \text{sech}(x^2) \frac{d}{dx}[x] \\ &= x[-\text{sech}(x^2) \tanh(x^2) \cdot \frac{d}{dx}(x^2)] + \text{sech}(x^2) \\ &= -2x^2 \text{sech} x^2 \tanh x^2 + \text{sech} x^2 \end{aligned}$$

(f) $y = \text{cosech}^2(x^2 + 1)$.

Solution

$$\begin{aligned} \frac{dy}{dx} &= 2 \text{cosech}(x^2 + 1) \cdot \frac{d}{dx}[\text{cosech}(x^2 + 1)] \\ &= 2 \text{cosech}(x^2 + 1) [-\text{cosech}(x^2 + 1) \coth(x^2 + 1)] \cdot \frac{d}{dx}(x^2 + 1) \\ &= -4x \text{cosech}^2(x^2 + 1) \coth(x^2 + 1) \end{aligned}$$

3. Find y' given the following functions.

(a) $y = \frac{1}{4} \sinh 2x - \frac{1}{2}x$.

Solution

$$\begin{aligned} y' &= \frac{1}{4} \frac{d}{dx}(\sinh 2x) - \frac{1}{2} \frac{d}{dx}(x) \\ &= \frac{1}{4} (2 \cosh 2x) - \frac{1}{2} = \frac{1}{2} \cosh 2x - \frac{1}{2} = \frac{1}{2} [\cosh 2x - 1] = \frac{1}{2} [2 \sinh^2 x] \\ &= \sinh^2 x \end{aligned}$$

(b) Find y' given $y = \ln(\tanh 2x)$.

Solution

$$\begin{aligned} y' &= \frac{d}{dx}[\ln(\tanh 2x)] = \frac{1}{\tanh 2x} \cdot \frac{d}{dx}(\tanh 2x) = \frac{1}{\tanh 2x} \operatorname{sech}^2 2x \cdot \frac{d}{dx}(2x) = \frac{2\operatorname{sech}^2 2x}{\tanh 2x} \\ &= \frac{2/\cosh^2 2x}{\sinh 2x/\cosh 2x} = \frac{2}{\sinh 2x \cosh 2x} = \frac{4}{2 \sinh 2x \cosh 2x} = \frac{4}{\sinh 4x} = 4\operatorname{cosech} 4x \end{aligned}$$

3.3 Inverse hyperbolic functions

$$\begin{aligned} \sinh^{-1} x &= \ln \left[x + \sqrt{1+x^2} \right] \text{ for all } x, & \operatorname{cosech}^{-1} x &= \ln \left[\frac{1 + \sqrt{1+x^2}}{x} \right] \text{ for } x \neq 0 \\ \cosh^{-1} x &= \ln \left[x + \sqrt{x^2-1} \right] \text{ for } x^2 \geq 1, & \operatorname{sech}^{-1} x &= \ln \left[\frac{1 + \sqrt{1-x^2}}{x} \right] \text{ for } 0 < x^2 \leq 1 \\ \tanh^{-1} x &= \frac{1}{2} \ln \left[\frac{1+x}{1-x} \right] \text{ for } x < 1, & \coth^{-1} x &= \frac{1}{2} \ln \left[\frac{x+1}{x-1} \right] \text{ for } x > 1 \end{aligned}$$

Example(s):

1. Derive the following:

(a) $\sinh^{-1} x = \ln \left[x + \sqrt{1+x^2} \right] \text{ for all } x.$

Solution

Let $y = \sinh^{-1} x \Rightarrow \sinh y = x \Rightarrow \frac{e^y - e^{-y}}{2} = x \Rightarrow e^y - e^{-y} - 2x = 0.$
 Multiplying by e^y yields

$$e^{y+y} - e^{y-y} - 2xe^y = 0 \Rightarrow e^{2y} - e^0 - 2xe^y = 0 \Rightarrow e^{2y} - 2xe^y - 1 = 0$$

This is a quadratic equation in e^y . Solving for e^y yields

$$\begin{aligned} e^y &= \frac{2x \pm \sqrt{4x^2 + 4}}{2} = \frac{2x \pm 2\sqrt{x^2 + 1}}{2} = x \pm \sqrt{x^2 + 1} \\ \Rightarrow \ln[e^y] &= \ln \left[x + \sqrt{x^2 + 1} \right] \Rightarrow y = \ln \left[x + \sqrt{x^2 + 1} \right] \end{aligned}$$

(b) $\operatorname{sech}^{-1} x = \ln \left[\frac{1 + \sqrt{1-x^2}}{x} \right] \text{ for } 0 < x \leq 1.$

Solution

Let $y = \operatorname{sech}^{-1} x \Rightarrow \operatorname{sech} y = x \Rightarrow \frac{2}{e^y + e^{-y}} = x \Rightarrow xe^y + xe^{-y} - 2 = 0.$
 Multiplying by e^y yields

$$xe^{2y} + xe^0 - 2e^y = 0 \Rightarrow xe^{2y} - 2e^y + x = 0$$

This is a quadratic equation in e^y . Solving for e^y yields

$$\begin{aligned} e^y &= \frac{2 \pm \sqrt{4 - 4x^2}}{2x} = \frac{2 \pm 2\sqrt{1-x^2}}{2x} = \frac{1 \pm \sqrt{1-x^2}}{x} \\ \Rightarrow \ln[e^y] &= \ln \left[\frac{1 + \sqrt{1-x^2}}{x} \right] \Rightarrow y = \ln \left[\frac{1 + \sqrt{1-x^2}}{x} \right] \end{aligned}$$

3.4 Derivatives of inverse hyperbolic functions

$$\begin{aligned}\frac{d}{dx}(\sinh^{-1} x) &= \frac{1}{\sqrt{1+x^2}}, & \frac{d}{dx}(\cosh^{-1} x) &= \frac{1}{\sqrt{x^2-1}}, & \frac{d}{dx}(\tanh^{-1} x) &= \frac{1}{1-x^2} \\ \frac{d}{dx}(\operatorname{cosech}^{-1} x) &= \frac{-1}{x\sqrt{1+x^2}}, & \frac{d}{dx}(\operatorname{sech}^{-1} x) &= \frac{-1}{x\sqrt{1-x^2}}, & \frac{d}{dx}(\coth^{-1} x) &= \frac{-1}{x\sqrt{x^2-1}}\end{aligned}$$

Example(s):

1. Show that $\frac{d}{dx}[\sinh^{-1} x] = \frac{1}{\sqrt{1+x^2}}$.

Proof. Let $y = \sinh^{-1} x \Rightarrow \sinh y = x$. Differentiating implicitly with respect to x yields

$$\cosh y \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{\cosh y} = \frac{1}{\sqrt{1+\sinh^2 y}} = \frac{1}{\sqrt{1+x^2}}$$

□

2. Find $\frac{dy}{dx}$ given that:

(a) $y = \tanh^{-1} x$.

Solution

$y = \tanh^{-1} x \Rightarrow \tanh y = x$. Differentiating implicitly yields

$$\operatorname{sech}^2 y \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{\operatorname{sech}^2 y} = \frac{1}{1 - \tanh^2 y} = \frac{1}{1 - x^2}. \quad \therefore \frac{dy}{dx} = \frac{1}{1 - x^2}$$

(b) $y = \cosh^{-1}(e^x)$.

Solution

$y = \cosh^{-1}(e^x) \Rightarrow \cosh y = e^x$. Differentiating implicitly yields

$$\sinh y \frac{dy}{dx} = e^x \Rightarrow \frac{dy}{dx} = \frac{e^x}{\sinh y} = \frac{e^x}{\sqrt{\cosh^2 y - 1}} = \frac{e^x}{\sqrt{e^{2x} - 1}}$$

(c) $y = 2 \tanh^{-1}[\tan(x/2)]$.

Solution

$y = 2 \tanh^{-1}[\tan(x/2)] \Rightarrow \tanh(y/2) = \tan(x/2)$. Differentiating implicitly with respect to x yields

$$\frac{1}{2} \operatorname{sech}^2(y/2) \frac{dy}{dx} = \frac{1}{2} \sec^2(x/2)$$

Making $\frac{dy}{dx}$ the subject yields

$$\begin{aligned}\frac{dy}{dx} &= \frac{\sec^2(x/2)}{\operatorname{sech}^2(y/2)} = \frac{\sec^2(x/2)}{1 - \tanh^2(y/2)} = \frac{\sec^2(x/2)}{1 - \tan^2(x/2)} \\ &= \frac{\sec^2(x/2)}{1 - \tan^2(x/2)} \cdot \frac{\cos^2(x/2)}{\cos^2(x/2)} = \frac{1}{\cos^2(x/2) - \sin^2(x/2)} \\ &= \frac{1}{\cos x} = \sec x\end{aligned}$$

(d) $y = \coth^{-1}(1/x)$.

Solution

$$y = \coth^{-1}(1/x) \Rightarrow \coth y = \frac{1}{x}. \text{ Differentiating implicitly yields}$$

$$-\operatorname{cosech}^2 y dy = -\frac{1}{x^2} dx \Rightarrow \frac{dy}{dx} = \frac{1}{x^2 \operatorname{cosech}^2 y} = \frac{1}{x^2 [\coth^2 y - 1]} = \frac{1}{x^2 \left[\frac{1}{x^2} - 1 \right]} = \frac{1}{1 - x^2}$$

(e) $y = \operatorname{sech}^{-1}(\cos x)$.

Solution

$$y = \operatorname{sech}^{-1}[\cos x] \Rightarrow \operatorname{sech} y = \cos x. \text{ Differentiating implicitly with respect to } x \text{ yields}$$

$$-\operatorname{sech} y \tanh y \frac{dy}{dx} = -\sin x$$

Making $\frac{dy}{dx}$ the subject yields

$$\begin{aligned} \frac{dy}{dx} &= \frac{\sin x}{\operatorname{sech} y \tanh y} = \frac{\sin x}{\cos x \tanh y} = \frac{\sin x}{\cos x \sqrt{1 - \operatorname{sech}^2 y}} \\ &= \frac{\sin x}{\cos x \sqrt{1 - \cos^2 x}} = \frac{\sin x}{\cos x \sin x} = \frac{1}{\cos x} = \sec x \end{aligned}$$

Exercise:

1. Find $\frac{dy}{dx}$ given that:

(a) $y = \operatorname{sech}^{-1}(\tan x)$.

[5 Marks]

(b) $y = \tanh^{-1}(3x/4)$.

[ans: $\frac{dy}{dx} = \frac{12}{16 - 9x^2}$]

(c) $y = \sinh^{-1}(\sqrt{x^2 - 1})$.

[ans: $\frac{dy}{dx} = \frac{1}{\sqrt{x^2 - 1}}$]

2. Show that if $y = \sinh^{-1}(\sin x)$, then $\frac{dy}{dx} = 1$.

LECTURE 5

4 Techniques of Integration

Definition 4.1 (Integration). *Integration is the reverse process of differentiation.*

Suppose $\frac{dy}{dx} = f(x)$. To obtain y , we integrate the function $f(x)$ with respect to the independent variable x . This is put in notation form as

$$y = \int f(x)dx + C,$$

where C is a constant of integration and $f(x)$ is the *integrand*. The following are the techniques of integration.

4.1 Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad \text{for } n \neq -1,$$

where C is a constant of integration.

Example(s):

$$(a) \int x^3 dx = \frac{x^{3+1}}{3+1} + C = \frac{x^4}{4} + C.$$

$$(b) \int x^{-7} dx = \frac{x^{-7+1}}{-7+1} + C = -\frac{x^{-6}}{6} + C.$$

$$(c) \int \sqrt{x} dx = \int x^{1/2} dx = \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C = \frac{2}{3}x^{3/2} + C.$$

$$(d) \int x(1-3x)dx = \int (x-3x^2)dx = \frac{x^2}{2} - 3 \cdot \frac{x^3}{3} = \frac{x^2}{2} - x^3 + C.$$

$$(e) \int (2x-3)^2 dx = \int (4x^2 - 12x + 9)dx = \frac{4}{3}x^3 - 6x^2 + 9x + C.$$

(f)

$$\begin{aligned} \int \frac{\sqrt{x} + \sqrt[3]{x} + 6}{x^2} dx &= \int \left(\frac{x^{1/2}}{x^2} + \frac{x^{1/3}}{x^2} + \frac{6}{x^2} \right) dx = \int \left(x^{-3/2} + x^{-5/3} + 6x^{-2} \right) dx \\ &= -2x^{-1/2} - \frac{3}{2}x^{-2/3} - 6x^{-1} + C \end{aligned}$$

$$(g) \int \frac{x^3}{x+1} dx.$$

Solution

Since the degree of the polynomial in the numerator is greater than that in the denominator, long division yields

$$\begin{array}{r} x^2 - x + 1 \\ x+1 \overline{) x^3} \\ \underline{-x^3 - x^2} \\ -x^2 \\ \underline{x^2 + x} \\ x \\ \underline{-x - 1} \\ -1 \end{array}$$

Therefore,

$$\int \frac{x^3}{x+1} dx = \int \left(x^2 - x + 1 - \frac{1}{x+1} \right) dx = \frac{x^3}{3} - \frac{x^2}{2} + x - \ln|x+1| + C$$

→ Note:

- For definite integrals such as $\int_a^b f(x)dx$, ignore the constant of integration since the limits of integration only act on the result. For example,

$$\int_1^2 x^2 dx = \left[\frac{x^3}{3} \right]_1^2 = \left[\frac{2^3}{3} \right] - \left[\frac{1^3}{3} \right] = \frac{8}{3} - \frac{1}{3} = \frac{7}{3}$$

Exercise:

1. The monthly sales at an import store are currently at \$10,000 but are expected to be declining at the rate of $S'(t) = -10t^{2/5}$ dollars t months from now. The store is profitable as long as the sales are above \$8,000 per month. For how long will the store remain profitable? [ans: $t = 55.9719$ months]

4.2 The u -substitution method

This technique requires that a new variable, say u , be introduced in the integrand to reduce the problem to a form in which the power rule of integration can be applied.

Example(s):

- (a) Evaluate $\int (2x+1)^{1/3} dx$.

Solution

Let $u = 2x + 1$. Differentiating with respect to x yields $\frac{du}{dx} = 2 \Rightarrow dx = \frac{du}{2}$. Substituting into the given integral, we get

$$\int (2x+1)^{1/3} dx = \int u^{1/3} \frac{du}{2} = \frac{1}{2} \int u^{1/3} du = \frac{1}{2} \left[\frac{3}{4} u^{4/3} \right] + C = \frac{3}{8} (2x+1)^{4/3} + C$$

- (b) Evaluate $\int \frac{x}{(1-x^2)^3} dx$.

Solution

Let $u = 1 - x^2$. Differentiating with respect to x yields $\frac{du}{dx} = -2x \Rightarrow dx = -\frac{du}{2x}$. Substituting into the given integral, we get

$$\int \frac{x}{(1-x^2)^3} dx = \int \frac{x}{u^3} \left(-\frac{du}{2x} \right) = -\frac{1}{2} \int u^{-3} du = -\frac{1}{2} \left[\frac{1}{-2} u^{-2} \right] + C = \frac{1}{4} (1-x^2)^{-2} + C$$

- (c) Evaluate $\int 5x(1+3x)^5 dx$.

Solution

Let $u = 1 + 3x$. Differentiating with respect to x yields $\frac{du}{dx} = 3$. Therefore, $dx = \frac{du}{3}$ and $x = \frac{1}{3}(u-1)$. Substituting into the given integral, we get

$$\begin{aligned} \int 5x(1+3x)^5 dx &= \int \frac{5}{3} (u-1) u^5 \left(\frac{du}{3} \right) = \frac{5}{9} \int (u^6 - u^5) du = \frac{5}{9} \left[\frac{1}{7} u^7 - \frac{1}{6} u^6 \right] + C \\ &= \frac{5}{9} \left[\frac{1}{7} (1+3x)^7 - \frac{1}{6} (1+3x)^6 \right] + C \end{aligned}$$

(d) Evaluate $\int \frac{e^{2x}}{(1 + e^{2x})^3} dx$.

Solution

Let $u = 1 + e^{2x}$. Differentiating with respect to x yields $\frac{du}{dx} = 2e^{2x} \Rightarrow dx = \frac{du}{2e^{2x}}$.
Substituting into the given integral, we get

$$\int \frac{e^{2x}}{(1 + e^{2x})^3} dx = \int \frac{e^{2x}}{u^3} \left(\frac{du}{2e^{2x}} \right) = \frac{1}{2} \int u^{-3} du = -\frac{1}{4} u^{-2} + C = -\frac{1}{4} (1 + e^{2x})^{-2} + C$$

Exercise:

1. Evaluate the following integrals

(a) $\int (7x - 2)^3 dx$. [hint: put $u = 7x - 2$, ans: $= \frac{1}{28} (7x - 2)^4 + C$]

(b) $\int 3x \sqrt{1 + 2x^2} dx$. [hint: put $u = 1 + 2x^2$, ans: $= \frac{1}{2} (1 + 2x^2)^{3/2} + C$]

(c) $\int \frac{1}{(x + 1)^2} dx$. [hint: put $u = x + 1$, ans: $= \frac{-1}{x + 1} + C$]

(d) $\int (3x + 5)^{-2} dx$. [hint: put $u = 3x + 5$, ans: $= -\frac{1}{3} (3x + 5)^{-1} + C$]

(e) $\int \frac{(\ln x)^2}{x} dx$. [hint: put $u = \ln x$, ans: $= \frac{1}{3} (\ln x)^3 + C$]

(f) $\int x(3 - 5x^2)^4 dx$. [hint: put $u = 3 - 5x^2$, ans: $= -\frac{1}{50} (3 - 5x^2)^5 + C$]

(g) $\int \frac{x + 1}{\sqrt[5]{x^2 + 2x + 7}} dx$. [hint: put $u = x^2 + 2x + 7$, ans: $= \frac{5}{8} (x^2 + 2x + 7)^{4/5} + C$]

(h) $\int x(2 + 3x)^4 dx$. [hint: put $u = 2 + 3x$, ans: $= \frac{1}{3} \left(\frac{(2 + 3x)^6}{18} - \frac{2(2 + 3x)^5}{15} \right) + C$]

(i) $\int x^2(1 + 4x^3)^3 dx$. [hint: put $u = 1 + 4x^3$, ans: $= \frac{1}{48} (1 + 4x^3)^4 + C$]

(j) $\int \frac{1}{\sqrt{x}(1 + \sqrt{x})^4} dx$. [hint: put $u = 1 + \sqrt{x}$, ans: $= -\frac{2}{3} (1 + \sqrt{x})^{-3} + C$]

(k) $\int \frac{e^{-x}}{\sqrt{1 - e^{-x}}} dx$. [hint: put $u = 1 - e^{-x}$, ans: $= 2\sqrt{1 - e^{-x}} + C$]

(l) $\int \frac{(1 + \ln 2x)^2}{x} dx$. [hint: put $u = 1 + \ln 2x$, ans: $= \frac{1}{3} (1 + \ln 2x)^3 + C$]

(m) $\int \sin^3 x \cos^5 x dx$. [hint: put $u = \cos x$, ans: $= \frac{1}{8} \cos^8 x - \frac{1}{6} \cos^6 x + C$]

(n) $\int \sin^5 x \cos^8 x dx$. [hint: put $u = \cos x$, ans: $= \frac{2}{11} \cos^{11} x - \frac{1}{13} \cos^{13} x - \frac{1}{9} \cos^9 x + C$]

(p) $\int x \sqrt{1 + x} dx$. [hint: put $u = 1 + x$, ans: $= \frac{2}{5} (1 + x)^{5/2} - \frac{2}{3} (1 + x)^{3/2} + C$]

2. If the marginal revenue function for output x is given by $MR = \frac{6}{(x + 2)^2} + 5$, find the total revenue function and the demand equation. [hint: $R(0) = 0$, ans: revenue $R(x) = -\frac{6}{x + 2} + 5x$, demand $p = \frac{R(x)}{x}$]

□ **The u -substitution based on the definition of integration as the reverse process of differentiation**

We consider the following forms of integrals.

4.2.1 Integrals of the form $\int f'(x)e^{f(x)}dx$

In this case, we let $u = f(x)$. Differentiating with respect to x yields $du = f'(x)dx$. Substituting yields

$$\int f'(x)e^{f(x)}dx = \int e^u du = e^u + c = e^{f(x)} + c,$$

where c is a constant of integration.

Example(s):

(a) Evaluate $\int e^{3x}dx$

Solution

Let $u = 3x$. Differentiating with respect to x , we get $\frac{du}{dx} = 3 \Rightarrow dx = \frac{du}{3}$. Substituting yields,

$$\int e^{3x}dx = \int e^u \frac{du}{3} = \frac{1}{3} \int e^u du = \frac{1}{3}e^u + c = \frac{1}{3}e^{3x} + c$$

(b) Evaluate $\int \frac{(5 - e^{5x})}{e^{3x}}dx$

Solution

Using the laws of indices, the given integrand can be rewritten as

$$\frac{(5 - e^{5x})}{e^{3x}} = 5e^{-3x} - e^{2x}$$

Integrating with respect to x yields

$$\int \frac{(5 - e^{5x})}{e^{3x}}dx = 5 \int e^{-3x}dx - \int e^{2x}dx = -\frac{5}{3}e^{-3x} - \frac{1}{2}e^{2x} + c$$

→ Note:

$$\square \text{ To evaluate } \int e^{-3x}dx, \text{ let } u = -3x \Rightarrow dx = -\frac{du}{3}.$$

$$\square \text{ To evaluate } \int e^{2x}dx, \text{ let } u = 2x \Rightarrow dx = \frac{du}{2}.$$

(c) Evaluate $\int x^2 e^{2x^3-4}dx$

Solution

Let $u = 2x^3 - 4$. Differentiating with respect to x , we get $\frac{du}{dx} = 6x^2 \Rightarrow x^2 dx = \frac{du}{6}$. Substituting yields,

$$\int x^2 e^{2x^3-4}dx = \int e^u \frac{du}{6} = \frac{1}{6} \int e^u du = \frac{1}{6}e^u + c = \frac{1}{6}e^{2x^3-4} + c$$

(d) Evaluate $\int (2x + 3)5^{x^2+3x+1}dx$

Solution

The function 5^{x^2+3x+1} can be rewritten as

$$5^{x^2+3x+1} = e^{\ln(5^{x^2+3x+1})} = e^{(x^2+3x+1)\ln(5)}$$

Therefore, the given integral becomes

$$I = \int (2x + 3)5^{x^2+3x+1} dx = \int (2x + 3)e^{(x^2+3x+1)\ln(5)} dx \quad (*)$$

Let $u = (x^2 + 3x + 1)\ln(5)$. Differentiating with respect to x , we get $\frac{du}{dx} = (2x + 3)\ln(5) \Rightarrow (2x + 3)dx = \frac{du}{\ln(5)}$. Substituting into equation $(*)$ yields,

$$I = \frac{1}{\ln(5)} \int e^u du = \frac{1}{\ln(5)} e^u + c = \frac{1}{\ln(5)} e^{(x^2+3x+1)\ln(5)} + c = \frac{1}{\ln(5)} 5^{x^2+3x+1} + c$$

Exercise:

1. Evaluate the following

(a) $\int \cos(4x)e^{\sin 4x} dx.$ [hint: let $u = \sin 4x$, ans: $= \frac{1}{4}e^{\sin 4x} + c$]

(b) $\int (x + 1)e^{2x^2+4x+5} dx.$ [hint: let $u = 2x^2 + 4x + 5$, ans: $= \frac{1}{4}e^{2x^2+4x+5} + c$]

(c) $\int \frac{(e^{2x} + 2e^x + 1)}{e^x} dx.$ [ans: $= e^x + 2x - e^{-x} + c$]

(d) $\int \frac{10^{\sqrt{x}}}{\sqrt{x}} dx.$ [hint: $10^{\sqrt{x}} = e^{\sqrt{x}\ln(10)}$ let $u = \sqrt{x}\ln(10)$, ans: $= \frac{1}{\ln(10)}10^{\sqrt{x}} + c$]

2. (a) Is it possible to evaluate $\int e^{x^2} dx$?

(b) Let $f(x) = \int_0^{x^2} e^{\sin t} dt$. Calculate $f'(x)$. [ans: $f'(x) = e^{\sin(x^2)} - 1$]

4.2.2 Integrals of the form $\int \frac{f'(x)}{f(x)} dx$

In this case, we let $u = f(x)$. Differentiating with respect to x yields $du = f'(x)dx$. Substituting yields

$$\int \frac{f'(x)}{f(x)} dx = \int \frac{1}{u} du = \ln |u| + c = \ln |f(x)| + c,$$

where c is a constant of integration.

Example(s):

(a) Evaluate $\int \frac{x-3}{x^2-6x+2} dx.$

Solution

Let $u = x^2 - 6x + 2$. Differentiating with respect to x yields $\frac{du}{dx} = 2x - 6 \Rightarrow dx = \frac{du}{2(x-3)}$.

Substituting into the given integral, we get

$$\int \frac{x-3}{x^2-6x+2} dx = \int \frac{(x-3)}{u} \cdot \frac{du}{2(x-3)} = \frac{1}{2} \int \frac{1}{u} du = \frac{1}{2} \ln |u| + C = \frac{1}{2} \ln |x^2 - 6x + 2| + C$$

(b) Evaluate $\int \tan x dx.$

Solution

Using the trigonometric identity $\tan x = \frac{\sin x}{\cos x}$, we have

$$\int \tan x dx = \int \frac{\sin x}{\cos x} dx$$

Let $u = \cos x$. Differentiating with respect to x yields $\frac{du}{dx} = -\sin x \Rightarrow dx = -\frac{du}{\sin x}$. Substituting into the given integral, we get

$$\int \frac{\sin x}{\cos x} dx = \int \frac{\sin x}{u} \left(-\frac{du}{\sin x} \right) = -\int \frac{1}{u} du = -\ln |u| + C = -\ln |\cos x| + C = \ln |\sec x| + C$$

Therefore,

$$\int \tan x dx = \ln |\sec x| + C$$

(c) Evaluate $\int \cot x dx$.

Solution

Using the trigonometric identity $\cot x = \frac{\cos x}{\sin x}$, we have

$$\int \cot x dx = \int \frac{\cos x}{\sin x} dx$$

Let $u = \sin x$. Differentiating with respect to x yields $\frac{du}{dx} = \cos x \Rightarrow dx = \frac{du}{\cos x}$. Substituting into the given integral, we get

$$\int \frac{\cos x}{\sin x} dx = \int \frac{\cos x}{u} \left(\frac{du}{\cos x} \right) = \int \frac{1}{u} du = \ln |u| + C = \ln |\sin x| + C$$

Therefore,

$$\int \cot x dx = \ln |\sin x| + C$$

(d) Evaluate $\int \sec x dx$.

Solution

Multiplying the numerator and denominator by $(\sec x + \tan x)$, we get

$$\int \sec x dx = \int \frac{\sec x (\sec x + \tan x)}{\sec x + \tan x} dx = \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx$$

Let $u = \sec x + \tan x$. Differentiating with respect to x yields

$$\frac{du}{dx} = \sec x \tan x + \sec^2 x \Rightarrow dx = \frac{du}{\sec x \tan x + \sec^2 x}$$

Substituting into the given integral, we get

$$\begin{aligned} \int \sec x dx &= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx = \int \frac{(\sec^2 x + \sec x \tan x)}{u} \cdot \frac{du}{(\sec x \tan x + \sec^2 x)} \\ &= \int \frac{1}{u} du = \ln |u| + C = \ln |\sec x + \tan x| + C \end{aligned}$$

Therefore,

$$\int \sec x dx = \ln |\sec x + \tan x| + C$$

(e) Evaluate $\int \operatorname{cosec} x dx$.

Solution

Multiplying the numerator and denominator by $(\operatorname{cosec} x + \cot x)$, we get

$$\int \operatorname{cosec} x dx = \int \frac{\operatorname{cosec} x (\operatorname{cosec} x + \cot x)}{\operatorname{cosec} x + \cot x} dx = \int \frac{\operatorname{cosec}^2 x + \operatorname{cosec} x \cot x}{\operatorname{cosec} x + \cot x} dx$$

Let $u = \operatorname{cosec} x + \cot x$. Differentiating with respect to x yields

$$\frac{du}{dx} = -\operatorname{cosec} x \cot x - \operatorname{cosec}^2 x \Rightarrow dx = -\frac{du}{(\operatorname{cosec} x \cot x + \operatorname{cosec}^2 x)}$$

Substituting into the given integral, we get

$$\begin{aligned} \int \frac{\operatorname{cosec}^2 x + \operatorname{cosec} x \cot x}{\operatorname{cosec} x + \cot x} dx &= \int \frac{\operatorname{cosec}^2 x + \operatorname{cosec} x \cot x}{u} \left(-\frac{du}{\operatorname{cosec} x \cot x + \operatorname{cosec}^2 x} \right) \\ &= -\int \frac{1}{u} du = -\ln |u| + C = -\ln |\operatorname{cosec} x + \cot x| + C \end{aligned}$$

Therefore,

$$\int \operatorname{cosec} x dx = -\ln |\operatorname{cosec} x + \cot x| + C$$

Exercise:

1. Evaluate the following

(a) $\int \frac{1}{5-3x} dx$. [hint: let $u = 5 - 3x$, ans: $= -\frac{1}{3} \ln |5 - 3x| + c$]

(b) $\int \frac{x}{1-3x^2} dx$. [hint: let $u = 1 - 3x^2$, ans: $= -\frac{1}{6} \ln |1 - 3x^2| + c$]

(c) $\int \frac{4x^2}{x^3-7} dx$. [hint: let $u = x^3 - 7$, ans: $= \frac{4}{3} \ln |x^3 - 7| + c$]

(d) $\int \frac{e^{3x}}{2+e^{3x}} dx$. [hint: let $u = 2 + e^{3x}$, ans: $= \frac{1}{3} \ln |2 + e^{3x}| + c$]

(e) $\int \frac{\cos 3x}{1+\sin 3x} dx$. [hint: let $u = 1 + \sin 3x$, ans: $= \frac{1}{3} \ln |1 + \sin 3x| + c$]

(f) $\int \frac{\sec^2 2x}{1+5 \tan 2x} dx$. [hint: let $u = 1 + 5 \tan 2x$, ans: $= \frac{1}{10} \ln |1 + 5 \tan 2x| + c$]

(g) $\int \frac{1}{x \ln |x|} dx$. [hint: let $u = \ln |x|$, ans: $= \ln(\ln |x|) + c$]

(h) $\int \frac{1}{1+\sqrt{2x}} dx$. [hint: let $u = 1 + \sqrt{2x}$, ans: $= 1 + \sqrt{2x} - \ln |1 + \sqrt{2x}| + c$]

(i) $\int \frac{\sqrt{x}}{1-x\sqrt{x}} dx$. [hint: let $u = 1 - x\sqrt{x} \Rightarrow \frac{du}{dx} = -\frac{3}{2}\sqrt{x}$, ans: $= -\frac{2}{3} \ln |1 - x\sqrt{x}| + c$]

(j) $\int \frac{x\sqrt{x}}{1+x^2\sqrt{x}} dx$. [hint: let $u = 1 + x^2\sqrt{x} \Rightarrow \frac{du}{dx} = \frac{5}{2}x\sqrt{x}$, ans: $= \frac{2}{5} \ln |1 + x^2\sqrt{x}| + c$]

(k) $\int \frac{5x}{(x+2)^2} dx$. [hint: let $u = x + 2$, ans: $= 5 \ln |x + 2| + \frac{10}{x+2} + c$]

4.2.3 Integrals of the form $\int g'(x) \cdot f(g(x))dx$

In this case, we let $u = g(x)$. Differentiating with respect to x yields $du = g'(x)dx$. Substituting yields

$$\int g'(x) \cdot f(g(x))dx = \int f(u)du + c,$$

where c is a constant of integration.

Example(s):

(a) Evaluate $\int x \sin(6 - 3x^2)dx$.

Solution

Let $u = 6 - 3x^2$. Differentiating with respect to x yields $\frac{du}{dx} = -6x \Rightarrow dx = -\frac{du}{6x}$. Substituting into the given integral, we get

$$\int x \sin(6 - 3x^2)dx = \int x \sin(u) \left(-\frac{du}{6x}\right) = -\frac{1}{6} \int \sin(u)du = \frac{1}{6} \cos(u) + C = \frac{1}{6} \cos(6 - 3x^2) + C$$

(b) Evaluate $\int \frac{\cos(\sqrt{x})}{\sqrt{x}}dx$.

Solution

Let $u = \sqrt{x}$. Differentiating with respect to x yields $\frac{du}{dx} = \frac{1}{2\sqrt{x}} \Rightarrow dx = 2\sqrt{x}du$. Substituting into the given integral, we get

$$\int \frac{\cos(\sqrt{x})}{\sqrt{x}}dx = \int \frac{\cos(u)}{\sqrt{x}} (2\sqrt{x}du) = 2 \int \cos(u)du = 2 \sin(u) + C = 2 \sin(\sqrt{x}) + C$$

(c) Evaluate $\int \sec^2(7x - 1)dx$.

Solution

Let $u = 7x - 1$. Differentiating with respect to x yields $\frac{du}{dx} = 7 \Rightarrow dx = \frac{du}{7}$. Substituting into the given integral, we get

$$\int \sec^2(7x - 1)dx = \int \sec^2(u) \cdot \frac{du}{7} = \frac{1}{7} \int \sec^2(u)du = \frac{1}{7} \tan(u) + C = \frac{1}{7} \tan(7x - 1) + C$$

(d) Evaluate $\int \operatorname{cosec}^2(4x)dx$.

Solution

Let $u = 4x$. Differentiating with respect to x yields $\frac{du}{dx} = 4 \Rightarrow dx = \frac{du}{4}$. Substituting into the given integral, we get

$$\int \operatorname{cosec}^2(4x)dx = \int \operatorname{cosec}^2(u) \cdot \frac{du}{4} = \frac{1}{4} \int \operatorname{cosec}^2(u)du = -\frac{1}{4} \cot(u) + C = -\frac{1}{4} \cot(4x) + C$$

(e) Evaluate $\int \tan(5x - 2)dx$.

Solution

Let $u = 5x - 2$. Differentiating with respect to x yields $\frac{du}{dx} = 5 \Rightarrow dx = \frac{du}{5}$. Substituting into the given integral, we get

$$\int \tan(5x - 2)dx = \int \tan(u) \cdot \frac{du}{5} = \frac{1}{5} \int \tan(u)du = \frac{1}{5} \ln |\sec u| + C = \frac{1}{5} \ln |\sec(5x - 2)| + C$$

(f) Evaluate $\int (x-1) \cot(x^2-2x-1) dx$.

Solution

Let $u = x^2 - 2x - 1$. Differentiating with respect to x yields $\frac{du}{dx} = 2x - 2 \Rightarrow dx = \frac{du}{2(x-1)}$. Substituting into the given integral, we get

$$\begin{aligned} \int (x-1) \cot(x^2-2x-1) dx &= \int (x-1) \cot(u) \cdot \frac{du}{2(x-1)} = \frac{1}{2} \int \cot(u) du = \frac{1}{2} \ln |\sin(u)| + C \\ &= \frac{1}{2} \ln |\sin(x^2-2x-1)| + C \end{aligned}$$

(g) Evaluate $\int 2x \sec(4x^2+5) dx$.

Solution

Let $u = 4x^2 + 5$. Differentiating with respect to x yields $\frac{du}{dx} = 8x \Rightarrow dx = \frac{du}{8x}$. Substituting into the given integral, we get

$$\begin{aligned} \int 2x \sec(4x^2+5) dx &= \int 2x \sec(u) \cdot \frac{du}{8x} = \frac{1}{4} \int \sec(u) du = \frac{1}{4} \ln |\sec(u) + \tan(u)| + C \\ &= \frac{1}{4} \ln |\sec(4x^2+5) + \tan(4x^2+5)| + C \end{aligned}$$

(h) Evaluate $\int \operatorname{cosec}(2x+3) dx$.

Solution

Let $u = 2x + 3$. Differentiating with respect to x yields $\frac{du}{dx} = 2 \Rightarrow dx = \frac{du}{2}$. Substituting into the given integral, we get

$$\begin{aligned} \int \operatorname{cosec}(2x+3) dx &= \int \operatorname{cosec}(u) \cdot \frac{du}{2} = \frac{1}{2} \int \operatorname{cosec}(u) du = -\frac{1}{2} \ln |\operatorname{cosec}(u) + \cot(u)| + C \\ &= -\frac{1}{2} \ln |\operatorname{cosec}(2x+3) + \cot(2x+3)| + C \end{aligned}$$

Exercise:

1. Evaluate the following

(a) $\int (x^2+1) \sin(x^3+3x-5) dx$

(b) $\int e^{2x} \sin(5-e^{2x}) dx$

(c) $\int \cos(4x+5) dx$

(d) $\int e^{-3x} \cos(4+e^{-3x}) dx$

(e) $\int \frac{\sec^2(\sqrt{x})}{\sqrt{x}} dx$

(f) $\int \sin(2x) \sec^2(1+\cos 2x) dx$

(g) $\int e^{3x} \operatorname{cosec}^2(1+e^{3x}) dx$

(h) $\int (x+1) \operatorname{cosec}^2(2x^2+4x-5) dx$

(i) $\int \frac{\tan(\sqrt{x})}{\sqrt{x}} dx$

(j) $\int (x^2-1) \tan(3x^3-9x-7) dx$

(k) $\int \frac{\cot(\sqrt{x})}{\sqrt{x}} dx$

(l) $\int \cot(2-x) dx$

(m) $\int e^{4x} \cot(1-e^{4x}) dx$

(n) $\int 2x^2 \sec(4\pi x^3+6) dx$

(p) $\int (x+2) \operatorname{cosec}(x^2+4x-3) dx$

4.2.4 Integrals of the form $\int f(x) \cdot f'(x)dx$

In this case, we let $u = f(x)$. Differentiating with respect to x yields $du = f'(x)dx$. Substituting yields

$$\int f(x) \cdot f'(x)dx = \int u du = \frac{u^2}{2} + c = \frac{1}{2} [f(x)]^2 + c,$$

where c is a constant of integration.

Example(s):

(a) Evaluate $\int \sin x \cos x dx$

Solution

Let $u = \sin x \Rightarrow du = \cos x dx$. Therefore,

$$\int \sin x \cos x dx = \int u du = \frac{u^2}{2} + c = \frac{\sin^2 x}{2} + c$$

(b) Evaluate $\int \frac{\ln x}{x} dx$

Solution

$\int \frac{\ln x}{x} dx = \int \ln x \cdot \frac{1}{x} dx$. Let $u = \ln x \Rightarrow du = \frac{1}{x} dx$. Therefore,

$$\int \frac{\ln x}{x} dx = \int u du = \frac{u^2}{2} + c = \frac{\ln^2 x}{2} + c$$

(c) Evaluate $\int \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx$

Solution

$\int \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx = \int \sin^{-1} x \cdot \frac{1}{\sqrt{1-x^2}} dx$. Let $u = \sin^{-1} x \Rightarrow du = \frac{1}{\sqrt{1-x^2}} dx$. Therefore,

$$\int \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx = \int u du = \frac{u^2}{2} + c = \frac{[\sin^{-1} x]^2}{2} + c$$

Exercise:

1. Evaluate $\int 2 \tan x \sec^2 x dx$

4.3 Trigonometric and hyperbolic substitutions

Consider the following general forms of radicals. An appropriate substitution should be identified to simplify the radical.

4.3.1 Integrals of the form $\int \frac{1}{x^2 + A^2} dx$

In this case, we put $x = A \tan \theta$ and $dx = A \sec^2 \theta d\theta$.

□ **Justification:** The given integral can be rewritten as $\frac{1}{A^2} \int \frac{1}{\left\{1 + \frac{x^2}{A^2}\right\}} dx$. Comparing $1 + \frac{x^2}{A^2}$ with the trigonometric identity $1 + \tan^2 \theta = \sec^2 \theta$, we have

$$\frac{x^2}{A^2} = \tan^2 \theta \Rightarrow x = A \tan \theta$$

Differentiating with respect to θ yields $\frac{dx}{d\theta} = A \sec^2 \theta$. Making dx the subject yields $dx = A \sec^2 \theta d\theta$.

→ Note: we use the identity for tangent if the denominator of the integrand doesn't have a square root and the sign is positive.

Example(s):

(a) Evaluate $\int \frac{1}{x^2 + 4} dx$

Solution

Using trigonometric substitution, put $x = 2 \tan \theta \Rightarrow dx = 2 \sec^2 \theta d\theta$. So,

$$\begin{aligned} \int \frac{1}{x^2 + 4} dx &= \int \frac{2 \sec^2 \theta d\theta}{4 \tan^2 \theta + 4} = \int \frac{2 \sec^2 \theta d\theta}{4 \sec^2 \theta} = \frac{1}{2} \int d\theta = \frac{1}{2} \theta + c \\ &= \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + c \end{aligned}$$

(b) Evaluate $\int \frac{1}{x^2 + 10x + 30} dx$

Solution

We first complete the square in the denominator to get

$$x^2 + 10x + 30 = x^2 + 10x + \left(\frac{10}{2}\right)^2 + 30 - \left(\frac{10}{2}\right)^2 = (x + 5)^2 + 5 \text{ [hint: } c = \left(\frac{b}{2}\right)^2 \text{]}$$

Hence, $\int \frac{1}{x^2 + 10x + 30} dx = \int \frac{1}{(x + 5)^2 + 5} dx$. Using trigonometric substitution, put

$$x + 5 = \sqrt{5} \tan \theta \Rightarrow dx = \sqrt{5} \sec^2 \theta d\theta$$

So,

$$\begin{aligned} \int \frac{1}{(x + 5)^2 + 5} dx &= \int \frac{\sqrt{5} \sec^2 \theta d\theta}{5 \tan^2 \theta + 5} = \int \frac{\sqrt{5} \sec^2 \theta d\theta}{5 \sec^2 \theta} = \frac{\sqrt{5}}{5} \int d\theta = \frac{1}{\sqrt{5}} \theta + c \\ &= \frac{1}{\sqrt{5}} \tan^{-1} \left(\frac{x + 5}{\sqrt{5}} \right) + c \end{aligned}$$

(c) Evaluate $\int \frac{1}{2x^2 + 12x + 32} dx$

Solution

We first complete the square in the denominator to get

$$2x^2 + 12x + 32 = 2(x^2 + 6x + 16) = 2 \left(x^2 + 6x + \left(\frac{6}{2}\right)^2 + 16 - \left(\frac{6}{2}\right)^2 \right) = 2 \left[(x + 3)^2 + 7 \right]$$

Hence, $\int \frac{1}{2x^2 + 12x + 32} dx = \frac{1}{2} \int \frac{1}{(x + 3)^2 + 7} dx$. Using trigonometric substitution, put

$$x + 3 = \sqrt{7} \tan \theta \Rightarrow dx = \sqrt{7} \sec^2 \theta d\theta$$

So,

$$\begin{aligned} \frac{1}{2} \int \frac{1}{(x + 3)^2 + 7} dx &= \frac{1}{2} \int \frac{\sqrt{7} \sec^2 \theta d\theta}{7 \tan^2 \theta + 7} = \frac{1}{2} \int \frac{\sqrt{7} \sec^2 \theta d\theta}{7 \sec^2 \theta} = \frac{\sqrt{7}}{2 \times 7} \int d\theta = \frac{1}{2\sqrt{7}} \theta + c \\ &= \frac{1}{2\sqrt{7}} \tan^{-1} \left(\frac{x + 3}{\sqrt{7}} \right) + c \end{aligned}$$

→ Note:

- For integrals of the form $\int \frac{1}{x^2 - A^2} dx$, we will use *partial fraction decomposition* after expressing the denominator as a product of two linear factors, i.e., $x^2 - A^2 = (x + A)(x - A)$.

Exercise:

1. Evaluate the following

$$\begin{aligned} \text{(a)} \quad & \int \frac{1}{4 + 3(x-1)^2} dx \quad [\text{hint: put } x-1 = \frac{2}{\sqrt{3}} \tan \theta, \text{ ans: } I = \frac{1}{2\sqrt{3}} \tan^{-1}(\frac{\sqrt{3}}{2}(x-1)) + c] \\ \text{(b)} \quad & \int \frac{1}{x^2 - 6x + 16} dx. \quad [\text{ans: } I = \frac{1}{\sqrt{7}} \tan^{-1}\left(\frac{x-3}{\sqrt{7}}\right) + c] \end{aligned}$$

4.3.2 Integrals of the form $\int \frac{1}{\sqrt{A^2 - x^2}} dx$

In this case, we put $x = A \sin \theta$ and $dx = A \cos \theta d\theta$.

- **Justification:** The given integral can be rewritten as $\frac{1}{A} \int \frac{1}{\sqrt{1 - \frac{x^2}{A^2}}} dx$. Comparing $1 - \frac{x^2}{A^2}$ with the trigonometric identity $1 - \sin^2 \theta = \cos^2 \theta$, we have

$$\frac{x^2}{A^2} = \sin^2 \theta \quad \Rightarrow \quad x = A \sin \theta$$

Differentiating with respect to θ yields $\frac{dx}{d\theta} = A \cos \theta$. Making dx the subject yields $dx = A \cos \theta d\theta$.

→ Note: we use the identity for sine (or cosine) if the denominator of the integrand has a square root and the sign is negative.

Example(s):

(a) Evaluate $\int \frac{1}{\sqrt{25 - x^2}} dx$

Solution

Using trigonometric substitution, put $x = 5 \sin \theta \Rightarrow dx = 5 \cos \theta d\theta$. So,

$$\begin{aligned} \int \frac{1}{\sqrt{25 - x^2}} dx &= \int \frac{5 \cos \theta d\theta}{\sqrt{25 - 25 \sin^2 \theta}} = \int \frac{5 \cos \theta d\theta}{\sqrt{25 \cos^2 \theta}} = \int \frac{5 \cos \theta d\theta}{5 \cos \theta} = \theta + c \\ &= \sin^{-1}\left(\frac{x}{5}\right) + c \end{aligned}$$

(b) Evaluate $\int \frac{1}{\sqrt{3 - 2x - x^2}} dx$

Solution

We first complete the square in the denominator to get

$$3 - 2x - x^2 = -(x^2 + 2x - 3) = -\left[x^2 + 2x + \left(\frac{2}{2}\right)^2 - 3 - \left(\frac{2}{2}\right)^2\right] = -[(x+1)^2 - 4] = 4 - (x+1)^2$$

Hence,

$$\int \frac{1}{\sqrt{3 - 2x - x^2}} dx = \int \frac{1}{\sqrt{4 - (x+1)^2}} dx$$

Using trigonometric substitution, put $x + 1 = 2 \sin \theta \Rightarrow dx = 2 \cos \theta d\theta$. So,

$$\begin{aligned} \int \frac{1}{\sqrt{4 - (x+1)^2}} dx &= \int \frac{2 \cos \theta d\theta}{\sqrt{4 - 4 \sin^2 \theta}} = \int \frac{2 \cos \theta d\theta}{\sqrt{4 \cos^2 \theta}} = \int d\theta = \theta + c \\ &= \sin^{-1} \left(\frac{x+1}{2} \right) + c \end{aligned}$$

Exercise:

1. Evaluate the following

(a) $\int \frac{1}{\sqrt{4 - 9x^2}} dx$ [hint: put $x = \frac{2}{3} \sin \theta$, ans: $I = \frac{1}{3} \sin^{-1}(\frac{3}{2}x) + c$]

(b) $\int \frac{1}{\sqrt{4 - 3(x+2)^2}} dx$ [hint: put $x + 2 = \frac{2}{\sqrt{3}} \sin \theta$, ans: $I = \frac{1}{\sqrt{3}} \sin^{-1}(\frac{\sqrt{3}}{2}(x+2)) + c$]

(c) $\int \frac{1}{\sqrt{5 + 4x - x^2}} dx$ [ans: $I = \sin^{-1}(\frac{x-2}{3}) + c$]

(d) $\int \frac{1}{\sqrt{-x^2 - 4x}} dx$ [ans: $I = \sin^{-1}(\frac{x+2}{2}) + c$]

(e) $\int_0^1 \sqrt{\frac{x}{2-x}} dx$ [hint: put $x = 2 \sin^2 \theta$, ans: $I = \frac{\pi}{2} - 1$]

(f) $\int \frac{x^2}{\sqrt{1-x^2}} dx$. [hint: put $x = \sin \theta$, ans: $I = \frac{1}{4} \sin^{-1} x - \frac{1}{4} x \sqrt{1-x^2} + c$]

4.3.3 Integrals of the form $\int \frac{1}{\sqrt{x^2 + A^2}} dx$

In this case, we put $x = A \sinh \theta$ and $dx = A \cosh \theta d\theta$.

□ **Justification:** The given integral can be rewritten as $\frac{1}{A} \int \frac{1}{\sqrt{1 + \frac{x^2}{A^2}}} dx$. Comparing $1 + \frac{x^2}{A^2}$ with the hyperbolic identity $1 + \sinh^2 \theta = \cosh^2 \theta$, we have

$$\frac{x^2}{A^2} = \sinh^2 \theta \Rightarrow x = A \sinh \theta$$

Differentiating with respect to θ yields $\frac{dx}{d\theta} = A \cosh \theta$. Making dx the subject yields $dx = A \cosh \theta d\theta$.

→ Note: we use the identity for hyperbolic sine (or hyperbolic cosine) if the denominator of the integrand has a square root.

Example(s):

(a) Evaluate $\int \frac{1}{\sqrt{x^2 + 4}} dx$

Solution

Using hyperbolic substitution, put $x = 2 \sinh \theta \Rightarrow dx = 2 \cosh \theta d\theta$. So,

$$\begin{aligned} \int \frac{1}{\sqrt{x^2 + 4}} dx &= \int \frac{2 \cosh \theta d\theta}{\sqrt{4 \sinh^2 \theta + 4}} = \int \frac{2 \cosh \theta d\theta}{\sqrt{4 \cosh^2 \theta}} = \int \frac{2 \cosh \theta d\theta}{2 \cosh \theta} = \theta + c \\ &= \sinh^{-1} \left(\frac{x}{2} \right) + c \end{aligned}$$

Exercise:

1. Evaluate the following

- (a) $\int \frac{1}{\sqrt{x^2 + 5x + 12}} dx.$ [ans: $I = \sinh^{-1} \left(\frac{x + \frac{5}{2}}{5.75} \right) + c$]
- (b) $\int \frac{1}{\sqrt{2x^2 + 8x + 15}} dx$
- (c) $\int \frac{x^2}{\sqrt{1 + x^2}} dx.$ [hint: put $x = \sinh \theta$, ans: $I = \frac{1}{4} \sinh(2 \sinh^{-1} x) - \frac{1}{2} \sinh^{-1} x + c$]

4.3.4 Integrals of the form $\int \frac{1}{\sqrt{x^2 - A^2}} dx$

In this case, we use either of the two substitutions below

- *Method 1:* put $x = A \cosh \theta$ and $dx = A \sinh \theta d\theta$
- *Method 2:* put $x = A \sec \theta$ and $dx = A \sec \theta \tan \theta d\theta$

□ **Justification:** The given integral can be rewritten as $\frac{1}{A} \int \frac{1}{\sqrt{\frac{x^2}{A^2} - 1}} dx.$

- *Method 1:* Comparing $\frac{x^2}{A^2} - 1$ with the hyperbolic identity $\cosh^2 \theta - 1 = \sinh^2 \theta$, we have

$$\frac{x^2}{A^2} = \cosh^2 \theta \Rightarrow x = A \cosh \theta$$

Differentiating with respect to θ yields $\frac{dx}{d\theta} = A \sinh \theta$. Making dx the subject yields

$$dx = A \sinh \theta d\theta$$

- *Method 2:* Comparing $\frac{x^2}{A^2} - 1$ with the trigonometric identity $\sec^2 \theta - 1 = \tan^2 \theta$, we have

$$\frac{x^2}{A^2} = \sec^2 \theta \Rightarrow x = A \sec \theta$$

Differentiating with respect to θ yields $\frac{dx}{d\theta} = A \sec \theta \tan \theta$. Making dx the subject yields

$$dx = A \sec \theta \tan \theta d\theta$$

Example(s):

- (a) Evaluate $\int \frac{1}{\sqrt{x^2 - 9}} dx$

Solution

□ *Method 1:* Using hyperbolic substitution, put $x = 3 \cosh \theta \Rightarrow dx = 3 \sinh \theta d\theta$. So,

$$\begin{aligned} \int \frac{1}{\sqrt{x^2 - 9}} dx &= \int \frac{3 \sinh \theta d\theta}{\sqrt{9 \cosh^2 \theta - 9}} = \int \frac{3 \sinh \theta d\theta}{\sqrt{9 \sinh^2 \theta}} = \int \frac{3 \sinh \theta d\theta}{3 \sinh \theta} = \theta + c \\ &= \cosh^{-1} \left(\frac{x}{3} \right) + c = \ln \left| \frac{x}{3} + \sqrt{\frac{x^2}{9} - 1} \right| + c \end{aligned}$$

□ *Method 2:* Using trigonometric substitution, put $x = 3 \sec \theta \Rightarrow dx = 3 \sec \theta \tan \theta d\theta$.
So,

$$\begin{aligned} \int \frac{1}{\sqrt{x^2 - 9}} dx &= \int \frac{3 \sec \theta \tan \theta d\theta}{\sqrt{9 \sec^2 \theta - 9}} = \int \frac{3 \sec \theta \tan \theta d\theta}{\sqrt{9 \tan^2 \theta}} = \int \frac{3 \sec \theta \tan \theta d\theta}{3 \tan \theta} = \int \sec \theta d\theta \\ &= \ln |\sec \theta + \tan \theta| + c = \ln \left| \sec \theta + \sqrt{\sec^2 \theta - 1} \right| + c \\ &= \ln \left| \frac{x}{3} + \sqrt{\frac{x^2}{9} - 1} \right| + c \end{aligned}$$

Exercise:

1. Show that $\int \frac{dx}{\sqrt{4x^2 - 9}} = \frac{1}{2} \ln \left[\frac{2x}{3} + \sqrt{\frac{4x^2}{9} - 1} \right] + c$. [hint: put $x = \frac{3}{2} \cosh \theta$ or $x = \frac{3}{2} \sec \theta$]

2. Evaluate the following

(a) $\int \frac{1}{\sqrt{x^2 + 6x + 1}} dx$ [ans: $I = \cosh^{-1} \left(\frac{x+3}{2\sqrt{2}} \right) + c$]

(b) $\int \frac{x^2}{\sqrt{x^2 - 4}} dx$ [ans: $I = 2 \cosh^{-1} \left(\frac{x}{2} \right) + \sinh \left(2 \cosh^{-1} \frac{x}{2} \right) + c$]

Exercise:

1. Evaluate the following integrals.

(a) $\int \frac{dx}{9x^2 - 25}$. [hint: put $x = \frac{5}{3} \coth \theta$, ans: $I = \frac{1}{30} \ln \left[\frac{3x-5}{3x+5} \right] + c$]

(b) $\int \sqrt{25 + x^2} dx$. [hint: put $x = \tan \theta$ then use integration by parts, ans:
 $I = \frac{5x}{2} \sec(\tan^{-1}(\frac{x}{5})) + \frac{25}{2} \ln[\sec(\tan^{-1}(\frac{x}{5})) + \frac{x}{5}] + c$]

(c) i) $\int \frac{\sqrt{x^2 - 36}}{x} dx$. [hint: put $x = 6 \sec \theta$, ans: $I = 6 \tan(\sec^{-1}(\frac{x}{6})) - 6 \sec^{-1}(\frac{x}{6}) + c$]

ii) $\int \frac{dx}{x\sqrt{4x^2 - 1}}$. [hint: put $x = \frac{1}{2} \sec \theta$, ans: $I = \sec^{-1}(2x) + c$]

iii) $\int \frac{dx}{(x-1)\sqrt{x^2 - 2x}}$. [hint: complete the square in the denominator then let
 $(x-1) = \sec \theta$, ans: $I = \sec^{-1}(x-1) + c$]

2. Show that:

(a) $\int \sqrt{x^2 + 4} dx = \frac{1}{2} x \sqrt{x^2 + 4} + 2 \sinh^{-1}(x/2) + c$. [hint: put $x = 2 \sinh \theta$, recall:
 $\sinh 2\theta = 2 \sinh \theta \cosh \theta = 2 \sinh \theta \sqrt{1 + \sinh^2 \theta}$]

(b) $\int \frac{dx}{x\sqrt{1-x^2}} = -\operatorname{sech}^{-1} x + c$. [hint: put $x = \operatorname{sech} \theta$]

LECTURE 7

4.4 Integration of trigonometric functions

4.4.1 Integrals involving products of sines and cosines

In this case, we rewrite the integrand using the following trigonometric identities.

$$\begin{aligned} \sin(A+B) &= \sin A \cos B + \cos A \sin B \\ \sin(A-B) &= \sin A \cos B - \cos A \sin B \end{aligned}$$

Adding yields

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

The restriction is that A is associated with sine and B is associated with cosine. Also,

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

Adding yields

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

Subtracting yields

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

Further, $\sin(-A) = -\sin A$ since sine is an odd function and $\cos(-A) = +\cos A$ since cosine is an even function.

Example(s):

$$1. \int \sin 4x \cos 2x dx = \frac{1}{2} \int (\sin 6x + \sin 2x) dx = -\frac{1}{12} \cos 6x - \frac{1}{4} \cos 2x + c$$

$$2. \int \cos 5x \sin 3x dx = \frac{1}{2} \int [\sin 8x + \sin(-2x)] dx = \frac{1}{2} \int [\sin 8x - \sin 2x] dx = -\frac{1}{16} \cos 8x + \frac{1}{4} \cos 2x + c$$

$$3. \int \cos 4x \cos 6x dx = \frac{1}{2} \int [\cos 10x + \cos(-2x)] dx = \frac{1}{2} \int [\cos 10x + \cos 2x] dx = \frac{1}{20} \sin 10x + \frac{1}{4} \sin 2x + c$$

$$4. \int \sin 5x \sin x dx = \frac{1}{2} \int (\cos 4x - \cos 6x) dx = \frac{1}{8} \sin 4x - \frac{1}{12} \sin 6x + c$$

Exercise:

1. Evaluate the following

$$(a) \int \sin 3x \cos 7x dx.$$

$$[\text{ans: } I = \frac{1}{8} \cos 4x - \frac{1}{20} \cos 10x + c]$$

$$(b) \int \sin 3x \sin 7x dx$$

$$(c) \int \cos 3x \cos 7x dx$$

$$(d) \int \cos 4u \sin 3u du$$

4.4.2 Integrals involving powers of sines and cosines

□ **Case 1:** Even powers of sines and cosines

In this case, we use the following trigonometric identities to express the integrand in terms of cosines (or sines) of multiples of x . From

$$\cos^2 \theta - \sin^2 \theta = \cos 2\theta \text{ and } \cos^2 \theta + \sin^2 \theta = 1$$

Solving the above equations simultaneously yields the following identities.

$$\text{i) } \cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta).$$

ii) $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta).$

Example(s):

1. Evaluate the following integrals

(a) $\int \sin^2 4x dx$ (b) $\int \cos^2 5x dx$ (c) $\int \cos^4 2x dx$ (d) $\int \cos 5x \sin^2 3x dx$

Solution

(a) $\int \sin^2 4x dx = \frac{1}{2} \int (1 - \cos 8x) dx = \frac{1}{2} \left(x - \frac{1}{8} \sin 8x \right) + c$

(b) $\int \cos^2 5x dx = \frac{1}{2} \int (1 + \cos 10x) dx = \frac{1}{2} \left(x + \frac{1}{10} \sin 10x \right) + c$

(c) Express $\cos^4 2x$ in terms of cosines (or sines) of multiple angles

$$\begin{aligned} \cos^4 2x &= (\cos^2 2x)^2 = \left[\frac{1}{2}(1 + \cos 4x) \right]^2 = \frac{1}{4}(1 + 2\cos 4x + \cos^2 4x) \\ &= \frac{1}{4} \left[1 + 2\cos 4x + \frac{1}{2}(1 + \cos 8x) \right] = \frac{1}{4} \left(\frac{3}{2} + 2\cos 4x + \frac{1}{2}\cos 8x \right) \end{aligned}$$

Integrating with respect to x yields

$$\begin{aligned} \int \cos^4 2x dx &= \frac{1}{4} \int \left(\frac{3}{2} + 2\cos 4x + \frac{1}{2}\cos 8x \right) dx \\ &= \frac{1}{4} \left(\frac{3}{2}x + \frac{1}{2}\sin 4x + \frac{1}{16}\sin 8x \right) + c \end{aligned}$$

(d) Express $\cos 5x \sin^2 3x$ in terms of cosines (or sines) of multiple angles

$$\begin{aligned} \cos 5x \sin^2 3x &= \cos 5x \left[\frac{1}{2}(1 - \cos 6x) \right] = \frac{1}{2}(\cos 5x - \cos 5x \cos 6x) \\ &= \frac{1}{2} \left[\cos 5x - \frac{1}{2}(\cos x + \cos 11x) \right] \\ &= \frac{1}{4}(2\cos 5x - \cos x - \cos 11x) \end{aligned}$$

Integrating with respect to x yields

$$\begin{aligned} \int \cos 5x \sin^2 3x dx &= \frac{1}{4} \int (2\cos 5x - \cos x - \cos 11x) dx \\ &= \frac{1}{4} \left(\frac{2}{5}\sin 5x - \sin x - \frac{1}{11}\sin 11x \right) + c \end{aligned}$$

Exercise:

1. Evaluate the following integrals

(a) $\int \sin^4 3x dx$ (c) $\int \cos^6 x dx$ (e) $\int \cos^6 x \sin 7x dx$
 (b) $\int \sin^6 x dx$ (d) $\int \sin 3x \cos^2 2x dx$ (f) $\int \sin^2 x \cos 3x dx$

□ **Case 2:** Odd powers of sines and cosines

In this case, we use the trigonometric identity $\cos^2 \theta + \sin^2 \theta = 1$ after breaking the power into a product of even and odd powers of sines and/or cosines.

Example(s):

1. Evaluate the following integrals

(a) $\int \sin^3 2x dx$

(b) $\int \cos^5 3x dx$

Solution

(a) The integrand can be rewritten as

$$\begin{aligned}\sin^3 2x &= \int \sin 2x \sin^2 2x = \sin 2x(1 - \cos^2 2x) \\ &= \sin 2x dx - \sin 2x \cos^2 2x\end{aligned}$$

Integrating with respect to x yields

$$\int \sin^3 2x dx = \int \sin 2x dx - \int \sin 2x \cos^2 2x dx \quad (*)$$

Now,

$$\int \sin 2x dx = -\frac{1}{2} \cos 2x$$

To evaluate $\int \sin 2x \cos^2 2x dx$, put $u = \cos 2x$. Differentiating with respect to x yields

$$\frac{du}{dx} = -2 \sin 2x \Rightarrow dx = -\frac{du}{2 \sin 2x}. \text{ Substituting yields}$$

$$\int \sin 2x \cos^2 2x dx = \int \sin 2x u^2 \left(-\frac{du}{2 \sin 2x}\right) = -\frac{1}{2} \int u^2 du = -\frac{1}{6} u^3 + c = -\frac{1}{6} \cos^3 2x$$

Therefore, equation (*) becomes

$$\int \sin^3 2x dx = -\frac{1}{2} \cos 2x + \frac{1}{6} \cos^3 2x + c$$

(b) The integrand can be rewritten as

$$\begin{aligned}\cos^5 3x &= \cos 3x (\cos^2 3x)^2 = \cos 3x (1 - \sin^2 3x)^2 = \cos 3x (1 - 2 \sin^2 3x + \sin^4 3x) \\ &= \cos 3x - 2 \cos 3x \sin^2 3x + \cos 3x \sin^4 3x\end{aligned}$$

Integrating with respect to x yields

$$\begin{aligned}\int \cos^5 3x dx &= \int \cos 3x dx - 2 \int \cos 3x \sin^2 3x dx + \int \cos 3x \sin^4 3x dx \\ &= \int \cos 3x dx - \frac{2}{3} \int u^2 du + \frac{1}{3} \int u^4 du, \quad \text{where } u = \sin 3x \\ &= \frac{1}{3} \sin 3x - \frac{2}{9} \sin^3 3x + \frac{1}{15} \sin^5 3x + c\end{aligned}$$

→ Note:

□ For integrals involving powers of $\tan \theta$ and $\sec \theta$, use the trigonometric identity $1 + \tan^2 \theta = \sec^2 \theta$.

□ For integrals involving powers of $\cot \theta$ and $\operatorname{cosec} \theta$, use the identity $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$.

Exercise:

1. Evaluate the following

$$(a) \int \tan^7 x dx. \quad [\text{ans: } I = \frac{1}{6} \tan^6 x - \frac{1}{4} \tan^4 x + \frac{1}{2} \tan^2 x + \ln(\cos x) + c]$$

Solution

□ **Method 1:** Using the trigonometric identity $1 + \tan^2 x = \sec^2 x$, we have

$$\int \tan^7 x dx = \int \tan^6 x \tan x dx = \int (\tan^2 x)^3 \tan x dx = \int (\sec^2 x - 1)^3 \tan x dx$$

Let $u = \sec x \Rightarrow du = \sec x \tan x dx$. Thus, $\tan x dx = \frac{du}{\sec x} = \frac{du}{u}$. Substituting yields

$$\begin{aligned} \int \tan^7 x dx &= \int \frac{(u^2 - 1)^3}{u} du = \int \frac{u^6 - 3u^4 + 3u^2 - 1}{u} du = \int \left(u^5 - 3u^3 + 3u - \frac{1}{u} \right) du \\ &= \frac{1}{6} u^6 - \frac{3}{4} u^4 + \frac{3}{2} u^2 - \ln(u) + c_1 \\ &= \frac{1}{6} \sec^6 x - \frac{3}{4} \sec^4 x + \frac{3}{2} \sec^2 x - \ln(\sec x) + c_1 \end{aligned}$$

□ **Method 2:** Using the trigonometric identity $1 + \tan^2 x = \sec^2 x$, we have

$$\begin{aligned} \tan^7 x &= \tan^2 x \tan^5 x = (\sec^2 x - 1) \tan^5 x = \tan^5 x \sec^2 x - \tan^5 x \\ \tan^5 x &= \tan^2 x \tan^3 x = (\sec^2 x - 1) \tan^3 x = \tan^3 x \sec^2 x - \tan^3 x \\ \tan^3 x &= \tan^2 x \tan x = (\sec^2 x - 1) \tan x = \tan x \sec^2 x - \tan x \end{aligned}$$

Back substitution yields

$$\begin{aligned} \tan^5 x &= \tan^3 x \sec^2 x - \tan x \sec^2 x + \tan x \\ \tan^7 x &= \tan^5 x \sec^2 x - \tan^3 x \sec^2 x + \tan x \sec^2 x - \tan x \\ &= (\tan^5 x - \tan^3 x + \tan x) \sec^2 x - \tan x \end{aligned}$$

Therefore,

$$\int \tan^7 x dx = \int (\tan^5 x - \tan^3 x + \tan x) \sec^2 x dx - \int \tan x dx$$

In the first integral, we let $u = \tan x \Rightarrow du = \sec^2 x dx$. Substituting yields

$$\begin{aligned} \int \tan^7 x dx &= \int (u^5 - u^3 + u) du - \int \frac{\sin x}{\cos x} dx \\ &= \frac{1}{6} u^6 - \frac{1}{4} u^4 + \frac{1}{2} u^2 + \ln(\cos x) + c \\ &= \frac{1}{6} \tan^6 x - \frac{1}{4} \tan^4 x + \frac{1}{2} \tan^2 x + \ln(\cos x) + c \end{aligned}$$

$$(b) \int \sec^3 x dx. \quad [\text{ans: } = \ln |\sec x + \tan x| + \frac{1}{4} \sinh(2 \sinh^{-1}(\tan x)) - \frac{1}{2} \sinh^{-1}(\tan x)]$$

2. Evaluate the following integrals

$$(a) \int \sin^5 3x dx \quad (b) \int \cos^3 8x dx \quad (c) \int \sin^7 x dx \quad (d) \int \cos^7 x dx$$

4.4.3 Integrals involving rational functions with sines and cosines in the denominator

In this case, we use the t -method of substitution. We consider the following two cases.

Case 1: Integrals of the form $\int \frac{1}{a + b \sin^2 x + c \cos^2 x} dx$

In this case, we put $t = \tan x$ so that

$$\square \quad \frac{dt}{dx} = \sec^2 x \Rightarrow dx = \frac{dt}{\sec^2 x} = \frac{dt}{1 + \tan^2 x} \Rightarrow dx = \frac{dt}{1 + t^2}.$$

$$\square \quad \sin x = \frac{t}{\sqrt{1+t^2}} \text{ and } \cos x = \frac{1}{\sqrt{1+t^2}} \text{ (from SOHCAHTOA)}$$

Example(s):

1. Solve $\int \frac{1}{3 + \cos^2 x} dx$

Solution

Put $t = \tan x \Rightarrow dx = \frac{dt}{1+t^2}$, and $\cos x = \frac{1}{\sqrt{1+t^2}}$. Hence,

$$\begin{aligned} \int \frac{1}{3 + \cos^2 x} dx &= \int \frac{1}{\left\{3 + \frac{1}{1+t^2}\right\} (1+t^2)} dt = \int \frac{(1+t^2)}{(4+3t^2)(1+t^2)} dt \\ &= \int \frac{1}{4+3t^2} dt. \text{ Again put } t = \frac{2}{\sqrt{3}} \tan \theta \Rightarrow dt = \frac{2}{\sqrt{3}} \sec^2 \theta d\theta \\ &= \frac{2}{\sqrt{3}} \int \frac{\sec^2 \theta}{[4 + 4 \tan^2 \theta]} d\theta = \frac{2}{\sqrt{3}} \int \frac{\sec^2 \theta}{4 \sec^2 \theta} d\theta = \frac{1}{2\sqrt{3}} \int d\theta \\ &= \frac{1}{2\sqrt{3}} \theta + c = \frac{1}{2\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{3}t}{2} \right) + c = \frac{1}{2\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{3} \tan x}{2} \right) + c \end{aligned}$$

2. Solve $\int \frac{1}{2 \sin^2 x + 4 \cos^2 x} dx$

Solution

Put $t = \tan x \Rightarrow dx = \frac{dt}{1+t^2}$, $\sin x = \frac{t}{\sqrt{1+t^2}}$ and $\cos x = \frac{1}{\sqrt{1+t^2}}$. Hence,

$$\begin{aligned} \int \frac{1}{2 \sin^2 x + 4 \cos^2 x} dx &= \int \frac{1}{\left\{2 \cdot \frac{t^2}{1+t^2} + 4 \cdot \frac{1}{1+t^2}\right\} (1+t^2)} dt = \int \frac{(1+t^2)}{(2t^2+4)(1+t^2)} dt \\ &= \int \frac{1}{2t^2+4} dt. \text{ Again put } t = \sqrt{2} \tan \theta \Rightarrow dt = \sqrt{2} \sec^2 \theta d\theta \\ &= \int \frac{\sqrt{2} \sec^2 \theta}{[4 \tan^2 \theta + 4]} d\theta = \int \frac{\sqrt{2} \sec^2 \theta}{4 \sec^2 \theta} d\theta = \frac{\sqrt{2}}{4} \int d\theta \\ &= \frac{1}{2\sqrt{2}} \theta + c = \frac{1}{2\sqrt{2}} \tan^{-1} \left(\frac{t}{\sqrt{2}} \right) + c = \frac{1}{2\sqrt{2}} \tan^{-1} \left(\frac{\tan x}{\sqrt{2}} \right) + c \end{aligned}$$

Case 2: Integrals of the form $\int \frac{1}{a + b \sin x + c \cos x} dx$

In this case, we put $t = \tan \left(\frac{x}{2} \right)$ so that

$$\square \quad \frac{dt}{dx} = \frac{1}{2} \sec^2 \left(\frac{x}{2} \right) \Rightarrow dx = \frac{2dt}{\sec^2 \left(\frac{x}{2} \right)} = \frac{2dt}{1 + \tan^2 \left(\frac{x}{2} \right)} \Rightarrow dx = \frac{2dt}{1+t^2}$$

$$\square \quad \sin \left(\frac{x}{2} \right) = \frac{t}{\sqrt{1+t^2}} \text{ and } \cos \left(\frac{x}{2} \right) = \frac{1}{\sqrt{1+t^2}} \text{ (from SOHCAHTOA).}$$

□ Therefore,

$$\sin x = 2 \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right) = 2 \cdot \frac{t}{\sqrt{1+t^2}} \cdot \frac{1}{\sqrt{1+t^2}} = \frac{2t}{1+t^2}$$

and

$$\cos x = \cos^2\left(\frac{x}{2}\right) - \sin^2\left(\frac{x}{2}\right) = \frac{1}{1+t^2} - \frac{t^2}{1+t^2} = \frac{1-t^2}{1+t^2}$$

Example(s):

1. Solve $\int \frac{1}{5+4\cos x} dx$

Solution

Put $t = \tan\left(\frac{x}{2}\right) \Rightarrow dx = \frac{2dt}{1+t^2}$, $\cos\left(\frac{x}{2}\right) = \frac{1}{\sqrt{1+t^2}}$, and $\cos x = \frac{1-t^2}{1+t^2}$. Hence,

$$\begin{aligned} \int \frac{1}{5+4\cos x} dx &= \int \frac{1}{\left\{5+4 \cdot \frac{1-t^2}{1+t^2}\right\}} \frac{2dt}{(1+t^2)} = 2 \int \frac{(1+t^2)}{(9+t^2)} \frac{2dt}{(1+t^2)} \\ &= 2 \int \frac{1}{9+t^2} dt. \text{ Again put } t = 3 \tan \theta \Rightarrow dt = 3 \sec^2 \theta d\theta \\ &= 2 \int \frac{3 \sec^2 \theta}{[9+9 \tan^2 \theta]} d\theta = 6 \int \frac{\sec^2 \theta}{9 \sec^2 \theta} d\theta = \frac{2}{3} \int d\theta \\ &= \frac{2}{3} \theta + c = \frac{2}{3} \tan^{-1}\left(\frac{t}{3}\right) + c = \frac{2}{3} \tan^{-1}\left(\frac{\tan\left(\frac{x}{2}\right)}{3}\right) + c \end{aligned}$$

2. Solve $\int \frac{1}{3\sin x + 4\cos x} dx$

Solution

Put $t = \tan\left(\frac{x}{2}\right) \Rightarrow dx = \frac{2dt}{1+t^2}$, $\cos\left(\frac{x}{2}\right) = \frac{1}{\sqrt{1+t^2}}$. Therefore, $\sin x = \frac{2t}{1+t^2}$ and $\cos x = \frac{1-t^2}{1+t^2}$. Hence, the given integral becomes

$$\begin{aligned} \int \frac{1}{3\sin x + 4\cos x} dx &= \int \frac{1}{\left\{3 \cdot \frac{2t}{1+t^2} + 4 \cdot \frac{1-t^2}{1+t^2}\right\}} \frac{2dt}{(1+t^2)} = \int \frac{2}{6t+4-4t^2} dt \\ &= \int \frac{-1}{2t^2-3t-2} dt = \int \frac{-1}{(2t+1)(t-2)} dt. \text{ (use partial fraction)} \\ &= \frac{1}{5} \int \left(\frac{2}{2t+1} - \frac{1}{t-2}\right) dt = \frac{1}{5} \ln\left(\frac{2t+1}{t-2}\right) + c \\ &= \frac{1}{5} \ln\left(\frac{2 \tan\left(\frac{x}{2}\right) + 1}{\tan\left(\frac{x}{2}\right) - 2}\right) + c \end{aligned}$$

Exercise:

1. Evaluate the following integrals

(a) $\int \frac{1}{1+\sin x - \cos x} dx$ [ans: $I = \ln\left(\frac{\tan\left(\frac{x}{2}\right)}{1+\tan\left(\frac{x}{2}\right)}\right) + c$]

(b) $\int \sec x dx$ [ans: $I = \ln(\sec x + \tan x) + c$]

(c) $\int \frac{\cos x}{2-\cos x} dx$ [hint: long div and t -subst., ans: $I = \frac{4}{\sqrt{3}} \tan^{-1}\left(\sqrt{3} \tan\left(\frac{x}{2}\right)\right) - x + c$]

$$(d) \int \frac{1}{1-3\sin x} dx \quad [\text{ans: } I = \frac{1}{\sqrt{8}} \ln \left(\frac{\tan(\frac{x}{2}) - 3 - \sqrt{8}}{\tan(\frac{x}{2}) - 3 + \sqrt{8}} \right) + c]$$

$$(e) \int \frac{1}{2+\cos x} dx \quad [\text{ans: } I = \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{\tan(\frac{x}{2})}{\sqrt{3}} \right) + c]$$

$$(f) \int \frac{1}{2-3\sin \theta} d\theta \quad [\text{ans: } I = \frac{4}{5} \ln \left| \frac{2\tan(\theta/2) - 3 - \sqrt{5}}{2\tan(\theta/2) - 3 + \sqrt{5}} \right| + c]$$

2. If $t = \tan(x/2)$, show that $\frac{dx}{dt} = \frac{2}{1+t^2}$ and $1 - \cos x = \frac{2t^2}{1+t^2}$. Hence, evaluate $\int \frac{1}{1-\cos x} dx$.
 [ans: $I = -\cot(x/2) + c$]

LECTURE 8

4.5 Integration by parts

The integration by parts formula, derived from the product rule of differentiation, is given as

$$\int u dv = uv - \int v du$$

Proof. Let u and v be functions of x . Differentiating the product uv with respect to x using the product rule of differentiation, we get

$$\frac{d(uv)}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

Rearranging yields

$$u \frac{dv}{dx} = \frac{d(uv)}{dx} - v \frac{du}{dx}$$

Integrating both sides with respect to x yields

$$\int u \frac{dv}{dx} dx = \int \frac{d(uv)}{dx} dx - \int v \frac{du}{dx} dx$$

Simplifying yields $\int u dv = uv - \int v du$. □

→ Note: the parts formula is used when we have a product of two functions and either function is not the derivative of the other. We have to decide which function to take as u and which to link to dv . The following steps are necessary when using the parts formula.

□ To identify u , use the LIATEC rule.

- **L** – Logarithmic function
- **I** – Inverse function
- **A** – Algebraic function
- **T** – Trigonometric function
- **E** – Exponential function
- **C** – Constant term

Thus u must be the function that we encounter first according to the LIATEC rule, from left to right (or from top to bottom). The remaining function(s) and dx form dv . For example, to evaluate:

$$(a) \int x^3 \ln x dx, \text{ put } u = \ln x \text{ and } dv = x^3 dx.$$

(b) $\int e^x \cos x dx$, put $u = \cos x$ and $dv = e^x dx$.

- ☐ Differentiate u with respect to x and make du the subject.
- ☐ Integrate dv with respect to x and disregard the constant of integration because a single constant will be added to the last step.
- ☐ Substitute the above results into the parts formula and simplify.
- ☐ Terminate the process if the last integrand can be integrated or repeat the process if the last integrand still contains product of functions or collect like terms if the last integrand is the same as the starting integrand.

→ Note:

- ☐ If the LIATEC rule fails, then u must be a group of functions that can be differentiated easily and dv must be a group of functions that can be integrated easily.

Example(s):

1. Evaluate $\int x^3 \ln x dx$

Solution

Put $u = \ln x$ and $dv = x^3 dx \Rightarrow du = \frac{1}{x} dx$ and $v = \frac{x^4}{4}$. Therefore,

$$\begin{aligned} \int x^3 \ln x dx &= uv - \int v du \\ &= \frac{x^4}{4} \ln x - \frac{1}{4} \int x^4 \cdot \frac{1}{x} dx = \frac{x^4}{4} \ln x - \frac{1}{4} \int x^3 dx \\ &= \frac{1}{4} x^4 \ln x - \frac{1}{16} x^4 + c \end{aligned}$$

2. Evaluate $\int x^2 \sin x dx$

Solution

Put $u = x^2$ and $dv = \sin x dx \Rightarrow du = 2x dx$ and $v = -\cos x$. Therefore,

$$\begin{aligned} \int x^2 \sin x dx &= uv - \int v du \\ &= -x^2 \cos x - \int (-\cos x) \cdot 2x dx = -x^2 \cos x + 2 \int x \cos x dx \end{aligned}$$

Since the last integrand still contains product of functions of x , we use another integration by parts. Put $u = x$ and $dv = \cos x dx \Rightarrow du = dx$ and $v = \sin x$. Therefore,

$$\begin{aligned} \int x \cos x dx &= uv - \int v du \\ &= x \sin x - \int \sin x dx = x \sin x + \cos x + c \end{aligned}$$

Hence, back substitution yields

$$\int x^2 \sin x dx = -x^2 \cos x + 2x \sin x + 2 \cos x + c_1$$

3. Evaluate $\int e^x \cos x dx$

Solution

Put $u = \cos x$ and $dv = e^x dx \Rightarrow du = -\sin x dx$ and $v = e^x$. Therefore,

$$\begin{aligned}\int e^x \cos x dx &= uv - \int v du \\ &= e^x \cos x - \int e^x \cdot (-\sin x dx) = e^x \cos x + \int e^x \sin x dx\end{aligned}$$

Since the last integrand still contains product of functions of x , we use another integration by parts. Put $u = \sin x$ and $dv = e^x dx \Rightarrow du = \cos x dx$ and $v = e^x$. Therefore,

$$\begin{aligned}\int e^x \sin x dx &= uv - \int v du \\ &= e^x \sin x - \int e^x \cos x dx\end{aligned}$$

Hence, back substitution yields

$$\int e^x \cos x dx = e^x \cos x + e^x \sin x - \int e^x \cos x dx$$

Since the last integrand is the same as the starting integrand, we collect like terms to get

$$2 \int e^x \cos x dx = e^x \cos x + e^x \sin x \Rightarrow \int e^x \cos x dx = \frac{1}{2} (e^x \cos x + e^x \sin x) + c$$

4. Evaluate $\int \ln x dx$

Solution

Put $u = \ln x$ and $dv = dx \Rightarrow du = \frac{1}{x} dx$ and $v = x$. Therefore,

$$\begin{aligned}\int \ln x dx &= uv - \int v du \\ &= x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - \int dx \\ &= x \ln x - x + c\end{aligned}$$

5. Evaluate $\int \sin^{-1} x dx$

Solution

Put $u = \sin^{-1} x$ and $dv = dx \Rightarrow du = \frac{1}{\sqrt{1-x^2}} dx$ and $v = x$. Therefore,

$$\begin{aligned}\int \sin^{-1} x dx &= uv - \int v du \\ &= x \sin^{-1} x - \int x \cdot \frac{1}{\sqrt{1-x^2}} dx = x \sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}} dx, \text{ [put } x = \cos \theta] \\ &= x \sin^{-1} x + \sqrt{1-x^2} + c\end{aligned}$$

Alternatively: to evaluate $\int \frac{x}{\sqrt{1-x^2}} dx$, use the u -substitution method by putting $u = 1 - x^2$.

6. Evaluate $\int \sin^2 x dx$

Solution

Express $\sin^2 x$ as a product of two functions. Thus,

$$\int \sin^2 x dx = \int \sin x \cdot \sin x dx$$

Since the LIATEC rule fails, we put dv to be the function that can be integrated easily. Thus, put $u = \sin x$ and $dv = \sin x dx \Rightarrow du = \cos x dx$ and $v = -\cos x$. Therefore,

$$\begin{aligned} \int \sin^2 x dx &= uv - \int v du \\ &= -\sin x \cos x - \int (-\cos x) \cdot \cos x dx = -\sin x \cos x + \int \cos^2 x dx \\ &= -\sin x \cos x + \int (1 - \sin^2 x) dx = -\sin x \cos x + \int dx - \int \sin^2 x dx \\ \Rightarrow \int \sin^2 x dx + \int \sin^2 x dx &= -\sin x \cos x + \int dx \\ \Rightarrow 2 \int \sin^2 x dx &= -\sin x \cos x + x + c \\ \therefore \int \sin^2 x dx &= \frac{x}{2} - \frac{\sin x \cos x}{2} + c_1 \end{aligned}$$

7. Evaluate $\int \sec^3 x dx$

Solution

Express $\sec^3 x$ as a product of two functions. Thus,

$$\int \sec^3 x dx = \int \sec x \cdot \sec^2 x dx$$

Since the LIATEC rule fails, we put dv to be the function that can be integrated easily. Thus, put $u = \sec x$ and $dv = \sec^2 x dx \Rightarrow du = \sec x \tan x dx$ and $v = \tan x$. Therefore,

$$\begin{aligned} \int \sec^3 x dx &= uv - \int v du \\ &= \sec x \tan x - \int \tan x \cdot \sec x \tan x dx = \sec x \tan x - \int \sec x \tan^2 x dx \\ &= \sec x \tan x - \int \sec x [\sec^2 x - 1] dx = \sec x \tan x + \int \sec x dx - \int \sec^3 x dx \\ \Rightarrow 2 \int \sec^3 x dx &= \sec x \tan x + \int \sec x dx \\ &= \sec x \tan x + \ln[\sec x + \tan x] + c \\ \therefore \int \sec^3 x dx &= \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln[\sec x + \tan x] + c_1 \end{aligned}$$

8. Evaluate $\int \frac{xe^x}{(1+x)^2} dx$

Solution

Since the LIATEC rule fails, we put $u = xe^x$ and $dv = \frac{1}{(1+x)^2} dx \Rightarrow du = e^x(1+x) dx$ and $v = -\frac{1}{(1+x)}$. Substituting into the parts formula yields

$$\begin{aligned} \int \frac{xe^x}{(1+x)^2} dx &= uv - \int v du \\ &= -\frac{xe^x}{(1+x)} + \int \frac{1}{(1+x)} \cdot e^x(1+x) dx = -\frac{xe^x}{(1+x)} + \int e^x dx \\ &= \left[-\frac{xe^x}{(1+x)} + e^x \right] + c = \frac{e^x}{(1+x)} + c \end{aligned}$$

Exercise:

1. Evaluate the following

$$(a) \int x^2 \sin x dx \quad [\text{ans: } I = -x^2 \cos x + 2x \sin x + 2 \cos x + c]$$

$$(b) \int x^3 e^{2x} dx \quad [\text{ans: } I = \frac{1}{8} e^{2x} (4x^3 - 6x^2 + 6x - 3) + c]$$

$$(c) \int \frac{\ln x}{\sqrt{x}} dx \quad [\text{ans: } I = 2\sqrt{x} (\ln x - 2) + c]$$

$$(d) \int 2x^2 \cos^{-1} x dx. \quad [\text{ans: } I = \frac{2}{3} x^3 \cos^{-1} x - \frac{2}{3} \sqrt{1-x^2} + \frac{2}{9} (1-x^2)^{3/2} + c]$$

$$(e) \int x^4 \ln x dx. \quad [\text{ans: } I = \frac{x^5}{25} (5 \ln x - 1) + c]$$

$$(f) \int x^3 e^{x^2} dx \quad [\text{hint: put } u = x^2, \text{ ans: } I = \frac{1}{2} x^2 e^{x^2} - \frac{1}{2} e^{x^2} + c]$$

$$(g) \int x^2 \ln(1+x^2) dx. \quad [\text{ans: } I = \frac{1}{9} (3x^3 \ln|1+x^2| - 2x^3 + 6x - 6 \tan^{-1} x) + c]$$

$$(h) \int x \cos^{-1} x dx. \quad [\text{ans: } I = \frac{1}{2} x^2 \cos^{-1} x + \frac{1}{4} \sin^{-1} x - \frac{1}{4} x \sqrt{1-x^2} + c]$$

$$2. \text{ Evaluate } \int_0^1 \frac{(\tan^{-1} x)^2}{1+x^2} dx \quad [\text{hint: put } u = (\tan^{-1} x)^2, dv = \frac{1}{1+x^2} dx, \text{ ans: } I = \frac{\pi^3}{192}, 7 \text{ Marks}]$$

$$3. \text{ Travis has the following information about the function } f(x): f(5) = 4, f(6) = 7 \text{ and } \int_5^6 e^{-7x} f(x) dx = 14. \text{ Evaluate } \int_5^6 e^{-7x} f'(x) dx. \quad [\text{ans: } I = 98]$$

LECTURE 9

4.6 Integration via partial fraction decomposition

The method of partial fraction decomposition is employed when integrating a rational function of the form $f(x) = \frac{p(x)}{q(x)}$, where $p(x)$ and $q(x)$ are polynomials and $q(x)$ consists of a product of linear and/or irreducible quadratic factors. The requirement is that the degree of $p(x)$ must be less than the degree of $q(x)$. The rational function $f(x)$ is then expressed as a sum of simpler rational functions that can be integrated.

→ Note:

- If the degree of $p(x)$ is greater than or equal to the degree of $q(x)$, then we first use long division to rewrite the integrand.

To decompose a rational function (in which the degree of $p(x)$ is less than the degree of $q(x)$) into partial fractions, we consider the following four cases.

- **Case 1:** when $q(x)$ consists of non-repeated linear factors.

The corresponding partial fraction decomposition is of the form

$$\frac{p(x)}{(a_1x+b_1)(a_2x+b_2)\cdots(a_nx+b_n)} = \frac{A_1}{(a_1x+b_1)} + \frac{A_2}{(a_2x+b_2)} + \cdots + \frac{A_n}{(a_nx+b_n)},$$

where $A_1, A_2, \dots, A_n$ are constants to be determined. For example, the partial fraction decomposition of $\frac{2x+1}{(x+1)(x-1)}$ is of the form

$$\frac{2x+1}{(x+1)(x-1)} = \frac{A_1}{(x+1)} + \frac{A_2}{(x-1)}$$

Multiplying through by the LCM, i.e., $(x+1)(x-1)$, we get

$$A_1(x-1) + A_2(x+1) = 2x+1 \quad (*)$$

To find the values of the unknown constants in equation (*), we use either of the following two methods.

- **Method 1:** We note that equation (*) is true for all x in the real number line. So we choose arbitrary values of x (especially the values of x that will make some terms vanish) and solve the resulting equation. Thus to eliminate the second term, we put $x = -1$ to get

$$-2A_1 = -1 \Rightarrow A_1 = \frac{1}{2}$$

To find A_2 , we put $x = 1$ (or any integer value of x different from -1) to get

$$2A_2 = 3 \Rightarrow A_2 = \frac{3}{2}$$

→ Note: it is advisable to choose integer values of x which are very close to zero (inclusive).

- **Method 2:** Expand the left-hand side of equation (*) and equate the corresponding coefficients of like powers of x to obtain a system of algebraic equations in the unknown constants. Thus, expanding equation (*) and simplifying yields

$$(A_1 + A_2)x + (-A_1 + A_2) = 2x + 1$$

Equating the corresponding coefficients of like powers of x yields

- Coefficients of x : $A_1 + A_2 = 2 \quad (i)$
- Constant terms: $-A_1 + A_2 = 1 \quad (ii)$

Solving equations (i) and (ii) simultaneously yields $A_1 = \frac{1}{2}$ and $A_2 = \frac{3}{2}$.

Therefore, replacing A_1 and A_2 in the given rational function yields

$$\frac{2x+1}{(x+1)(x-1)} = \frac{1}{2} \left(\frac{1}{x+1} + \frac{3}{x-1} \right)$$

- **Case 2:** when $q(x)$ consists of linear factors, at least one of which is repeated.

This is treated in the same manner as case 1 but a repeated factor gives rise to partial fractions of the form

$$\frac{p(x)}{(a_1x+b_1)(a_2x+b_2)\cdots(a_nx+b_n)(cx+d)^k} = \frac{A_1}{(a_1x+b_1)} + \frac{A_2}{(a_2x+b_2)} + \cdots + \frac{A_n}{(a_nx+b_n)} \\ + \frac{B_1}{(cx+d)} + \frac{B_2}{(cx+d)^2} + \cdots + \frac{B_k}{(cx+d)^k},$$

where $A_1, A_2, \dots, A_n; B_1, B_2, \dots, B_k$ are constants to be determined. For example, the partial fraction decomposition of $\frac{3x+1}{(x-1)^2(x-2)}$ is of the form

$$\frac{3x+1}{(x-1)^2(x-2)} = \frac{A_1}{(x-1)} + \frac{A_2}{(x-1)^2} + \frac{A_3}{(x-2)}$$

Multiplying through by the LCM, i.e., $(x-1)^2(x-2)$, we get

$$A_1(x-1)(x-2) + A_2(x-2) + A_3(x-1)^2 = 3x+1 \quad (*)$$

To find the values of the unknown constants in equation (*), we use either of the following two methods.

□ **Method 1:** To eliminate the first term, we put $x = 1$ to get

$$-A_2 = 4 \Rightarrow A_2 = -4$$

To find A_1 and A_3 , we choose two arbitrary integer values of x different from 1. Thus,

- Put $x = 2$ and $A_2 = -4$ to get $A_1(1)(0) + (-4)(0) + A_3(1) = 7 \Rightarrow A_3 = 7$
- Put $x = 0$, $A_2 = -4$ and $A_3 = 7$ to get $A_1(-1)(-2) + (-4)(-2) + (7)(1) = 1 \Rightarrow A_1 = -7$

□ **Method 2:** Expanding equation (*) and simplifying yields

$$(A_1 + A_3)x^2 + (-3A_1 + A_2 - 2A_3)x + (2A_1 - 2A_2 + A_3) = 3x + 1$$

Equating the corresponding coefficients of like powers of x yields

- Coefficients of x^2 : $A_1 + A_3 = 0$ (i)
- Coefficients of x : $-3A_1 + A_2 - 2A_3 = 3$ (ii)
- Constant terms: $2A_1 - 2A_2 + A_3 = 1$ (iii)

Solving equations (i), (ii) and (iii) simultaneously yields $A_1 = -7$, $A_2 = -4$ and $A_3 = 7$.

Therefore, replacing A_1 , A_2 and A_3 in the given rational function yields

$$\frac{3x+1}{(x-1)^2(x-2)} = -\frac{7}{(x-1)} - \frac{4}{(x-1)^2} + \frac{7}{(x-2)}$$

→ Note: x^k , for $k = 2, 3, 4, \dots$, is a repeated linear factor. E.g., $\frac{18}{x^2(x-3)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-3}$.

□ **Case 3:** when $q(x)$ contains irreducible quadratic factors, but none is repeated.

The corresponding partial fraction decomposition is of the form

$$\frac{p(x)}{(a_1x+b_1)(a_2x+b_2)\cdots(a_nx+b_n)(cx^2+dx+e)} = \frac{A_1}{(a_1x+b_1)} + \frac{A_2}{(a_2x+b_2)} + \cdots + \frac{A_n}{(a_nx+b_n)} + \frac{Bx+C}{(cx^2+dx+e)},$$

where $A_1, A_2, \dots, A_n$; B, C are constants to be determined.

→ Note: a quadratic factor is a quadratic expression that cannot be factorized further (or an irreducible quadratic expression), e.g., $(x^2 + 1)$, $(x^2 + 3x + 5)$, etc.

For example, the partial fraction decomposition of $\frac{x^2-1}{(x+2)(x^2+1)}$ is of the form

$$\frac{x^2-1}{(x+2)(x^2+1)} = \frac{A_1}{(x+2)} + \frac{Bx+C}{(x^2+1)}$$

Multiplying through by the LCM, i.e., $(x+2)(x^2+1)$, we get

$$A_1(x^2+1) + (Bx+C)(x+2) = x^2-1 \quad (*)$$

To find the values of the unknown constants in equation (*), we use either of the following two methods.

□ **Method 1:** To eliminate the second term, we put $x = -2$ to get

$$5A_1 = 3 \Rightarrow A_1 = \frac{3}{5}$$

To find B and C , we choose two arbitrary integer values of x different from -2. Thus,

- Put $x = 0$ and $A_1 = \frac{3}{5}$ to get $\frac{3}{5} + 2C = -1 \Rightarrow C = -\frac{4}{5}$

- Put $x = 1$, $A_1 = \frac{3}{5}$ and $C = -\frac{4}{5}$ to get $(\frac{3}{5})(2) + (B - \frac{4}{5})(3) = 0 \Rightarrow B = \frac{2}{5}$

□ **Method 2:** Expanding equation (*) and simplifying yields

$$(A_1 + B)x^2 + (2B + C)x + (A_1 + 2C) = x^2 - 1$$

Equating the corresponding coefficients of like powers of x yields

- Coefficients of x^2 : $A_1 + B = 1$ (i)
- Coefficients of x : $2B + C = 0$ (ii)
- Constant terms: $A_1 + 2C = -1$ (iii)

Solving equations (i), (ii) and (iii) simultaneously yields $A_1 = \frac{3}{5}$, $B = \frac{2}{5}$ and $C = -\frac{4}{5}$.

Therefore, replacing A_1, B and C in the given rational function yields

$$\frac{x^2 - 1}{(x + 2)(x^2 + 1)} = \frac{1}{5} \left[\frac{3}{(x + 2)} + \frac{2x - 4}{(x^2 + 1)} \right]$$

□ **Case 4:** when $q(x)$ contains at least one repeated irreducible quadratic factor.

The corresponding partial fraction decomposition is of the form

$$\frac{p(x)}{(a_1x + b_1)(a_2x + b_2) \cdots (a_nx + b_n)(cx^2 + dx + e)^k} = \frac{A_1}{(a_1x + b_1)} + \frac{A_2}{(a_2x + b_2)} + \cdots + \frac{A_n}{(a_nx + b_n)} \\ + \frac{B_1x + C_1}{(cx^2 + dx + e)} + \frac{B_2x + C_2}{(cx^2 + dx + e)^2} + \cdots + \frac{B_kx + C_k}{(cx^2 + dx + e)^k},$$

where $A_1, A_2, \dots, A_n; B, C, D, E$ are constants to be determined. For example, the partial fraction decomposition of $\frac{x^3 + 1}{(x + 1)(x^2 + 1)^2}$ is of the form

$$\frac{x^3 + 1}{(x + 1)(x^2 + 1)^2} = \frac{A_1}{(x + 1)} + \frac{Bx + C}{(x^2 + 1)} + \frac{Dx + E}{(x^2 + 1)^2}$$

Multiplying through by the LCM, i.e., $(x + 1)(x^2 + 1)^2$, we get

$$A_1(x^2 + 1)^2 + (Bx + C)(x + 1)(x^2 + 1) + (Dx + E)(x + 1) = x^3 + 1 \quad (*)$$

To find the values of the unknown constants in equation (*), we use either of the following two methods.

□ **Method 1:** To eliminate the second term, we put $x = -1$ to get

$$4A_1 = 0 \Rightarrow A_1 = 0$$

To find B, C, D and E , we choose four arbitrary integer values of x different from -1. Thus,

- Put $x = 0$ and $A_1 = 0$ to get $C + E = 1$ (i)
- Put $x = 1$ and $A_1 = 0$ to get $2B + 2C + D + E = 1$ (ii)
- Put $x = -2$ and $A_1 = 0$ to get $10B - 5C + 2D - E = -7$ (iii)
- Put $x = 2$ and $A_1 = 0$ to get $10B + 5C + 2D + E = 3$ (iv)

Solving equations (i) to (iv) simultaneously yields $B = 0, C = 1, D = -1$ and $E = 0$.

□ **Method 2:** Expanding equation (*) and simplifying yields

$$(A_1 + B)x^4 + (B + C)x^3 + (2A_1 + B + C + D)x^2 + (B + C + D + E)x + (A_1 + C + E) = x^3 + 1$$

Equating the corresponding coefficients of like powers of x yields

- Coefficients of x^4 : $A_1 + B = 0$ (i)
- Coefficients of x^3 : $B + C = 1$ (ii)
- Coefficients of x^2 : $2A_1 + B + C + D = 0$ (iii)

- Coefficients of x : $B + C + D + E = 0$ (iv)
- Constant terms: $A_1 + C + E = 1$ (v)

Solving equations (i) to (v) simultaneously yields $A_1 = 0, B = 0, C = 1, D = -1$ and $E = 0$.

Therefore, replacing A_1, B, C, D and E in the given rational function yields

$$\frac{x^3 + 1}{(x + 1)(x^2 + 1)^2} = \frac{1}{(x^2 + 1)} - \frac{x}{(x^2 + 1)^2}$$

Example(s):

1. Evaluate $\int \frac{3x^3 - 18x^2 + 29x - 4}{(x + 1)(x - 2)^3} dx$

Solution

Clearly, the degree of the polynomial in the numerator is less than the degree of the polynomial in the denominator. Let

$$\frac{3x^3 - 18x^2 + 29x - 4}{(x + 1)(x - 2)^3} = \frac{A}{(x + 1)} + \frac{B}{(x - 2)} + \frac{C}{(x - 2)^2} + \frac{D}{(x - 2)^3}$$

Multiply both sides by $(x + 1)(x - 2)^3$ to get

$$A(x - 2)^3 + B(x + 1)(x - 2)^2 + C(x + 1)(x - 2) + D(x + 1) = 3x^3 - 18x^2 + 29x - 4$$

- Put $x = 2 \Rightarrow D = 2$
- Put $x = -1 \Rightarrow A = 2$
- Put $x = 0 \Rightarrow 2B - C = 5$ (i)
- Put $x = 1 \Rightarrow B - C = 4$ (ii)

Solving (i) and (ii) simultaneously yields $B = 1, C = -3$. Substituting the values of A, B, C and D into the given rational function yields

$$\frac{3x^3 - 18x^2 + 29x - 4}{(x + 1)(x - 2)^3} = \frac{2}{(x + 1)} + \frac{1}{(x - 2)} - \frac{3}{(x - 2)^2} + \frac{2}{(x - 2)^3}$$

Integrating both sides with respect to x , we have

$$\begin{aligned} \int \frac{3x^3 - 18x^2 + 29x - 4}{(x + 1)(x - 2)^3} dx &= \int \left(\frac{2}{(x + 1)} + \frac{1}{(x - 2)} - \frac{3}{(x - 2)^2} + \frac{2}{(x - 2)^3} \right) dx \\ &= \int \frac{2}{x + 1} dx + \int \frac{1}{x - 2} dx - \int \frac{3}{(x - 2)^2} dx + \int \frac{2}{(x - 2)^3} dx \\ &= 2 \ln |x + 1| + \ln |x - 2| + \frac{3}{x - 2} - \frac{2}{(x - 2)^2} + c \end{aligned}$$

2. Evaluate $\int \frac{x^2 - x - 21}{(2x - 1)(x^2 + 4)} dx$

Solution

Clearly, the degree of the polynomial in the numerator is less than the degree of the polynomial in the denominator. Let

$$\frac{x^2 - x - 21}{(2x - 1)(x^2 + 4)} = \frac{A}{(2x - 1)} + \frac{Bx + C}{(x^2 + 4)}$$

Multiply both sides by $(2x - 1)(x^2 + 4)$ to get

$$A(x^2 + 4) + (Bx + C)(2x - 1) = x^2 - x - 21$$

$$\begin{aligned}\square \text{ Put } x = \frac{1}{2} &\Rightarrow A = -5 \\ \square \text{ Put } x = 0 &\Rightarrow C = 1 \\ \square \text{ Put } x = 1 &\Rightarrow B = 3\end{aligned}$$

Substituting the values of A, B and C into the given rational function yields

$$\frac{x^2 - x - 21}{(2x - 1)(x^2 + 4)} = -\frac{5}{(2x - 1)} + \frac{3x + 1}{(x^2 + 4)}$$

Integrating both sides with respect to x , we have

$$\begin{aligned}\int \frac{x^2 - x - 21}{(2x - 1)(x^2 + 4)} dx &= \int \left(\frac{-5}{2x - 1} + \frac{3x + 1}{x^2 + 4} \right) dx \\ &= -\frac{5}{2} \int \frac{2}{2x - 1} dx + \frac{3}{2} \int \frac{2x}{x^2 + 4} dx + \int \frac{1}{x^2 + 4} dx \\ &= -\frac{5}{2} \ln |2x - 1| + \frac{3}{2} \ln |x^2 + 4| + \int \frac{1}{x^2 + 4} dx\end{aligned}$$

Now, to evaluate $\int \frac{1}{x^2 + 4} dx$ put $x = 2 \tan \theta \Rightarrow dx = 2 \sec^2 \theta d\theta$. So,

$$\begin{aligned}\int \frac{1}{x^2 + 4} dx &= \int \frac{2 \sec^2 \theta d\theta}{4 \tan^2 \theta + 4} = \int \frac{2 \sec^2 \theta d\theta}{4 \sec^2 \theta} = \frac{1}{2} \int d\theta = \frac{1}{2} \theta + c \\ &= \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + c\end{aligned}$$

Hence,

$$\int \frac{x^2 - x - 21}{(2x - 1)(x^2 + 4)} dx = -\frac{5}{2} \ln |2x - 1| + \frac{3}{2} \ln |x^2 + 4| + \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + c$$

3. Evaluate $\int \frac{x^2}{x^3 - 3x^2 - 9x + 27} dx$

Solution

Clearly, the degree of the polynomial in the numerator is less than the degree of the polynomial in the denominator. Factorizing the denominator, we get $x^3 - 3x^2 - 9x + 27 = (x - 3)^2(x + 3)$. Thus, the integrand becomes

$$\frac{x^2}{(x - 3)^2(x + 3)} = \frac{A}{(x - 3)} + \frac{B}{(x - 3)^2} + \frac{C}{(x + 3)}$$

Multiply both sides by $(x - 3)^2(x + 3)$ to get

$$A(x - 3)(x + 3) + B(x + 3) + C(x - 3)^2 = x^2$$

$$\square \text{ Put } x = 3 \Rightarrow B = \frac{3}{2}.$$

$$\square \text{ Put } x = -3 \Rightarrow C = \frac{1}{4}$$

$$\square \text{ Put } x = 0 \Rightarrow -9A + 3B + 9C = 0 \Rightarrow A = \frac{3}{4}$$

Substituting the values of A, B and C into the given rational function yields

$$\frac{x^2}{(x - 3)^2(x + 3)} = \frac{1}{4} \left[\frac{3}{(x - 3)} + \frac{6}{(x - 3)^2} + \frac{1}{(x + 3)} \right]$$

Integrating both sides with respect to x , we have

$$\begin{aligned}\therefore \int \frac{x^2}{x^3 - 3x^2 - 9x + 27} dx &= \frac{1}{4} \int \left[\frac{3}{(x - 3)} + \frac{6}{(x - 3)^2} + \frac{1}{(x + 3)} \right] dx \\ &= \frac{3}{4} \int \frac{1}{x - 3} dx + \frac{3}{2} \int \frac{1}{(x - 3)^2} dx + \frac{1}{4} \int \frac{1}{x + 3} dx \\ &= \frac{3}{4} \ln |x - 3| - \frac{3}{2} \cdot \frac{1}{x - 3} + \frac{1}{4} \ln |x + 3| + c\end{aligned}$$

4. Evaluate $\int \frac{2x^3 + x^2 - 6x + 7}{x^2 + x - 6} dx$

Solution

Since the degree of the polynomial in the numerator is greater than the degree of the polynomial in the denominator, we perform long division to get

$$\begin{array}{r} 2x - 1 \\ x^2 + x - 6 \overline{) 2x^3 + x^2 - 6x + 7} \\ \underline{- 2x^3 - 2x^2 + 12x} \\ -x^2 + 6x + 7 \\ \underline{x^2 + x - 6} \\ 7x + 1 \end{array}$$

Therefore, the integrand can be written as

$$\frac{2x^3 + x^2 - 6x + 7}{x^2 + x - 6} = 2x - 1 + \frac{7x + 1}{x^2 + x - 6}$$

Integrating both sides with respect to x yields

$$\int \frac{2x^3 + x^2 - 6x + 7}{x^2 + x - 6} dx = \int (2x - 1) dx + \int \frac{7x + 1}{x^2 + x - 6} dx \quad (*)$$

$$\square \int (2x - 1) dx = x^2 - x + c_1$$

$\square$ To evaluate $\int \frac{7x + 1}{x^2 + x - 6} dx$, we use partial fraction decomposition. Factorizing the denominator, we get $x^2 + x - 6 = (x - 2)(x + 3)$. Thus, the integrand becomes

$$\frac{7x + 1}{(x - 2)(x + 3)} = \frac{A}{x - 2} + \frac{B}{x + 3}$$

Multiply both sides by $(x - 2)(x + 3)$ to get

$$A(x + 3) + B(x - 2) = 7x + 1$$

$$\square \text{ Put } x = 2 \Rightarrow A = 3$$

$$\square \text{ Put } x = -3 \Rightarrow B = 4$$

Therefore,

$$\frac{7x + 1}{x^2 + x - 6} = \frac{3}{x - 2} + \frac{4}{x + 3}$$

Integrating both sides with respect to x yields

$$\int \frac{7x + 1}{x^2 + x - 6} dx = \int \left(\frac{3}{x - 2} + \frac{4}{x + 3} \right) dx = 3 \ln |x - 2| + 4 \ln |x + 3| + c_2$$

Therefore, equation $(*)$ becomes $\int \frac{2x^3 + x^2 - 6x + 7}{x^2 + x - 6} dx = x^2 - x + 3 \ln |x - 2| + 4 \ln |x + 3| + c$, where $c = c_1 + c_2$.

5. Evaluate $\int \frac{x^2}{(x^2 - 1)} dx$

Solution

Since the degree of the polynomial in the numerator is equal to the degree of the polynomial in the denominator, we perform long division to get

$$\begin{array}{r} 1 \\ x^2 - 1 \overline{) x^2} \\ \underline{- x^2 + 1} \\ 1 \end{array}$$

Therefore, the integrand can be written as

$$\frac{x^2}{x^2 - 1} = 1 + \frac{1}{x^2 - 1}$$

Integrating both sides with respect to x yields

$$\int \frac{x^2}{(x^2 - 1)} dx = \int 1 dx + \int \frac{1}{(x^2 - 1)} dx \quad (*)$$

$$\square \int 1 dx = x + c_1$$

$\square$ To evaluate $\int \frac{1}{(x^2 - 1)} dx$, we use partial fraction decomposition. Factorizing the denominator, we get $x^2 - 1 = (x - 1)(x + 1)$. Thus, the integrand becomes

$$\frac{1}{(x - 1)(x + 1)} = \frac{A}{(x - 1)} + \frac{B}{(x + 1)}$$

Multiply both sides by $(x - 1)(x + 1)$ to get

$$A(x + 1) + B(x - 1) = 1$$

$$\square \text{ Put } x = 1 \Rightarrow A = \frac{1}{2}$$

$$\square \text{ Put } x = -1 \Rightarrow B = -\frac{1}{2}$$

Therefore,

$$\frac{1}{x^2 - 1} = \frac{1}{2} \left[\frac{1}{x - 1} - \frac{1}{x + 1} \right]$$

Integrating both sides with respect to x yields

$$\begin{aligned} \int \frac{1}{x^2 - 1} dx &= \frac{1}{2} \int \left[\frac{1}{x - 1} - \frac{1}{x + 1} \right] dx = \frac{1}{2} [\ln |x - 1| - \ln |x + 1|] + c_2 \\ &= \frac{1}{2} \ln \left| \frac{x - 1}{x + 1} \right| + c_2 \end{aligned}$$

Therefore, equation $(*)$ becomes $\int \frac{x^2}{x^2 - 1} dx = x + \frac{1}{2} \ln \left| \frac{x - 1}{x + 1} \right| + c$, where $c = c_1 + c_2$.

Exercise:

- (a) Decompose $\frac{x^2 - 29x + 5}{(x - 4)^2(x^2 + 3)}$ into partial fractions. Hence, evaluate $\int \frac{x^2 - 29x + 5}{(x - 4)^2(x^2 + 3)} dx$.
[ans: $\frac{x^2 - 29x + 5}{(x - 4)^2(x^2 + 3)} = \frac{1}{(x - 4)} - \frac{5}{(x - 4)^2} + \frac{2 - x}{(x^2 + 3)}$, $I = \ln |x - 4| + \frac{5}{x - 4} - \frac{1}{2} \ln |x^2 + 3| + \frac{2}{\sqrt{3}} \tan^{-1}(\frac{x}{\sqrt{3}}) + c$]

(b) Decompose $\frac{x^3 + 10x^2 + 3x + 36}{(x - 1)(x^2 + 4)^2}$ into its partial fractions. Hence, evaluate the integral $\int \frac{x^3 + 10x^2 + 3x + 36}{(x - 1)(x^2 + 4)^2} dx$.
[ans: $\frac{x^3 + 10x^2 + 3x + 36}{(x - 1)(x^2 + 4)^2} = \frac{2}{(x - 1)} - \frac{2x + 1}{(x^2 + 4)} + \frac{x}{(x^2 + 4)^2}$,
 $I = 2 \ln |x - 1| - \ln |x^2 + 4| - \frac{1}{2} \tan^{-1}(\frac{x}{2}) - \frac{1}{2(x^2 + 4)} + c$]

(c) Decompose $\frac{x^4 - 5x^3 + 6x^2 - 18}{x^3 - 3x^2}$ into its partial fractions. Hence, evaluate the integral $\int \frac{x^4 - 5x^3 + 6x^2 - 18}{x^3 - 3x^2} dx$.
[ans: $\frac{x^4 - 5x^3 + 6x^2 - 18}{x^3 - 3x^2} = x - 2 + \frac{2}{x} + \frac{6}{x^2} - \frac{2}{(x - 3)}$,
 $I = \frac{1}{2}x^2 - 2x + 2 \ln |x| - \frac{6}{x} - 2 \ln |x - 3| + c$]

(d) Evaluate $\int \frac{x^4 + x^3 + x^2 + 1}{x^2 + x - 2} dx$. [ans: $I = \frac{1}{3}x^3 + 3x + \frac{4}{3}\ln|x-1| - \frac{13}{3}\ln|x+2| + c$]

2. Evaluate the following integrals

(a) $\int \frac{1}{3-2x-x^2} dx$. [ans: $I = \frac{1}{4}\ln\left|\frac{x+3}{x-1}\right| + c$]

(b) $\int \frac{a^2}{(x+a)(x^2+2a^2)} dx$, where $a \in \mathbb{R}$. [ans: $I = \frac{1}{3\sqrt{2}}\tan^{-1}\left(\frac{x}{a\sqrt{2}}\right) + \frac{1}{6}\ln\left|\frac{(x+a)^2}{x^2+a^2}\right| + c$]

(c) $\int \frac{1}{x^3+1} dx$. [hint: $x^3+1 = (x+1)^3 - 3x(x+1) = (x+1)(x^2-x+1)$, ans:
 $I = \frac{1}{6}\ln\left|\frac{4(x+1)^2}{3(x^2-x+1)}\right| + \frac{\sqrt{3}}{3}\tan^{-1}\left(\frac{2x-1}{\sqrt{3}}\right) + c$]

(d) $\int \frac{4x^2+1}{x(2x-1)^2} dx$. [ans: $I = \ln|x| - \frac{2}{2x-1} + c$]

(e) $\int \frac{4x^2-x+12}{x(x^2+4)} dx$. [ans: $I = 3\ln|x| - \frac{1}{2}\ln|x^2+4| - \frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right) + c$]

(f) $\int \frac{x^4-2x^3+3x^2-x+3}{x^3-2x^2+3x} dx$. [ans: $I = \frac{x^2}{2} + \ln\left|\frac{x}{\sqrt{x^2-2x+3}}\right| + c$, 8 Marks]

(g) $\int \frac{x^4-x^3-x-1}{x^3-x^2} dx$. [ans: $I = \frac{x^2}{2} + \frac{1}{x} + 2\ln\left|\frac{x-1}{x}\right| + c$, 5 Marks]

(h) $\int \frac{1}{x(x^2+1)^2} dx$. [ans: $I = \ln\left|\frac{x}{\sqrt{x^2+1}}\right| + \frac{1}{2(x^2+1)} + c$]

(i) $\int \frac{10x-2x^2}{(x-1)^2(x+3)} dx$. [ans: $I = \ln\left|\frac{x-1}{(x+3)^3}\right| - \frac{2}{x-1} + c$]

(j) $\int \frac{16}{x^2+6x+5} dx$. [ans: $I = 4\ln\left|\frac{x+1}{x+5}\right| + c$]

(k) $\int \frac{3x^2-7x-2}{x^3-x} dx$. [ans: $I = 2\ln|x| + 4\ln|x+1| - 3\ln|x-1| + c$]

(l) $\int \frac{1}{x(x^4+1)} dx$. [hint: $\frac{1}{x(x^4+1)} = \frac{A}{x} + \frac{Bx^3+Cx^2+Dx+E}{(x^4+1)}$, ans:
 $I = \ln|x| - \frac{1}{4}\ln|x^4+1| + c$]

(m) $\int \frac{x^2+1}{x^3+2x^2+x} dx$. [hint: $x^3+2x^2+x = x(x+1)^2$, ans: $I = \ln|x| + \frac{2}{(x+1)} + c$]

(n) $\int \frac{4x^4+x+1}{x^5+x^4} dx$. [hint: $\frac{4x^4+x+1}{x^4(x+1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{D}{x^4} + \frac{E}{(x+1)}$, ans:
 $I = -\frac{1}{3x^3} + 4\ln|x+1| + c$]

(p) $\int \frac{x^2+4}{3x^3-4x^2-4x} dx$. [ans: $I = \frac{1}{2}\ln|x-2| + \frac{5}{6}\ln|3x+2| - \ln|x| + c$]

(q) $\int \frac{x^4-x^3-x-1}{x^3-x^2} dx$. [ans: $I = \frac{x^2}{2} + 2\ln|x| - \frac{1}{x} - 2\ln|x-1| + c$]

3. Evaluate

(a) $\int \frac{\sqrt{9-4x^2}}{x} dx$. [hint: put $u^2 = 9-4x^2$ then use partial fraction. 5 Marks]

(b) $\int \frac{1}{x^3+1} dx$. [hint:
 $(x+1)^3 = x^3+3x^2+3x+1$ so $x^3+1 = (x+1)^3 - 3x(x+1) = (x+1)(x^2-x+1)$, ans:
 $= \frac{1}{3}\ln|x+1| - \frac{1}{6}\ln(x^2-x+1) + \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x-1}{\sqrt{3}}\right) + c$]

LECTURE 10

5 Solution of first order ODEs by separation of variables method

An ordinary differential equation (ODE) is an equation that contains derivatives of one or more dependent variables with respect to one independent variable. For example,

$$\frac{d^2y}{dx^2} + x \frac{dy}{dx} + 3y = 2xe^{x^2}$$

The *order* of an ODE is the order of the highest derivative involved in the equation. The *degree* of an ODE is the greatest power of the highest derivative involved in the equation. A first order ODE in two variables, say x and y , takes the general form

$$M(x, y)dx + N(x, y)dy = 0$$

If in a first order ODE it is possible to collect all the functions of x and dx on one side and all the functions of y and dy on the other side, we say that the variables are separable. Thus, the above equation becomes

$$P(x)dx + Q(y)dy = 0$$

To find the general solution, we integrate both sides to get

$$\int P(x)dx + \int Q(y)dy = C,$$

where C is a constant of integration (or arbitrary constant).

Example(s):

- Find the general solution of the following differential equations

$$(a) \quad \frac{dy}{dx} = \frac{2x}{y+1}.$$

Solution

The given equation can be rewritten as $(y+1)dy = 2xdx$. Integrating yields

$$\int (y+1)dy = \int 2xdx \quad \Rightarrow \quad \frac{y^2}{2} + y = x^2 + c$$

$$(b) \quad \frac{dy}{dx} = (1+x)(1+y).$$

Solution

The given equation can be rewritten as $\frac{dy}{1+y} = (1+x)dx$. Integrating yields

$$\int \frac{dy}{1+y} = \int (1+x)dx \quad \Rightarrow \quad \ln|1+y| = x + \frac{x^2}{2} + c$$

$$(c) \quad \frac{dy}{dx} = e^{3x-2y} + x^2e^{-2y}.$$

Solution

The given equation can be rewritten as

$$\frac{dy}{dx} = e^{3x}e^{-2y} + x^2e^{-2y} = e^{-2y}(e^{3x} + x^2) \quad \Rightarrow \quad e^{2y}dy = (e^{3x} + x^2)dx$$

Integrating yields

$$\int e^{2y}dy = \int (e^{3x} + x^2)dx \quad \Rightarrow \quad \frac{1}{2}e^{2y} = \frac{1}{3}e^{3x} + \frac{1}{3}x^3 + c$$

(d) $\frac{dy}{dx} = \frac{y^2 + xy^2}{x^2y - x^2}.$

Solution

The given equation can be rewritten as

$$\frac{dy}{dx} = \frac{y^2(1+x)}{x^2(y-1)} \Rightarrow \frac{y-1}{y^2} dy = \frac{x+1}{x^2} dx$$

Integrating yields

$$\int \frac{y-1}{y^2} dy = \int \frac{x+1}{x^2} dx \Rightarrow \int \left(\frac{1}{y} - \frac{1}{y^2} \right) dy = \int \left(\frac{1}{x} + \frac{1}{x^2} \right) dx \Rightarrow \ln y + \frac{1}{y} = \ln x - \frac{1}{x} + c$$

2. Solve the initial value problem $\frac{dy}{dx} = 2xy^2$, $y(0) = \frac{1}{2}$.

Solution

The given equation can be rewritten as $\frac{dy}{y^2} = 2xdx$. Integrating yields

$$\int \frac{dy}{y^2} = \int 2xdx \Rightarrow -\frac{1}{y} = x^2 + c \Rightarrow y = \frac{-1}{x^2 + c} \quad (*)$$

The initial condition $y(0) = \frac{1}{2}$ means that when $x = 0$, $y = \frac{1}{2}$. Substituting into the general solution (*), we get

$$\frac{1}{2} = \frac{-1}{0^2 + c} \Rightarrow c = -2$$

Therefore, the particular solution of the given ODE is $y = \frac{-1}{x^2 - 2}$.

→ Note:

□ An initial value problem (IVP) is a differential equation along with initial condition. The initial condition is used to find the value of the arbitrary constant, c .

Exercise:

1. Solve the following differential equations.

(a) $\frac{dy}{dx} = \frac{y^2 - 1}{x}$. [ans: $\frac{1}{2} \ln \left(\frac{y-1}{y+1} \right) = \ln x + c$]

(b) $xy \frac{dy}{dx} = \frac{x^2 + 1}{y + 1}$. [ans: $\frac{y^3}{3} + \frac{y^2}{2} = \frac{x^2}{2} + \ln x + c$]

(c) $x \frac{dy}{dx} = y + xy$. [ans: $y = Axe^x$]

(d) $y \tan x \frac{dy}{dx} = (4 + y^2) \sec^2 x$. [ans: $y^2 = A \tan^2 x - 4$]

(e) $\frac{dy}{dx} = \frac{y}{x}$. [ans: $y = Ax$]

(f) $\frac{\sin x}{1+y} \frac{dy}{dx} = \cos x$. [ans: $y = A \sin x - 1$]

(g) $2xdx + (1 - x^2) \cot y dy = 0$. [ans: $\sin y = c(1 - x^2)$]

(h) $\frac{dy}{dx} = \frac{xy}{x^2 + 1}$, $y(0) = 1$. [ans: $y^2 = x^2 + 1$]

(i) $y \tan x \frac{dy}{dx} = (4 + y^2) \sec^2 x$. [ans: $y = \sqrt{c \tan^2 x - 4}$]

2. A company is using Newspaper advertising to introduce a new product to a community of 50,000 people. It was assumed that the rate at which people learn about the new product is proportional to the number of people who have not heard about it. If no individual is aware of the product initially and 5000 people were aware about it after 10 days of the campaign, [hint: $\frac{dN}{dt} \propto (50000 - N)$]

- i) How many people will be aware of the product after 20 days?
- ii) When will half of the community be aware of the product?

6 Numerical Integration

Many functions such as $\sqrt{1+x^3}$ or e^{x^2} do not have antiderivatives that are easily found. For such functions, the fundamental theorem of calculus is not useful and we can only approximate definite integrals involving these functions. Thus, we need to approximate the area of the region R bounded by the curve $y = f(x)$, the lines $x = a$ and $x = b$, and the x -axis, i.e.,

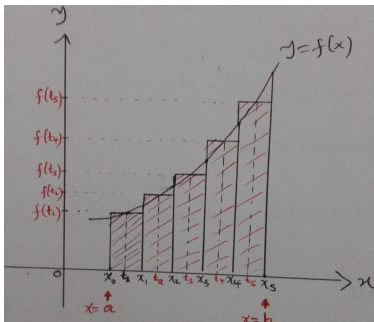
$$A = \int_a^b f(x) dx$$

The numerical techniques used to approximate this integral are mid-ordinate rule, trapezoidal rule and Simpson's rule.

→ Note:

- If $f(x)$ involves trigonometric and hyperbolic functions, then set the *radian mode* in your calculator before you perform the computations.

6.1 Mid-ordinate rule



In mid-ordinate rule, we subdivide the region R into N strips of equal width $h = \frac{b-a}{N}$ and then approximate the strips with rectangles of length $f(t_{i+1})$, where t_{i+1} is the mid-point of the subinterval $[x_i, x_{i+1}]$, i.e., $t_{i+1} = \frac{x_i + x_{i+1}}{2}$. The total area is the sum of the areas of the resulting rectangles, i.e.,

$$\int_a^b f(x) dx \approx \sum_{i=0}^{N-1} \{h f(t_{i+1})\}$$

Expanding the summation and simplifying yields the mid-ordinate rule as

$$\int_a^b f(x) dx \approx h [f(t_1) + f(t_2) + f(t_3) + \cdots + f(t_N)] = h [\text{sum of mid-ordinates}]$$

→ Note: The bound for the truncation error (E) in using the mid-ordinate rule is given by

$$|E| \leq \frac{|b-a|}{24} h^2 \max_{a \leq x \leq b} |f''(x)|$$

Example(s):

1. i) Approximate $\int_1^4 \frac{1}{x} dx$ correct to 4 d.p using the mid-ordinate rule with 7 ordinates.

ii) Obtain a bound for the truncation error in this approximation.

Solution

i) $f(x) = \frac{1}{x}, a = 1, b = 4, N = 6, h = \frac{b-a}{N} = \frac{4-1}{6} = \frac{1}{2}.$

t	$\frac{5}{4}$	$\frac{7}{4}$	$\frac{9}{4}$	$\frac{11}{4}$	$\frac{13}{4}$	$\frac{15}{4}$
$f(t)$	$\frac{4}{5}$	$\frac{4}{7}$	$\frac{4}{9}$	$\frac{4}{11}$	$\frac{4}{13}$	$\frac{4}{15}$

Therefore, the approximate value of the integral is given by

$$\int_1^4 \frac{1}{x} dx \approx h [\text{sum of mid-ordinates}] = \frac{1}{2} \left[\frac{4}{5} + \frac{4}{7} + \frac{4}{9} + \frac{4}{11} + \frac{4}{13} + \frac{4}{15} \right] = 1.3769 \text{ sq. units}$$

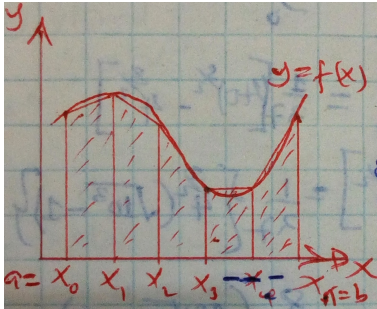
ii) The truncation error bound in using mid-ordinate rule is given by

$$|E| \leq \frac{|b-a|}{24} h^2 \max_{a \leq x \leq b} |f''(x)|$$

Here, $a = 1, b = 4, N = 6, f(x) = \frac{1}{x}$. Thus, $f''(x) = \frac{2}{x^3}$ and $h = (b-a)/N = (4-1)/6 = 0.5$. Therefore,

$$|E| \leq \frac{|4-1|}{24} (0.5)^2 \max_{1 \leq x \leq 4} \left| \frac{2}{x^3} \right| = \frac{3}{24} (0.5)^2 \left| \frac{2}{(1)^3} \right| = 0.0625$$

6.2 Trapezoidal rule



In trapezoidal rule, we subdivide the region R into N strips of equal width $h = \frac{b-a}{N}$ and then draw a line between point $(x_i, f(x_i))$ and $(x_{i+1}, f(x_{i+1}))$, for $i = 0, 1, 2, \dots, N-1$. Since the shape of each strip resembles that of a trapezium, the formula for finding the area of a trapezium is used to estimate the area of each strip. Thus, the total area is the sum of the areas of the trapezia, i.e.,

$$\int_a^b f(x) dx \approx \sum_{i=0}^{N-1} \left\{ \frac{h}{2} [f(x_i) + f(x_{i+1})] \right\}$$

Expanding the summation and simplifying yields the trapezoidal rule as

$$\int_a^b f(x) dx \approx \frac{h}{2} [f_0 + 2(f_1 + f_2 + \dots + f_{N-2} + f_{N-1}) + f_N]$$

→ Note:

□ $f_0, f_1, f_2, \dots, f_N$ are called *ordinates*. Thus, the number of ordinates exceeds the number of strips by 1, i.e., if the number of strips is N , then the number of ordinates will be $(N+1)$.

□ The bound for the truncation error (E) in using the trapezoidal rule is given by

$$|E| \leq \frac{|b-a|}{12} h^2 \max_{a \leq x \leq b} |f''(x)|$$

Example(s):

1. i) Approximate $\int_1^4 \frac{1}{x} dx$ correct to 4 d.p using the trapezoidal rule with 7 ordinates.
- ii) Obtain a bound for the truncation error in this approximation.

Solution

	x_0	x_1	x_2	x_3	x_4	x_5	x_6
x	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3	$\frac{7}{2}$	4
$f(x)$	1	$\frac{2}{3}$	$\frac{1}{2}$	$\frac{2}{5}$	$\frac{1}{3}$	$\frac{2}{7}$	$\frac{1}{4}$
	f_0	f_1	f_2	f_3	f_4	f_5	f_6

$$f(x) = \frac{1}{x}, \quad a = 1, b = 4, \quad N = 6, \quad \text{and}$$

$$h = \frac{b-a}{N} = \frac{4-1}{6} = \frac{1}{2}.$$

Therefore,

$$\begin{aligned} \int_1^4 \frac{1}{x} dx &\approx \frac{h}{2} [f_0 + 2(f_1 + f_2 + f_3 + f_4 + f_5) + f_6] \\ &= \frac{1}{4} \left[1 + 2 \left(\frac{2}{3} + \frac{1}{2} + \frac{2}{5} + \frac{1}{3} + \frac{2}{7} \right) + \frac{1}{4} \right] = 1.4054 \text{ sq. units} \end{aligned}$$

- ii) The truncation error bound in using trapezoidal rule is given by

$$|E| \leq \frac{|b-a|}{12} h^2 \max_{a \leq x \leq b} |f''(x)|$$

Here, $a = 1, b = 4, N = 6, f(x) = \frac{1}{x}$. Thus, $f''(x) = \frac{2}{x^3}$ and $h = (b-a)/N = (4-1)/6 = 0.5$.
Therefore,

$$|E| \leq \frac{|4-1|}{12} (0.5)^2 \max_{1 \leq x \leq 4} \left| \frac{2}{x^3} \right| = \frac{3}{12} (0.5)^2 \left| \frac{2}{(1)^3} \right| = 0.125$$

Exercise:

1. Use trapezoidal rule with 9 ordinates to estimate $\int_2^4 \frac{5x}{2 + \ln 10x} dx$ correct to 6 d.p.
2. Estimate the following integrals correct to 6 d.p using trapezoidal rule with 8 subdivisions.

$$(a) \int_2^6 \frac{x}{1+x^2} dx$$

$$(b) \int_0^1 10e^{-x^2} dx$$

$$(c) \int_2^4 \sqrt{1 + \cos 4x} dx$$

$$(d) \int_{\pi/3}^{\pi/2} \sin(x^2) dx$$

$$(e) \int_3^6 \frac{7x + \ln |3x|}{2 + e^{2x}} dx$$

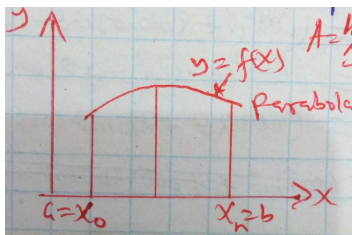
$$(f) \int_2^4 \frac{5x + 100}{2 + e^{2x}} dx$$

3. Evaluate $\int_0^1 \frac{e^{2x} + 1}{4 + x^2} dx$ correct to 4dp using trapezoidal rule with $h = \frac{1}{5}$. [ans: = 0.9510]

6.3 Simpson's rule

This approximation procedure uses parabolic arcs rather than line segments to approximate portions of the graph $y = f(x)$. It uses the fact that if x_0, x_1 and x_2 are three consecutive ordinates, then the definite integral of the quadratic function passing through the points (x_0, y_0) , (x_1, y_1) and (x_2, y_2) is equal to $\frac{h}{3}(y_0 + 4y_1 + y_2)$. We subdivide the interval $[a, b]$ into N even strips. We then approximate the given integral over each pair of strips to get the Simpson's (1/3) rule as:

$$\begin{aligned}\int_a^b f(x)dx &\approx \frac{h}{3} [(f_0 + 4f_1 + f_2) + (f_2 + 4f_3 + f_4) + \cdots + (f_{N-2} + 4f_{N-1} + f_N)] \\ &= \frac{h}{3} [f_0 + 4(f_1 + f_3 + \cdots + f_{N-1}) + 2(f_2 + f_4 + \cdots + f_{N-2}) + f_N]\end{aligned}$$

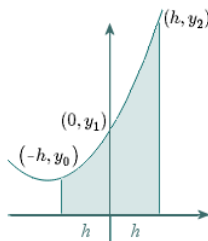


→ Note: The bound for the truncation error (E) in using the Simpsons 1/3 rule is given by

$$|E| \leq \frac{(b-a)}{180} h^4 \max_{a \leq x \leq b} |f^{(4)}(x)|$$

□ Prove Simpson's 1/3 rule (**Assignment**)

Consider approximating the area bounded by the parabola $f(x) = ax^2 + bx + c$ between $x = -h$ and $x = h$ and the x -axis as shown below.



$$A_1 \approx \int_{-h}^h (ax^2 + bx + c)dx = \left[a\frac{x^3}{3} + b\frac{x^2}{2} + cx \right]_{-h}^h = \frac{h}{3} [2ah^2 + 6c]$$

The parabola passes through the points $(-h, y_0)$, $(0, y_1)$ and (h, y_2) . Substituting these x and y values into the equation of the parabola, we get $y_0 = ah^2 - bh + c$, $y_1 = c$, $y_2 = ah^2 + bh + c \Rightarrow y_0 - 2y_1 + y_2 = 2ah^2$. Therefore,

$$A_1 = \int_{-h}^h f(x)dx = \frac{h}{3} [y_0 - 2y_1 + y_2 + 6y_1] = \frac{h}{3} [y_0 + 4y_1 + y_2]$$

The parabola passing through the next set of three points will have an area given by

$$A_2 = \frac{h}{3} [y_2 + 4y_3 + y_4]$$

Thus,

$$A_1 + A_2 = \frac{h}{3} [y_0 + 4(y_1 + y_3) + 2y_2 + y_4]$$

We could keep going by creating more and more segments, and adding the areas as we go along and we would obtain the Simpson's rule above.

Example(s):

1. i) Approximate $\int_1^4 \frac{1}{x} dx$ correct to 4 d.p using the Simpson's 1/3 rule with 7 ordinates.

- ii) Obtain a bound for the truncation error in this approximation.

Solution

i)

	x_0	x_1	x_2	x_3	x_4	x_5	x_6
x	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3	$\frac{7}{2}$	4
$f(x)$	1	$\frac{2}{3}$	$\frac{1}{2}$	$\frac{2}{5}$	$\frac{1}{3}$	$\frac{2}{7}$	$\frac{1}{4}$
	f_0	f_1	f_2	f_3	f_4	f_5	f_6

$$f(x) = \frac{1}{x}, a = 1, b = 4, N = 6, \text{ and}$$

$$h = \frac{b-a}{N} = \frac{4-1}{6} = \frac{1}{2}.$$

Therefore,

$$\begin{aligned} \int_1^4 \frac{1}{x} dx &\approx \frac{h}{2} [f_0 + 4(f_1 + f_3 + f_5) + 2(f_2 + f_4) + f_6] \\ &= \frac{1}{6} \left[1 + 4 \left(\frac{2}{3} + \frac{2}{5} + \frac{2}{7} \right) + 2 \left(\frac{1}{2} + \frac{1}{3} \right) + \frac{1}{4} \right] = 1.3877 \text{ sq. units} \end{aligned}$$

- ii) The truncation error bound in using Simpson's 1/3 rule is given by

$$|E| \leq \frac{|b-a|}{180} h^4 \max_{a \leq x \leq b} |f^{(4)}(x)|$$

Here, $a = 1, b = 4, N = 6, f(x) = \frac{1}{x}$. Thus, $f''(x) = \frac{24}{x^5}$ and $h = (b-a)/N = (4-1)/6 = 0.5$. Therefore,

$$|E| \leq \frac{|4-1|}{180} (0.5)^4 \max_{1 \leq x \leq 4} \left| \frac{24}{x^5} \right| = \frac{3}{180} (0.5)^4 \left| \frac{24}{(1)^5} \right| = 0.025$$

Exercise:

1. Use Simpson's rule with $h = 0.125$ to approximate $\int_0^1 \frac{1}{1+x} dx$ correct to 4dp. [9 Marks]
2. Use Simpson's rule with $N = 8$ to approximate the following integrals correct to 6 d.p.

(a) $\int_1^5 \frac{7 \cos 2x}{x} dx$

(b) $\int_2^3 \frac{e^{2x}}{5 + e^x} dx$

(c) $\int_2^4 x \ln(2x) dx$

(c) $\int_0^{\pi/4} \sqrt{5 - \sin^2 \theta} d\theta$

3. Find the approximate value of $\int_0^{\pi} \sqrt{\sin \theta} d\theta$ using Simpson's rule with six subintervals.

LECTURE 11

7 Applications of Definite Integrals

7.1 Area between a curve and the x -axis

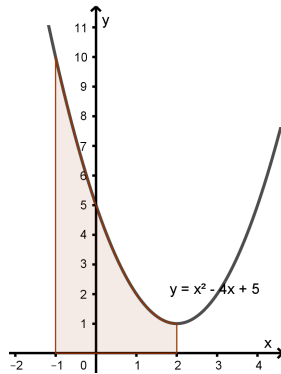
The area of the region bounded by the graph of a non-negative continuous function $y = f(x)$, the x -axis and the lines $x = a$ and $x = b$ is given by

$$A = \int_a^b f(x) dx$$

Example(s):

- Find the area under the curve $y = x^2 - 4x + 5$ between $x = -1$ and $x = 2$.

Solution



$$\begin{aligned} A &= \int_{-1}^2 (x^2 - 4x + 5) dx \\ &= \left[\frac{x^3}{3} - 2x^2 + 5x \right]_{-1}^2 = 12 \text{ square units} \end{aligned}$$

Exercise:

- Find the area between the curves $y = 1 - x^2$, $x = 2$ and $x = 5$. [ans: $A = -36$]

7.2 Area between two curves

If f and g are continuous functions with $f(x) \geq g(x)$ for all x in the interval $[a, b]$, then the area of the region between the curves $y = f(x)$ and $y = g(x)$ from $x = a$ to $x = b$ is given by

$$A = \int_a^b [f(x) - g(x)] dx$$

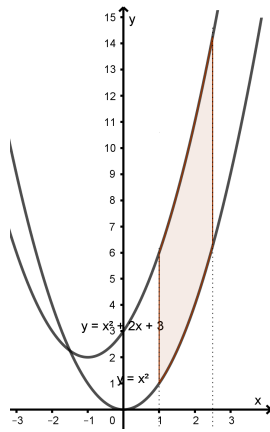
→ Note:

- ☐ If a region is determined by curves that intersect, then the x -coordinates of the intersection points give the limits of integration.
- ☐ If the formula for a bounding curve changes at one or more points, we partition the region into subregions that correspond to the formula changes and apply the area formula to each subregion.

Example(s):

- Find the area between the curves $y = x^2$ and $y = x^2 + 2x + 3$ from $x = 1$ to $x = \frac{5}{2}$.

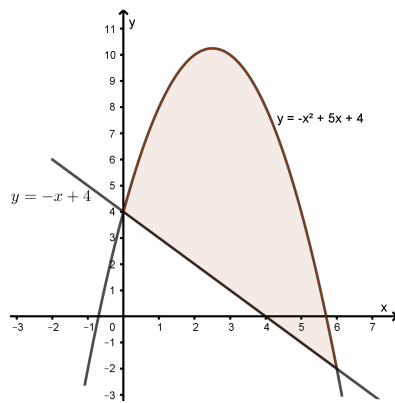
Solution



$$\begin{aligned}
 A &= \int_1^{5/2} [(x^2 + 2x + 3) - x^2] dx \\
 &= \int_1^{5/2} [2x + 3] dx = [x^2 + 3x]_1^{5/2} = 9\frac{3}{4} \text{ square units}
 \end{aligned}$$

2. Find the area of the region enclosed by the parabola $y = 4 + 5x - x^2$ and the line $y = 4 - x$.

Solution

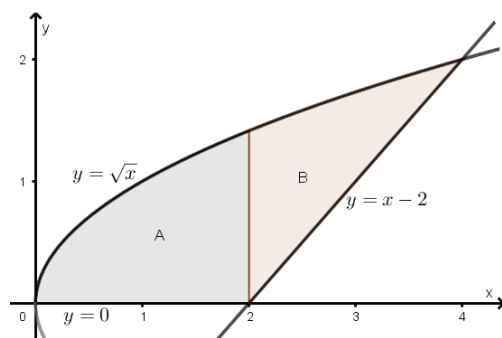


The curves intersect when $4 + 5x - x^2 = 4 - x$.
The roots are $x = 0$ or $x = 6$, which are the limits of integration.

$$\begin{aligned}
 A &= \int_0^6 [(4 + 5x - x^2) - (4 - x)] dx \\
 &= \int_0^6 [6x - x^2] dx = \left[3x^2 - \frac{x^3}{3} \right]_0^6 = 36 \text{ square units}
 \end{aligned}$$

3. Find the area of the region in the first quadrant that is bounded above by $y = \sqrt{x}$ and below by the x -axis and the line $y = x - 2$.

Solution



The curves intersect when $\sqrt{x} = x - 2$. The roots are $x = 1$ or $x = 4$. The sketch shows that the region's upper boundary is the graph of $y = \sqrt{x}$. The lower boundary changes from $y = 0$ (between $x = 0$ and $x = 2$) to $y = x - 2$ (between $x = 2$ and $x = 4$). We partition the region at $x = 2$ into subregions A and B and find the area of each subregion.

$$\begin{aligned}
 \text{Total area} &= \int_0^2 \sqrt{x} dx + \int_2^4 [\sqrt{x} - (x - 2)] dx = \left[\frac{2}{3} x^{3/2} \right]_0^2 + \left[\frac{2}{3} x^{3/2} - \frac{x^2}{2} + 2x \right]_2^4 \\
 &= \left[\frac{2}{3} \sqrt{8} \right] + \left[\left(\frac{16}{3} - 8 + 8 \right) - \left(\frac{2}{3} \sqrt{8} - 2 + 4 \right) \right] = \frac{16}{3} - 2 \\
 &= \frac{10}{3} \text{ square units}
 \end{aligned}$$

$$\text{Alternatively, total area} = \int_0^4 \sqrt{x} dx - \int_2^4 (x - 2) dx = \frac{10}{3} \text{ square units.}$$

Exercise:

1. Find the areas of the regions enclosed by the following curves

(a) $y = \sec^2 x$ and $y = \sin x$ between $x = 0$ and $x = \pi/4$. [ans: $A = \frac{\sqrt{2}}{2}$]

(b) $y = 2 \sin x$ and $y = \sin 2x$ between $x = 0$ and $x = \pi$.

(c) $y = \sec^2 x$ and $y = 8 \cos x$ between $x = -\pi/3$ and $x = \pi/3$.

2. Find the areas of the regions enclosed by the following curves

(a) $y = 2 - x^2$ and $y = -x$. [ans: $A = \frac{9}{2}$]

(b) $y = 2x - x^2$ and $y = -3$

(c) $y = 7 - 2x^2$ and $y = x^2 + 4$

(d) $y = x^4$ and $y = 8x$

(e) $y = x^4 - 4x^2 + 4$ and $y = x^2$

(f) $y = |x^2 - 4|$ and $y = \frac{1}{2}x^2 + 4$

(g) $x = y^3 - y^2$ and $x = 2y$

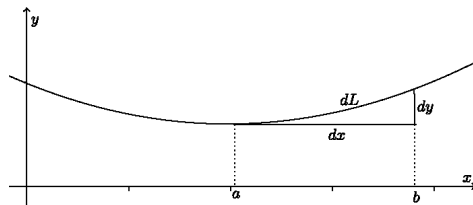
(h) $x + y^2 = 0$ and $x + 3y^2 = 2$

(i) $y = x^2 + 5x$ and $y = 3 - x^2$ [ans: $= 343/24$]

3. Find the area of the region in the first quadrant bounded by the line $y = x$, the line $x = 2$, the curve $y = \frac{1}{x^2}$, and the x -axis.

7.3 Arc length

Consider a curve defined by $y = f(x)$ for $x \in [a, b]$.



Let dx and dy denote the small change in x and y , respectively. By Pythagoras theorem, the small change in its length L is given by

$$(dL)^2 = (dx)^2 + (dy)^2 \quad \Rightarrow \quad dL = \sqrt{(dx)^2 + (dy)^2}$$

Therefore, $dL = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$. Integrating from $x = a$ to $x = b$ yields

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx,$$

which is the length L of the arc joining the points $(a, f(a))$ and $(b, f(b))$ along the curve $y = f(x)$.

→ Note: If the curve is defined in terms of the parameter t , i.e., if $x = x(t)$ and $y = y(t)$, for $t \in [t_1, t_2]$, then the arc length L is given by

$$L = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Example(s):

1. Find the length of the curve $y = \frac{x^3}{6} + \frac{1}{2x}$ from $x = 1$ to $x = 2$.

Solution

Differentiating the given curve with respect to x yields

$$\frac{dy}{dx} = \frac{x^2}{2} - \frac{1}{2x^2}$$

Squaring both sides, we obtain

$$\left(\frac{dy}{dx}\right)^2 = \left(\frac{x^2}{2} - \frac{1}{2x^2}\right)^2 = \frac{x^4}{4} - \frac{1}{2} + \frac{1}{4x^4} = \frac{1}{4}\left(x^4 - 2 + \frac{1}{x^4}\right)$$

Thus,

$$1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{1}{4}\left(x^4 - 2 + \frac{1}{x^4}\right) = \frac{1}{4}\left(x^4 + 2 + \frac{1}{x^4}\right) = \frac{1}{4}\left(x^2 + \frac{1}{x^2}\right)^2$$

Hence,

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \frac{1}{2}\left(x^2 + \frac{1}{x^2}\right)$$

Therefore, the arc length is

$$L = \int_1^2 \sqrt{1 + \left[\frac{dy}{dx}\right]^2} dx = \frac{1}{2} \int_1^2 \left(x^2 + \frac{1}{x^2}\right) dx = \frac{1}{2} \left[\frac{x^3}{3} - \frac{1}{x}\right]_1^2 = \frac{17}{12} \text{ units}$$

2. Find the length of the curve $x = 2 \cos^3 \theta$, $y = 2 \sin^3 \theta$ between the points corresponding to $\theta = 0$ and $\theta = \frac{\pi}{2}$.

Solution

Differentiating the given curve with respect to θ , yields

$$\frac{dx}{d\theta} = -6 \cos^2 \theta \sin \theta \quad \text{and} \quad \frac{dy}{d\theta} = 6 \sin^2 \theta \cos \theta$$

Squaring both sides and adding, we obtain

$$\begin{aligned} \left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 &= (-6 \cos^2 \theta \sin \theta)^2 + (6 \sin^2 \theta \cos \theta)^2 = 36 \cos^4 \theta \sin^2 \theta + 36 \cos^2 \theta \sin^4 \theta \\ &= 36 \cos^2 \theta \sin^2 \theta \end{aligned}$$

Hence,

$$\sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} = 6 \cos \theta \sin \theta = 3 \sin(2\theta)$$

Therefore, the arc length is

$$\begin{aligned} L &= \int_0^{\pi/2} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta = 3 \int_0^{\pi/2} \sin 2\theta d\theta = 3 \left[-\frac{\cos 2\theta}{2}\right]_0^{\pi/2} \\ &= -3 \left[\frac{\cos 2\theta}{2}\right]_0^{\pi/2} = -3 \left[\frac{-1}{2} - \frac{1}{2}\right] = 3 \text{ units} \end{aligned}$$

3. Find the length of the curve $y^2 = x \left(1 - \frac{x}{3}\right)^2$, measured from the origin to the ordinate where $x = 4$.

Solution

Differentiating the given curve with respect to x yields

$$2y \frac{dy}{dx} = \left(1 - \frac{x}{3}\right)^2 - \frac{2x}{3} \left(1 - \frac{x}{3}\right) = \left(1 - \frac{x}{3}\right)(1 - x)$$

Squaring both sides, we obtain

$$\begin{aligned} 4y^2 \left(\frac{dy}{dx}\right)^2 &= \left(1 - \frac{x}{3}\right)^2 (1 - x)^2 \\ \Rightarrow \left(\frac{dy}{dx}\right)^2 &= \frac{\left(1 - \frac{x}{3}\right)^2 (1 - x)^2}{4y^2} = \frac{\left(1 - \frac{x}{3}\right)^2 (1 - x)^2}{4x \left(1 - \frac{x}{3}\right)^2} = \frac{(1 - x)^2}{4x} \end{aligned}$$

Thus,

$$1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{(1 - x)^2}{4x} = \frac{4x + (1 - x)^2}{4x} = \frac{4x + 1 - 2x + x^2}{4x} = \frac{x^2 + 2x + 1}{4x} = \frac{(x + 1)^2}{4x}$$

Hence,

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \frac{(x + 1)}{2\sqrt{x}} = \frac{1}{2} (x^{1/2} + x^{-1/2})$$

Therefore, the arc length is

$$L = \int_0^4 \sqrt{1 + \left[\frac{dy}{dx}\right]^2} dx = \frac{1}{2} \int_0^4 (x^{1/2} + x^{-1/2}) dx = \frac{1}{2} \left[\frac{2}{3} x^{3/2} + 2x^{1/2} \right]_0^4 = \frac{14}{3} \text{ units}$$

4. Show that the length of the curve $y = x^{3/2}$ from (1,1) to (4,8) is $\frac{1}{27}[80\sqrt{10} - 13\sqrt{13}]$.

Solution

Differentiating the given curve with respect to x yields

$$\frac{dy}{dx} = \frac{3}{2} x^{1/2}$$

Squaring both sides, we obtain

$$\left(\frac{dy}{dx}\right)^2 = \frac{9x}{4}$$

Thus,

$$1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{9x}{4} = \frac{4 + 9x}{4}$$

Hence,

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \frac{1}{2} \sqrt{4 + 9x}$$

Therefore, the arc length is

$$\begin{aligned} L &= \int_1^4 \sqrt{1 + \left[\frac{dy}{dx}\right]^2} dx = \frac{1}{2} \int_1^4 \sqrt{4 + 9x} dx, \text{ (put } u = 4 + 9x \Rightarrow du = 9dx) \\ &= \frac{1}{2} \int_{13}^{40} u^{1/2} \cdot \frac{du}{9} = \frac{1}{18} \left[\frac{2}{3} u^{3/2} \right]_{13}^{40} = \frac{1}{27} [80\sqrt{10} - 13\sqrt{13}] \text{ units} \end{aligned}$$

Exercise:

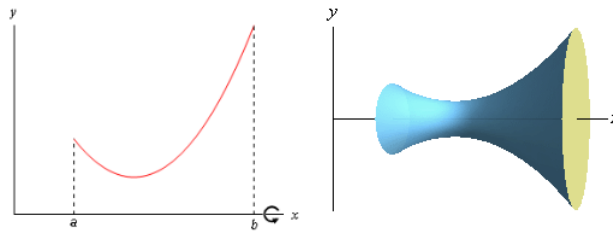
1. Show that the length of the segment of the curve $y = x^{3/2}$ from $(0,0)$ to $(4,8)$ is $\frac{8}{27}[10\sqrt{10} - 1]$.
2. Find the length of the curve $x = 5(2t - \sin 2t)$, $y = 10 \sin^2 t$ between $t = 0$ to $t = \pi$. [ans: $L = 40$ units]
3. Find the length of the curve $(y + 1)^2 = 4x^3$ between $x = 1$ and $x = 2$. [ans: $L = \frac{2}{27}(19\sqrt{19} - 10\sqrt{10})$ units]

7.4 Areas of surfaces of revolution

A surface of revolution is formed when a curve is rotated about a line.

□ *Case 1:* Revolution about the x -axis

Consider revolving the curve $y = f(x)$, $a \leq x \leq b$, about the x -axis, where $f(x)$ is positive and has a continuous derivatives on the interval $[a, b]$. E.g.



The area of the surface generated is given by $dS = 2\pi(\text{radius})(\text{arc length})$, i.e.,

$$dS = 2\pi y dL = 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Integrating both sides, we get

$$S = 2\pi \int_a^b y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

□ *Case 2:* Revolution about the y -axis

Consider revolving the curve $x = f(y)$, $c \leq y \leq d$, about the y -axis, where $f(y)$ is positive and has a continuous derivatives on the interval $[c, d]$. The area of the surface generated is given by

$$dS = 2\pi x dL = 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

Integrating both sides, we get

$$S = 2\pi \int_c^d x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

□ *Case 3:* Revolution about the line $y = a$

Consider revolving the curve $y = f(x)$, $x_0 \leq x \leq x_1$, about the line $y = a$, where $f(x)$ is positive and has a continuous derivatives on the interval $[x_0, x_1]$. The area of the surface generated is given by

$$dS = 2\pi(y - a)dL = 2\pi(y - a) \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Integrating both sides, we get

$$S = 2\pi \int_{x_0}^{x_1} (y - a) \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

→ Note: $a = 0$ corresponds to case 1.

□ *Case 4:* Revolution about the $x = b$

Consider revolving the curve $x = f(y)$, $y_0 \leq y \leq y_1$, about the $x = b$, where $f(y)$ is positive and has a continuous derivatives on the interval $[y_0, y_1]$. The area of the surface generated is given by

$$dS = 2\pi x dL = 2\pi(x - b) \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

Integrating both sides, we get

$$S = 2\pi \int_{y_0}^{y_1} (x - b) \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

→ Note: $b = 0$ corresponds to case 2.

Example(s):

1. Find the area of the surface obtained by rotating the arc $y = \sqrt{4 - x^2}$ about the x-axis given that $-1 \leq x \leq 1$.

Solution

We have

$$\frac{dy}{dx} = \frac{1}{2}(4 - x^2)^{-1/2}(-2x) = \frac{-x}{\sqrt{4 - x^2}}$$

So, the surface area is

$$\begin{aligned} S &= 2\pi \int_{-1}^1 y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = 2\pi \int_{-1}^1 \sqrt{4 - x^2} \sqrt{1 + \frac{x^2}{4 - x^2}} dx \\ &= 2\pi \int_{-1}^1 \sqrt{4 - x^2} \frac{2}{\sqrt{4 - x^2}} dx = 4\pi \int_{-1}^1 dx = 4\pi(2) = 8\pi \end{aligned}$$

2. The arc of the parabola $y = x^2$ from (1,1) to (2,4) is rotated about the y-axis. Find the area of the resulting surface. [ans: $A = \frac{\pi}{6}(17\sqrt{17} - 5\sqrt{5})$]

Solution

We have

$$x = \sqrt{y} \text{ and } \frac{dx}{dy} = \frac{1}{2\sqrt{y}}$$

So, the surface area is

$$\begin{aligned} S &= 2\pi \int_1^4 x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy \\ &= 2\pi \int_1^4 \sqrt{y} \sqrt{1 + \frac{1}{4y}} dy = \pi \int_1^4 \sqrt{4y + 1} dy, \quad \text{put } u = 4y + 1 \Rightarrow du = 4dy \\ &= \frac{\pi}{4} \int_5^{17} \sqrt{u} du = \frac{\pi}{6}(17\sqrt{17} - 5\sqrt{5}) \end{aligned}$$

3. Show that the area of the surface generated by rotating the curve $y = e^x$, $0 \leq x \leq 1$, about the x -axis is given by

$$S = \pi \left[e\sqrt{1+e^2} + \ln|e + \sqrt{1+e^2}| - \sqrt{2} - \ln|\sqrt{2}+1| \right]$$

Solution

We have

$$y = e^x \text{ and } \frac{dy}{dx} = e^x$$

So, the surface area is

$$\begin{aligned} S &= 2\pi \int_0^1 y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = 2\pi \int_0^1 e^x \sqrt{1 + e^{2x}} dx, \text{ put } u = e^x \Rightarrow du = e^x dx \\ &= 2\pi \int_1^e \sqrt{1 + u^2} du, \text{ put } u = \tan \theta \Rightarrow du = \sec^2 \theta d\theta \\ &= 2\pi \int_{\pi/4}^{\tan^{-1} e} \sec^3 \theta d\theta = 2\pi \int_{\pi/4}^{\tan^{-1} e} \sec \theta (1 + \tan^2 \theta) d\theta = 2\pi \int_{\pi/4}^{\tan^{-1} e} [\sec \theta \tan^2 \theta + \sec \theta] d\theta \\ &= 2\pi \cdot \frac{1}{2} \left[\sec \theta \tan \theta + \ln |\sec \theta + \tan \theta| \right]_{\pi/4}^{\tan^{-1} e} \\ &= \pi \left[\sec \tan^{-1}(e) \tan \tan^{-1}(e) + \ln |\sec \tan^{-1}(e) + \tan \tan^{-1}(e)| \right. \\ &\quad \left. - \sec(\pi/4) \tan(\pi/4) - \ln |\sec(\pi/4) + \tan(\pi/4)| \right] \\ &= \pi \left[e \sec \tan^{-1}(e) + \ln |\sec \tan^{-1}(e) + e| - \sqrt{2} - \ln |\sqrt{2} + 1| \right]. \text{ Recall that } \sec^2 \alpha = 1 + \tan^2 \alpha \\ &= \pi \left[e\sqrt{1 + \tan^2(\tan^{-1} e)} + \ln |\sqrt{1 + \tan^2(\tan^{-1} e)} + e| - \sqrt{2} - \ln |\sqrt{2} + 1| \right] \\ &= \pi \left[e\sqrt{1 + e^2} + \ln |e + \sqrt{1 + e^2}| - \sqrt{2} - \ln |\sqrt{2} + 1| \right] \end{aligned}$$

Exercise:

- Find the area of the surface generated when the curve $y^2 = 4x$ between $x = 0$ and $x = 3$ is rotated about the x -axis.
[ans: $S = 56\pi/3$ square units]
- Find the area of the surface generated when the curve $y^2 = 4x$ between $x = 0$ and $x = 3$ is rotated about the line $y = 1$.
[ans: $S =$ square units]

LECTURE 12

7.5 Volumes of solids of revolution

Solids of revolution are solids whose shapes can be generated by revolving plane regions about a straight line called the *axis of revolution*.

7.5.1 Disk method

In disk method, the axis of revolution is the boundary of the plane region and the cross sections are taken perpendicular to the axis of revolution.

□ *Case 1:* Rotation about the x -axis

The volume (V) of the solid generated by revolving the region bounded by the curve $y = R(x)$ and the x -axis, for $a \leq x \leq b$, about the x -axis is given by

$$dV = \pi[R(x)]^2 dx \quad \Rightarrow \quad V = \pi \int_a^b [R(x)]^2 dx$$

→ Note: the function $R(x)$ is the radius of the disk.

□ *Case 2:* Rotation about the y -axis

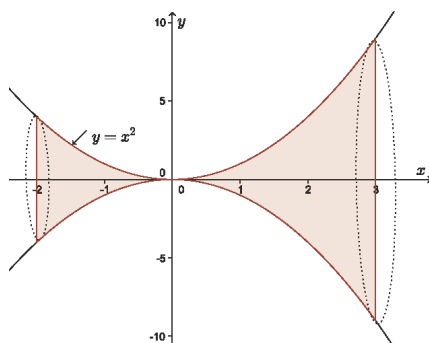
If the region bounded by the curve $x = R(y)$ and the y -axis, for $c \leq y \leq d$, is revolved about the y -axis, then its volume (V) is given by

$$V = \pi \int_c^d [R(y)]^2 dy$$

Example(s):

- (a) Find the volume of the solid generated by revolving the region bounded by the line $y = x^2$ and the x -axis between $x = -2$ and $x = 3$ about the x -axis.

Solution

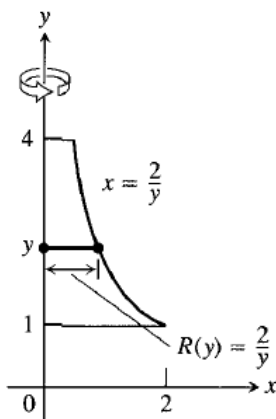


Since the x -axis is a boundary of the region, we use the disk method. The disk radius is $R(x) = x^2$ (i.e., perpendicular distance from the axis of revolution) and the disk thickness is dx . The volume of the solid generated is

$$\begin{aligned} V &= \pi \int_{-2}^3 [R(x)]^2 dx = \pi \int_{-2}^3 x^4 dx \\ &= \pi \left[\frac{x^5}{5} \right]_{-2}^3 = \pi \left[\frac{243}{5} - \left(\frac{-32}{5} \right) \right] = 55\pi \end{aligned}$$

- (b) Find the volume of the solid generated by revolving the region between the y -axis and the curve $x = \frac{2}{y}$, $1 \leq y \leq 4$, about the y -axis.

Solution



Since the y -axis is a boundary of the region, we use the disk method. The disk radius is $R(y) = 2/y$ and the disk thickness is dy . The volume of the solid generated is

$$\begin{aligned} V &= \pi \int_1^4 [R(y)]^2 dy = \pi \int_1^4 \left(\frac{2}{y} \right)^2 dy = \pi \int_1^4 \frac{4}{y^2} dy \\ &= 4\pi \left[-\frac{1}{y} \right]_1^4 = 4\pi \left[-\frac{1}{4} + 1 \right] = 3\pi \end{aligned}$$

Exercise:

1. Find the volume of the solid generated by revolving the region bounded by the curve $x = 2/(y+1)$ and the lines $x = 0, y = 0, y = 3$ about

i) The x -axis [ans: $V = \pi \int_0^{0.5} (3)^2 dx + \pi \int_{0.5}^2 \left(\frac{2}{x} - 1\right)^2 dx = [27/2 - 4 \ln(4)]\pi$]

ii) The y -axis [ans: $V = \pi \int_0^3 \left(\frac{2}{y+1}\right)^2 dy = 3\pi$]

2. (a) The region between the curve $y = \sqrt{x}$, $0 \leq x \leq 4$, and the x -axis is revolved about the x -axis to generate a solid. Find its volume. [ans: $V = 8\pi$]

(b) Find the volume of the solid generated by revolving the region bounded by the parabola $y = \sqrt{x}$ and the lines $y = 1, x = 4$ about the line $y = 1$. [ans: $V = \pi \int_1^4 (\sqrt{x} - 1)^2 dx = \frac{7\pi}{6}$]

(c) Find the volume of the solid generated if the region bounded by the curves $y = 2x - x^2$ and $y = 0$ is revolved about the x -axis. [ans: $= 16\pi/15$]

3. Find the volume of the solid generated by revolving the region bounded by the parabola $x = y^2 + 1$ and the line $x = 5$ about the line $x = 5$. [ans: $V = \pi \int_{-2}^2 [3 - (y^2 + 1)]^2 dy = \frac{112\pi}{15}$]

4. The region in the second quadrant bounded above by the curve $y = -x^3$, below by the x -axis, and on the left by the line $x = -2$ is revolved about the line $x = -2$ to generate a solid. Find the volume of the solid.

5. Find the volume of the sphere $x^2 + y^2 + z^2 = r^2$ for $r > 0$. [hint: revolve $y = \sqrt{r^2 - x^2}$ about the x -axis between $-r \leq x \leq r$, ans: $V = \frac{4}{3}\pi r^3$]

7.5.2 Washer method

In washer method, the axis of revolution is not a boundary of the plane region (or the axis doesn't cross the plane region) and the cross sections are taken perpendicular to the axis of revolution. Think of washer as a disk with a hole in it, i.e., the center is empty.

□ *Case 1:* Rotation about the x -axis

The volume (V) of the solid generated by revolving the region bounded by $y = R(x)$ and $y = r(x)$, $a \leq x \leq b$, where $R(x) \geq r(x)$, about the x -axis is given by

$$V = \pi \int_a^b ([R(x)]^2 - [r(x)]^2) dx$$

→ Note: if $r(x) = 0$, we get the disk method. Thus, the disk method is a special case of the washer method.

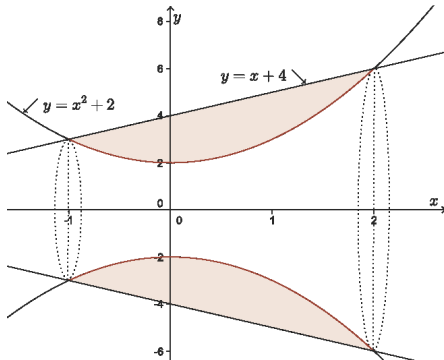
□ *Case 2:* Rotation about the y -axis

If the region bounded by $x = R(y)$ and $x = r(y)$, $c \leq y \leq d$, where $R(y) \geq r(y)$ is revolved about the y -axis, then its volume (V) is given by

$$V = \pi \int_c^d ([R(y)]^2 - [r(y)]^2) dy$$

Example(s):

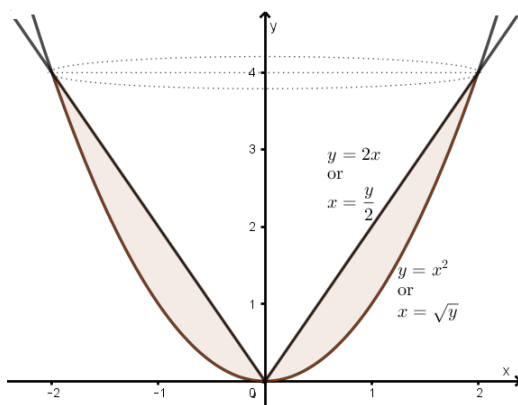
1. Find the volume of the solid generated by revolving the region bounded by $y = x^2 + 2$ and $y = x + 4$ about the x-axis.

Solution

The outer radius is $R(x) = x + 4$ and the inner radius is $r(x) = x^2 + 2$. The curve $y = x^2 + 2$ and the line $y = x + 4$ intersect at $x = -1$ and $x = 2$. Since the x-axis is not a boundary of the region, we use the washer method. The volume of the solid generated is given by

$$\begin{aligned}
 V &= \pi \int_a^b ([R(x)]^2 - [r(x)]^2) dx = \pi \int_{-1}^2 ([x + 4]^2 - [x^2 + 2]^2) dx \\
 &= \pi \int_{-1}^2 (x^2 + 8x + 16 - x^4 - 4x^2 - 4) dx = \pi \int_{-1}^2 (-x^4 - 3x^2 + 8x + 12) dx \\
 &= \pi \left[-\frac{x^5}{5} - x^3 + 4x^2 + 12x \right]_{-1}^2 = \pi \left[\frac{128}{5} - \left(\frac{-34}{5} \right) \right] = \frac{162\pi}{5}
 \end{aligned}$$

2. The region bounded by the parabola $y = x^2$ and the line $y = 2x$ in the first quadrant is revolved about the y-axis to generate a solid. Find the volume of the solid.

Solution

The outer radius is $R(y) = \sqrt{y}$ and the inner radius is $r(y) = \frac{y}{2}$. The parabola $y = x^2$ and the line $y = 2x$ intersect at points $(0,0)$ and $(2,4)$. Since the y-axis is not a boundary of the region, we use the washer method. The volume of the solid generated is given by

$$\begin{aligned}
 V &= \pi \int_c^d ([R(y)]^2 - [r(y)]^2) dy = \pi \int_0^4 [(\sqrt{y})^2 - \left(\frac{y}{2}\right)^2] dy \\
 &= \pi \int_0^4 \left[y - \frac{y^2}{4} \right] dy = \pi \left[\frac{y^2}{2} - \frac{y^3}{12} \right]_0^4 = \pi \left[(8 - 0) - \left(\frac{16}{3} - 0 \right) \right] \\
 &= \frac{8\pi}{3}
 \end{aligned}$$

Exercise:

1. The region bounded above by the curve $y = \sqrt{x}$ and below by the line $y = x$ is revolved about the line x -axis to generate a solid. Find the volume of the solid. [ans: $V =$]
2. The region in the first quadrant enclosed by the parabola $y = x^2$, the y -axis, and the line $y = 1$ is revolved about the line $x = 3/2$ to generate a solid. Find the volume of the solid. [ans: $V = \pi \int_0^1 \left[\left(\frac{3}{2}\right)^2 - \left(\frac{3}{2} - \sqrt{y}\right)^2 \right] dy = \frac{3\pi}{2}$]
3. The region in the second quadrant bounded above by the curve $y = -x^3$, below by the x -axis, and on the left by the line $x = -1$ is revolved about the line $x = -2$ to generate a solid. Find the volume of the solid.
4. The region in the first quadrant bounded on the left by the circle $x^2 + y^2 = 3$, on the right by the line $x = \sqrt{3}$, and above by the line $y = \sqrt{3}$ is revolved about the line y -axis to generate a solid. Find the volume of the solid. [ans: $V = \frac{3\pi}{2}$]
5. Find the volume of the solid generated by revolving the region enclosed by the triangle with vertices $(1,0)$, $(2,1)$ and $(1,1)$ about
 - i) The x -axis [ans:]
 - ii) The y -axis [ans:]

7.5.3 Cylindrical shell method

In cylindrical shell method, the cross sections of the solid generated are taken parallel to the axis of revolution. Cylindrical shells sometimes work better than washers. To implement the shell method, follow these steps

- i) Sketch the region and draw a line segment across it parallel to the axis of revolution. Label the segment's length (shell height), perpendicular distance from the axis of revolution (shell radius), and width (shell thickness).
- ii) Find the limits of integration, i.e., the extents of all the shells across the region.
- iii) Integrate the product $2\pi(\text{shell radius})(\text{shell height})$ with respect to the appropriate variable (x or y) to find the volume of the solid of revolution.

□ *Case 1:* Revolution about the y -axis

The volume (V) of the solid generated by revolving the region bounded by the curve $y = f(x)$ and the x -axis, $a \leq x \leq b$, where $f(x) \geq 0$ and $a \geq 0$, about the y -axis is given by

$$V = 2\pi \int_a^b x f(x) dx$$

→ Note: the volume of each shell is $dV = 2\pi(\text{shell radius})(\text{shell height})(\text{shell thickness})$, where the shell radius is the perpendicular distance from the axis of revolution to the shell.

□ *Case 2:* Revolution about the x -axis

If the region bounded by $x = f(y)$ and the y -axis on the interval $[c, d]$, where $f(y) \geq 0$ and $c \geq 0$, is revolved about the x -axis, then its volume (V) is given by

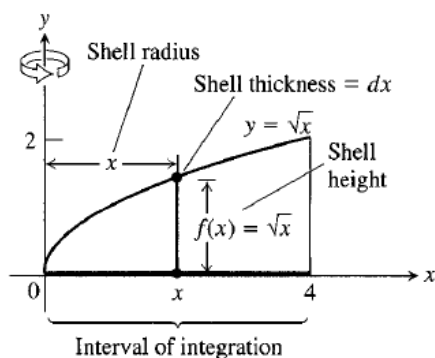
$$V = 2\pi \int_c^d y f(y) dy$$

Example(s):

- Find the volume of the solid generated by revolving the region bounded by the curve $y = \sqrt{x}$, the x -axis, and the line $x = 4$ about
 - the y -axis.
 - the x -axis.

Solution

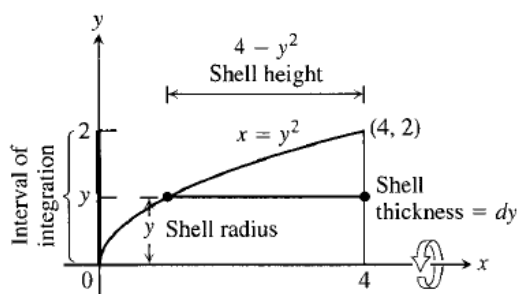
(a)



The radius of the shell is x , the height of the shell is $\sqrt{x}$ and the shell thickness is dx . The volume of the solid generated is

$$\begin{aligned}
 V &= 2\pi \int_0^4 x\sqrt{x}dx = 2\pi \int_0^4 x^{3/2}dx \\
 &= 2\pi \left[\frac{2}{5}x^{5/2} \right]_0^4 = \frac{128\pi}{5}
 \end{aligned}$$

(b)



The radius of the shell is y , the height of the shell is $4 - y^2$ and the shell thickness is dy . The volume of the solid generated is

$$\begin{aligned}
 V &= 2\pi \int_0^2 y(4 - y^2)dy \\
 &= 2\pi \left[2y^2 - \frac{1}{4}y^4 \right]_0^2 = 8\pi
 \end{aligned}$$

Exercise:

- Find the volume of the solid generated by revolving the region bounded by $y = x^2$ and the x -axis on the interval $[1, 3]$ about the y -axis. [ans: $V = 40\pi$]
- The region in the first quadrant bounded by the parabola $y = x^2$, the y -axis, and the line $y = 1$ is revolved about the line $x = 2$ to generate a solid. Find the volume of the solid. [ans: $V = \pi \int_0^1 (2 - x)(1 - x^2)dx = \frac{13\pi}{6}$]
- Find the volume of the solid generated by revolving the region bounded by the curve $y = 1/x$ and the lines $y = 0$, $x = 1/2$, $x = 2$ about
 - The y -axis. [ans: $V = 3\pi$]
 - The x -axis. [ans: $V = \frac{3\pi}{2}$]
 - The line $x = -1$.
 - The line $y = -1$.

LECTURE 13

8 Complex Numbers

A complex number (z) is a number that can be expressed in the form $z = x + iy$, where x is the real part and y is the imaginary part of the complex number z , i.e., $x = \operatorname{Re}(z)$ and $y = \operatorname{Im}(z)$. The “iota” i is defined as $i = \sqrt{-1}$. The set of real numbers is a subset of the set of complex numbers, i.e., $\mathbb{R} \subset \mathbb{C}$ since a real number is a complex number whose imaginary part is zero. Now,

$$i = \sqrt{-1}, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1$$

Example(s):

1. Work out i^{79} .

Solution

$$i^{79} = i^{76} \times i^3 = (i^4)^{19} \times i^3 = 1^{19} \times -i = -i$$

2. Solve $x^2 - x + 1 = 0$.

Solution

$$x = \frac{1 \pm \sqrt{1-4}}{2} = \frac{1 \pm \sqrt{-3}}{2} = \frac{1 \pm \sqrt{3} \times \sqrt{-1}}{2} = \frac{1}{2} \pm i \frac{\sqrt{3}}{2}$$

8.1 Equality of complex numbers

Two complex numbers are said to be equal if and only if their corresponding parts are equal, i.e., if $z_1 = x + iy$ and $z_2 = a + ib$, then $z_1 = z_2$ if and only if $x = a$ and $y = b$.

8.2 Conjugate of a complex number

If $z = x + iy$, then $\bar{z} = x - iy$ is the conjugate of z .

Exercise:

1. Prove the following

$$(a) \quad \overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$$

$$(b) \quad \overline{z_1 \cdot z_2} = \bar{z}_1 \cdot \bar{z}_2$$

2. Evaluate $\frac{i^{20} + i^5 - i^{16}}{i^{15} - 2i + i^{18}}$

$$[\text{ans: } = \frac{3+i}{10}, 4 \text{ Marks}]$$

3. Evaluate the following

$$(a) \quad (2i - 1)^2 \left[\frac{4}{1-i} + \frac{2-i}{1+i} \right] \quad [5 \text{ Marks}]$$

$$(b) \quad \frac{1}{2} \left[\frac{z}{\bar{z}} + \frac{\bar{z}}{z} \right] \text{ given that } z = \sqrt{3} - i2 \quad [4 \text{ Marks}]$$

8.3 Addition and subtraction

If $z_1 = a + ib$ and $z_2 = c + id$, then

$$z_1 + z_2 = a + ib + c + id = (a + c) + i(b + d)$$

and

$$z_1 - z_2 = a + ib - c - id = (a - c) + i(b - d)$$

8.4 Multiplication and division

If $z_1 = a + ib$ and $z_2 = c + id$, then

$$z_1 \cdot z_2 = (a + ib)(c + id) = ac + iad + ibc - bd = (ac - bd) + i(ad + bc)$$

and

$$\frac{z_1}{z_2} = \frac{a + ib}{c + id} = \frac{(a + ib)(c - id)}{(c + id)(c - id)} = \frac{ac + bd + i(bc - ad)}{c^2 + d^2}$$

Example(s):

1. Express $\frac{2 + i}{3 + 2i}$ in the form $a + bi$.

Solution

$$\frac{2 + i}{3 + 2i} = \frac{(2 + i)(3 - 2i)}{(3 + 2i)(3 - 2i)} = \frac{6 - 4i + 3i + 2}{9 + 4} = \frac{8 - i}{13} = \frac{8}{13} - \frac{1}{13}i$$

Exercise:

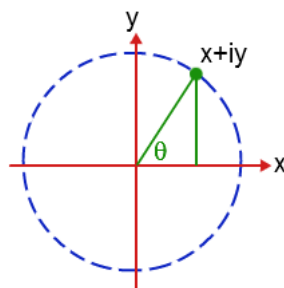
1. Express the following in the form $a + bi$.

$$(a) \frac{(2 + 3i)}{i(4 - 5i)} + \frac{2}{i} \quad [\text{ans:} = \frac{22}{41} - \frac{75}{41}i]$$

$$(b) \frac{(1 + 3i)}{-2i(i - 5)} \quad [\text{ans:} = \frac{4}{13} - \frac{1}{26}i]$$

8.5 Graphical representation of a complex number

A complex number can be represented in the Cartesian plane in which the x -axis is the real axis and y -axis is the imaginary axis. The resulting complex plane is called an *Argand diagram*.



Exercise:

1. Draw the following complex numbers on an Argand diagram

- (a) $4 + i3$
- (b) $-5 - i4$
- (c) $1 - i$
- (d) $-2 + i$

8.6 Modulus and argument of a complex number

8.6.1 Modulus of a complex number

If $z = x + iy$, then the modulus of z is $|z| = r = \sqrt{x^2 + y^2}$ (by Pythagoras theorem). Thus, the modulus of a complex number is the distance between the origin and the point representing the complex number in the Argand diagram.

Exercise::

1. Show that $z \cdot \bar{z} = |z|^2$

2. Prove the following

(a) $|z_1 \cdot z_2| = |z_1||z_2|$

(b) $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$

3. Find the modulus of $z = \frac{(2-i)(5+12i)}{(1+2i)^3}$

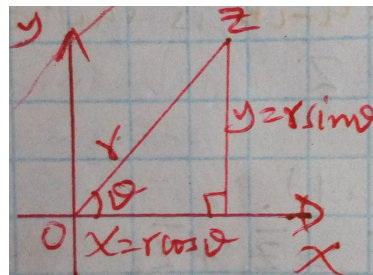
8.6.2 Argument of a complex number

The argument of a complex number, denoted by $\arg(z)$ or θ , is the angle between the positive real axis and the line joining the origin and the complex number measured anticlockwise from the real axis. From trigonometry,

$$\theta = \tan^{-1} \left(\frac{y}{x} \right)$$

8.7 Polar form of a complex number

Just as the Cartesian form of a complex number uses $Re(z)$ and $Im(z)$, the polar form of a complex number uses r and θ .



$$0 \leq \theta \leq 2\pi$$

Consider the complex number $z = x + iy$. From trigonometry, $\cos \theta = \frac{x}{r} \Rightarrow x = r \cos \theta$ and $\sin \theta = \frac{y}{r} \Rightarrow y = r \sin \theta$. Therefore, $z = r \cos \theta + ir \sin \theta = r(\cos \theta + i \sin \theta)$. This is the polar form of z . The argument of z is $\arg(z) = \theta = \tan^{-1} \left(\frac{y}{x} \right)$ and its modulus is $r = |z| = \sqrt{x^2 + y^2}$.

8.8 Euler's identity

It is expressed as

$$e^{i\theta} = \cos \theta + i \sin \theta$$

Therefore, $z = re^{i\theta}$ is the exponential form of a complex number.

$$\cos \theta = \cos(\theta + 2\pi k) \text{ and } \sin \theta = \sin(\theta + 2\pi k), \text{ where } k = 0, 1, 2, \dots$$

Hence, $z = re^{i\theta+2\pi k}$.

→ Note that $e^{-i\theta} = \cos \theta - i \sin \theta$

Example(s):

1. Express $e^{1-i\frac{\pi}{4}}$ in the form $a + ib$.

$$[\text{ans:} = \frac{e}{\sqrt{2}}(1-i)]$$

Solution

$$e^{1-i\frac{\pi}{4}} = e \cdot e^{-i\frac{\pi}{4}} = e \left[\cos \left(\frac{\pi}{4} \right) - i \sin \left(\frac{\pi}{4} \right) \right] = e \left[\frac{1}{\sqrt{2}} - i \frac{1}{\sqrt{2}} \right] = \frac{e}{\sqrt{2}}(1-i)$$

Exercise:

1. If x is real, show that $(2+i)e^{(1+3i)x} + (2-i)e^{(1-3i)x}$ is also real. [ans: $= e^x(4 \cos 3x - 2 \sin 3x)$]

8.9 De Moivre's theorem

To raise a complex number to the n^{th} power, the modulus is raised to the n^{th} power and the argument is multiplied by n , i.e., if $z = r(\cos \theta + i \sin \theta)$, then

$$z^n = r^n(\cos n\theta + i \sin n\theta)$$

→ Note: for complex numbers in the general form $z = x + iy$, it may be necessary to first convert it to polar form before applying the De Moivre's theorem.

Example(s):

1. (a) Using De Moivre's theorem, show that $(\cos \theta + i \sin \theta)^4 = \cos 4\theta + i \sin 4\theta$. Hence or otherwise, prove that [7 Marks]

$$\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$$

Solution

$$\begin{aligned} (\cos \theta + i \sin \theta)^4 &= [(\cos \theta + i \sin \theta)^2]^2 = [\cos^2 \theta - \sin^2 \theta + i2 \cos \theta \sin \theta]^2 \\ &= [\cos 2\theta + i \sin 2\theta]^2 = \cos^2 2\theta - \sin^2 2\theta + i2 \cos 2\theta \sin 2\theta \\ &= \cos 4\theta + i \sin 4\theta \end{aligned}$$

Also from binomial theorem, we have

$$\begin{aligned} (\cos \theta + i \sin \theta)^4 &= \cos^4 \theta + i4 \cos^3 \theta \sin \theta - 6 \cos^2 \theta \sin^2 \theta - i4 \cos \theta \sin^3 \theta + \sin^4 \theta \\ &= (\cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta) + i(4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta) \end{aligned}$$

Equating the corresponding parts we get:

$$\begin{aligned} \cos 4\theta &= \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta \\ \sin 4\theta &= 4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta \end{aligned}$$

Hence,

$$\begin{aligned} \tan 4\theta &= \frac{\sin 4\theta}{\cos 4\theta} = \frac{4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta}{\cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta}, \text{ divide through by } \cos^4 \theta \text{ to get} \\ &= \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta} \end{aligned}$$

- (b) Expand $\cos^3 \theta$ and $\sin^3 \theta$ in terms of sines and cosines of multiples of θ .

Solution

$$\text{Let } z = \cos \theta + i \sin \theta \Rightarrow \frac{1}{z} = z^{-1} = \cos(-\theta) + i \sin(-\theta) = \cos \theta - i \sin \theta. \text{ So,}$$

$$z + \frac{1}{z} = 2 \cos \theta \text{ and } z - \frac{1}{z} = i2 \sin \theta$$

Also by the De Moivre's theorem, we have

$$z^n = \cos n\theta + i \sin n\theta \Rightarrow \frac{1}{z^n} = z^{-n} = \cos(-n\theta) + i \sin(-n\theta) = \cos n\theta - i \sin n\theta$$

So,

$$z^n + \frac{1}{z^n} = 2 \cos n\theta \text{ and } z^n - \frac{1}{z^n} = i2 \sin n\theta$$

Hence,

$$(2 \cos \theta)^3 = \left(z + \frac{1}{z}\right)^3 = z^3 + 3z^2 \cdot \frac{1}{z} + 3z \cdot \frac{1}{z^2} + \frac{1}{z^3} = \left(z^3 + \frac{1}{z^3}\right) + 3\left(z + \frac{1}{z}\right)$$

$$\therefore 8 \cos^3 \theta = 2 \cos 3\theta + 6 \cos \theta \Rightarrow \cos^3 \theta = \frac{1}{4}[\cos 3\theta + 3 \cos \theta]$$

and

$$(i2 \sin \theta)^3 = \left(z - \frac{1}{z}\right)^3 = z^3 - 3z^2 \cdot \frac{1}{z} + 3z \cdot \frac{1}{z^2} - \frac{1}{z^3} = \left(z^3 - \frac{1}{z^3}\right) + 3\left(z - \frac{1}{z}\right)$$

$$\therefore -i8 \sin^3 \theta = i2 \sin 3\theta + i6 \sin \theta \Rightarrow \sin^3 \theta = -\frac{1}{4}[\sin 3\theta + 3 \sin \theta]$$

2. Find the cube root of $-1 + i$

Solution

Let $z = -1 + i \Rightarrow r = |z| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$ and $\theta = \tan^{-1}\left(\frac{1}{-1}\right) = \frac{3\pi}{4}$. So, $z = \sqrt{2} \left[\cos\left(\frac{3\pi}{4}\right) + i \sin\left(\frac{3\pi}{4}\right)\right] = \sqrt{2} \left[\cos\left(\frac{3\pi}{4} + 2\pi k\right) + i \sin\left(\frac{3\pi}{4} + 2\pi k\right)\right]$, where $k = 0, \pm 1, \pm 2, \dots$. Hence by the De Moivre's theorem, we have

$$z^{\frac{1}{3}} = 2^{1/6} \left[\cos\left(\frac{3\pi}{4} + 2\pi k\right) + i \sin\left(\frac{3\pi}{4} + 2\pi k\right)\right]^{1/3} = 2^{1/6} \left[\cos\left(\frac{\pi}{4} + \frac{2\pi}{3}k\right) + i \sin\left(\frac{\pi}{4} + \frac{2\pi}{3}k\right)\right]$$

$$\text{Put } k = 0 \Rightarrow z_1 = 2^{1/6} \left[\cos\left(\frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{4}\right)\right]$$

$$\text{Put } k = 1 \Rightarrow z_2 = 2^{1/6} \left[\cos\left(\frac{11\pi}{12}\right) + i \sin\left(\frac{11\pi}{12}\right)\right]$$

$$\text{Put } k = 2 \Rightarrow z_3 = 2^{1/6} \left[\cos\left(\frac{19\pi}{12}\right) + i \sin\left(\frac{19\pi}{12}\right)\right]$$

Exercise:

- Express $z = -3 - i4$ in polar form. Hence, evaluate $\sqrt[6]{-3 - i4}$ [11 Marks]
- Write $\cos 4\theta$ in terms of powers of $\cos \theta$ only [ans: $\cos 4\theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1$]
- Express $\sin^6 x$ as a series of terms which are cosines of angles that are multiples of x . [ans: $\sin^6 x = \frac{1}{32}(10 - 15 \cos 2x + 6 \cos 4x - \cos 6x)$]
- If $z = x + iy$, find the equation of the locus defined by $\arg(z^2) = -\frac{\pi}{4}$ [ans: $y^2 = x^2 + 2xy$]

Solution

By the De Moivre's theorem, $z^2 = r^2(\cos 2\theta + i \sin 2\theta) \Rightarrow \arg(z^2) = 2\theta = -\frac{\pi}{4}$. Taking the tangent both sides yields: $\tan 2\theta = \tan(-\pi/4) = -1$. But $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$. So,

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} = -1 \Rightarrow 2 \tan \theta = \tan^2 \theta - 1$$

$$\text{But } \tan \theta = \frac{y}{x} \Rightarrow \frac{2y}{x} = \frac{y^2}{x^2} - 1 \Rightarrow y^2 = x^2 + 2xy.$$